

From Adoption to Innovation: State-Dependent Technology Policy in Developing Countries*

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Abstract

When should policymakers transition from adoption-led to innovation-led growth? Using South Korea's historical technology-transfer data, we document three facts: firms benefit more from adoption when the technology gap is larger; sellers charge higher fees as the gap narrows; and adoption generates knowledge diffusion to non-adopters. We build a two-country growth model consistent with these facts. As the gap narrows, productivity gains from adoption diminish and adoption fees rise, eroding the effectiveness of adoption subsidies. Adoption's contribution to Korea's TFP growth falls from 31% to 5%, while innovation's rises from 2% to 24% over 1973–2023. Korea's state-dependent shift from adoption to innovation subsidies improved welfare by 6.2%. Higher initial import tariffs can further improve welfare by inducing foreign sellers to charge lower fees.

Keywords: Technology adoption, technology diffusion, innovation, competition

JEL Codes: F12, F16, L11, N15, O40

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1. INTRODUCTION

Technology adoption is central to growth in developing countries (Parente and Prescott, 1994; Comin and Hobijn, 2010), while innovation is the primary engine of growth in developed countries (Romer, 1990; Grossman and Helpman, 1991; Aghion and Howitt, 1992). The 2024 World Development Report highlights the transition from adoption to innovation as critical for escaping the middle-income trap and achieving sustainable growth (World Bank, 2024). Yet few countries have successfully navigated this transition, and little is known about how to do so. When should policymakers shift from adoption-led to innovation-led growth? Which mix of technology and trade policies can best facilitate this transition?

We study South Korea (henceforth Korea) from the 1970s to the 1990s, an ideal setting for three reasons. First, Korea achieved extraordinary catch-up growth, known as the growth miracle (Lucas, 1993), and emerged as one of the world’s most innovative countries.¹ Second, the Korean government pursued a state-dependent technology policy that initially subsidized adoption and later shifted toward domestic innovation as the country closed the technology gap with the global frontier.² Third, our historical firm-to-firm technology-transfer dataset covers the universe of adoption contracts with fee information and is linked to firm-level outcomes and patent records, allowing us to track each firm’s adoption and innovation investments over several decades of rapid growth.

Our contributions are twofold. First, drawing on our new historical technology-transfer dataset, we document three stylized facts: (i) the advantage of backwardness (Gerschenkron, 1962), (ii) strategic pricing in technology transfers, and (iii) knowledge diffusion to non-adopting domestic firms. Second, we develop and estimate a two-country endogenous growth model of adoption and innovation, consistent with these three facts, and use it for policy evaluation and counterfactual analysis.

Our first stylized fact is the advantage of backwardness: performance improvements after adoption are larger for firms further behind the frontier. Using a matching-based difference-in-differences design, we match first-time adopters (treated) to observably similar never- or not-yet-adopting firms (control), split matched pairs by the median pre-adoption technology gap (measured as productivity relative to the global frontier), and estimate heterogeneous effects across the two groups. We find that firms further behind the frontier experience larger

¹The share of patents by Korean firms in the US Patent Office was less than 0.01% in the 1970s, but increased to 7% in the 2010s, making Korea the third-largest patenting country after the US and Japan.

²Similar policies have been implemented more recently in other developing countries. For example, Brazil shifted from adoption-centered to innovation-centered policies in 2001 (de Souza, 2021), and China emphasized adoption via FDI early on before moving toward an innovation agenda in its 13th Five-Year Plan (2016).

post-adoption growth in size and labor productivity, with no pre-trends. We supplement these baseline results with the winners vs. losers design of [Choi and Shim \(forthcoming\)](#), which compares successful adopters (“winners”) to firms whose contracts were canceled by the foreign seller (“losers”). This design is more robust to selection bias because both groups revealed intent to adopt, though it yields fewer matches. The findings are qualitatively similar across both designs. By contrast, we find no such gap-dependent heterogeneous effects for innovation events, defined as first-time patenting.

The second fact is strategic pricing. Sellers charge higher fees when buyers are closer to the frontier, because a buyer’s marginal productivity gain poses a larger competitive threat when gaps are small. We document this negative relationship between adoption fees and the technology gap in our data. The pattern is robust to controlling for firm size and a rich set of contract- and technology-related quality characteristics selected via LASSO, and is stronger in sectors with higher competitive pressure, which further supports strategic pricing.

The third fact is knowledge diffusion to non-adopters. Using a matching-based difference-in-differences design, we compare observationally similar foreign sellers, those that enter a technology-transfer contract for the first time (treated) and those that have never or not yet done so (control). After the contract, citations to a treated seller’s patents rise among Korean firms that never engaged in any transfer contracts, with no pre-trends. Because these firms had no direct interaction with the seller, this pattern supports diffusion. Placebo tests confirm this interpretation: firms in other countries show no comparable citation spike, ruling out seller-level time-varying unobservable shocks as an explanation. A more granular patent-level analysis, which allows us to absorb patent-level time-invariant unobservables and restrict our focus to patents that were filed before the event, further confirms this finding. Patent-level heterogeneity analysis shows that diffusion concentrates in higher-quality technologies and in patents directly cited by firms engaged in adoption contracts.

To build intuition, we first study a static, closed-economy model featuring the advantage of backwardness (Fact 1) and knowledge diffusion (Fact 3). The economy has two firm types, leaders and fringe firms, each producing a unique CES variety, and only leaders engage in adoption or innovation. Both activities have exogenously given step sizes and their successes occur probabilistically, with higher success probabilities requiring larger costs. Fringe firms benefit through diffusion, which likewise occurs probabilistically.

We derive closed-form optimal subsidies and show they favor activities with larger productivity step sizes, remaining positive even without diffusion. This reflects a surplus appropriability problem: firms capture only a fraction of the surplus they generate. Because this fraction falls as the step size grows, larger steps imply more severe underinvestment and

larger required subsidies. Consequently, when firms lag far behind the frontier, adoption delivers larger productivity gains than innovation and thus warrants heavier subsidies, but the optimal mix shifts toward innovation as firms approach the frontier and adoption's step-size advantage erodes. Higher diffusion raises the optimal subsidy rates for both activities but narrows the gap between the rates, as diffusion yields smaller marginal gains when the leader's productivity gain is already large.

To conduct counterfactual analysis, we construct a quantitative two-country step-by-step growth model that extends the simple model in three directions: (i) open-economy interactions, (ii) strategic pricing (Fact 2) through endogenous adoption fees via buyer-seller bargaining, and (iii) transitional dynamics where the advantage of backwardness endogenously declines as the technology gap narrows.

Leaders can upgrade productivity either by adopting foreign technologies or through their own innovation, with knowledge diffusion extending these gains beyond leaders to fringe firms (Fact 3). Adoption and innovation differ in three key respects: only adoption exhibits the advantage of backwardness (Fact 1); it cannot raise a domestic firm's productivity above a foreign seller's level, whereas innovation faces no such ceiling; and it entails a fixed adoption fee. This adoption fee is set by Nash bargaining between buyers and sellers. While sellers receive a one-time fee, successful adoption erodes their future profits by narrowing the technology gap and intensifying global competition. Two opposing forces shape how fees vary with the gap. A narrower gap weakens the advantage of backwardness, generating smaller gains for buyers and pushing fees down. Yet even a marginal productivity gain can substantially shift global market shares, inducing sellers to seek compensation and pushing fees up. When the latter competition effect dominates, fees rise as the gap narrows (Fact 2).

In an open-economy setup, strategic profit-shifting motives for policy intervention arise, and trade policy interacts with technology policy. Higher tariffs reduce sellers' export profits, lowering their outside option, pushing adoption fees down, and thereby increasing adoption rates. When a country is initially far from the frontier, higher tariffs, combined with the advantage of backwardness, can improve welfare despite raising domestic price levels.

We calibrate the model to firm-level data by simulating moments along the transition path from the initial state to a balanced-growth path and matching them to their data counterparts. In particular, we infer the advantage of backwardness and the rate of domestic diffusion by targeting Facts 1 and 3. The calibrated model also reproduces four untargeted moments. It matches the observed decline in the adoption expenditure share and the negative correlation between adoption fees and technology gaps (Fact 2), the latter as a consequence of strategic pricing. It also reproduces rising firm concentration driven by adoption-induced

productivity gains, consistent with [Choi et al. \(2023\)](#), and rising export intensity as productivity gains expand firm exports, consistent with Korea's export-led growth ([Ventura, 1997](#)). These moments validate the model's key mechanisms and capture salient features of Korea's growth experience.

Using the calibrated model, we decompose aggregate TFP growth into contributions from adoption, innovation, domestic diffusion, and international diffusion. Adoption accounts for 31% of TFP growth in 1973, but its contribution falls to 5% by 2023 as Korea approaches the frontier. Innovation follows the opposite trajectory, rising from 2% to 24% as the technology gap narrows. The decomposition highlights the importance of adoption in Korea's early catch-up growth.

Next, we evaluate Korea's state-dependent technology policy against two alternatives that hold the subsidy-to-GDP ratio fixed, one allocating the entire budget to adoption subsidies and the other to innovation subsidies. The actual policy raises welfare by 6.22% relative to the no-subsidy scenario, outperforming both the adoption-only (4.62%) and innovation-only (3.63%) policies.

We also solve for the optimal subsidy policy under two scenarios. In the first (constrained optimum), the government maximizes welfare by optimizing the adoption-innovation allocation while holding the subsidy-to-GDP ratio fixed at the actual level, reflecting the government's fiscal capacity. The resulting allocation closely resembles the actual policy, yielding a modest welfare gain of 6.51%, modestly above the 6.22% gain of the actual policy. In the second (unconstrained optimum), we remove the budget constraint and allow the government to freely choose subsidy rates. The qualitative pattern is similar, with adoption subsidies front-loaded and declining over time. Although the welfare gain rises to 10.96%, achieving it requires a subsidy-to-GDP ratio 213% higher than under the actual policy.

Absent the actual policy, Korea's TFP convergence toward Japan would have been 11% lower. Holding Japan's global profit share constant through a lump-sum transfer from Korea reduces the welfare gain by 13%, the share attributable to profit shifting from Japan to Korea.

To examine how technology policies interact with import tariffs, we compute the optimal tariff given the actual subsidy schedule. The optimal path starts at 22% and gradually falls to zero by 2023. Initially, higher tariffs lower foreign adoption fees and increase adoption rates. As Korea catches up with the frontier, however, adoption becomes less important, and higher tariffs instead raise consumer prices and weaken innovation incentives by reducing competitive pressure. The optimal tariff yields additional welfare gains of 0.2 percentage points over the actual policy, while jointly optimal tariffs and subsidies yield 0.4 percentage points.

Our paper contributes to several strands of the literature. First, we contribute to the macro literature on innovation and growth in developing countries (e.g., [Vandenbussche et al., 2006](#); [Hsieh and Klenow, 2014](#); [König et al., 2022](#); [Peters, 2020](#); [Akcigit et al., 2021](#); [de Souza, 2021](#); [Buera and Fattal-Jaef, 2025](#); [Choi et al., 2025](#)). The advantage of backwardness was first proposed by [Gerschenkron \(1962\)](#) and further formalized by [Acemoglu et al. \(2006\)](#), who show that policy should shift from adoption to innovation as a country approaches the frontier, because the advantage of backwardness declines and innovation induces stronger selection of high-skilled managers. Our main contribution is to provide the first micro-level evidence of this mechanism using novel historical microdata and to develop a quantitative framework, disciplined by these data and reduced-form estimates, for studying the transition from adoption- to innovation-led growth. [Acemoglu et al. \(2018\)](#), [Akcigit et al. \(2021\)](#), [Akcigit et al. \(2022\)](#), and [Aghion et al. \(2023\)](#) study innovation policies for developed economies in closed-economy settings. Our open-economy setup allows us to study how trade and technology policies interact, which matters for developing countries that are small, trade-dependent, and reliant on foreign sources for advanced technologies.

Second, our paper is related to the literature on step-by-step growth models ([Aghion et al., 2001, 2005](#); [Olmstead-Rumsey, 2022](#); [Liu et al., 2022](#); [Sui, 2023](#); [Cavenaile et al., 2023](#); [Akcigit et al., 2024](#)). We extend the open-economy framework of [Akcigit et al. \(forthcoming\)](#) in two directions: introducing adoption of foreign technologies and endogenous adoption fees via Nash bargaining following [Benhabib et al. \(2021\)](#) and [Hopenhayn and Shi \(2020\)](#), and modeling two domestic firms with diffusion between them as in [Akcigit and Ates \(2023\)](#) and [Cavenaile et al. \(2025\)](#). Consistent with the central theme of this literature that technology investment depends on gaps and competitive pressure, we contribute by studying how the advantage of backwardness and strategic competition jointly shape productivity gains, adoption fee pricing, and the relative effectiveness of adoption versus innovation.

Third, this paper contributes to the literature on dynamic models of trade, innovation, and knowledge diffusion (e.g., [Grossman and Helpman, 1993](#); [Atkeson and Burstein, 2010](#); [Impullitti, 2010](#); [Santacreu, 2015](#); [Impullitti and Licandro, 2018](#); [Buera and Oberfield, 2020](#); [Rachapalli, 2021](#); [Hsieh et al., 2023](#); [Perla et al., 2021](#); [Sampson, 2016, 2023](#); [Cai et al., 2022](#); [de Souza et al., 2025](#); [Moscona and Sastry, 2025](#)). Among recent contributions, [Santacreu \(2025a\)](#) studies dynamic gains from trade agreements with IP provisions, [Milicevic et al. \(2026\)](#) examines R&D policy coordination in the EU, and [Hémous et al. \(2025\)](#) analyzes optimal patent policy in the global economy. Whereas prior papers infer international diffusion from patents and citations ([Eaton and Kortum, 1999](#); [Keller, 2004](#); [Aghion et al., 2024](#)), trade flows ([Lind and Ramondo, 2023](#)), or licensing payments ([Santacreu, 2025b](#)), we use direct

firm-level measures with price information.

Lastly, this paper relates to the macro literature on South Korea’s growth miracle (e.g., [Lucas, 1993](#); [Ventura, 1997](#); [Buera and Shin, 2013](#); [Connolly and Yi, 2015](#); [Choi and Levchenko, 2025](#); [Kim et al., 2021](#); [Choi et al., 2023](#)). Our companion paper [Choi and Shim \(forthcoming\)](#) studies how the HCI Drive in the 1970s promoted technology adoption in heavy manufacturing, while this paper studies the subsequent economy-wide shift from adoption to innovation subsidies during the 1980s–1990s, when Korea transitioned to high-income status with growing innovation. As in [Itskhoki and Moll \(2019\)](#), optimal policy in our model is dynamic and state-dependent, though here state-dependence arises from the technology gap rather than financial frictions. [Chung and Lee \(2015\)](#) use the same technology-transfer data to show correlations between adoption and subsequent patenting post-1980s. We instead document different empirical facts and embed them in a quantitative model for counterfactual analysis, drawing on a historical dataset extending back to the 1970s, when Korean firms were most active in adoption.

Section 2 presents the data and background, Section 3 three stylized facts, Section 4 the simple model, Section 5 the quantitative framework and calibration, and Section 6 the counterfactuals. Section 7 concludes.

2. BACKGROUND AND DATA

2.1 Data

We construct our main dataset by merging firm-level technology-adoption records, patent data, and firm outcomes (employment, capital, sales, and exports). The dataset covers manufacturing firms between 1970 and 1993. Appendix A.1 describes its construction in detail.

Our data build on the firm-to-firm technology-transfer dataset of [Choi and Shim \(forthcoming\)](#), expanded in this paper to include information on foreign counterparties and reported adoption fees, with an extended sample window.³ The resulting dataset covers the universe of official technology transfers between Korean and foreign firms from 1970 to 1993, with 8,346 contracts, roughly 75% of which can be matched to firm outcomes.⁴ About 95%

³From 1962 to 1993, under the Foreign Capital Inducement Act, Korean firms were required to submit documents for transactions with foreign entities. Because of balance-of-payments concerns, the authorities strictly regulated unauthorized contracts. The National Archives of Korea preserved these records, which form the basis of our dataset. Appendix Figure A.1 shows an example document.

⁴The dataset records only official, documented measures of technology adoption from foreign firms. Other channels, such as diffusion via foreign direct investment (FDI), transfers between Korean firms, or informal adoption through reverse engineering, are not captured. That said, FDI played a limited role in Korea’s industrialization because of government regulation of FDI ([Kim, 1997](#), pp. 42–43), estimated intra-domestic adoption

Table 1: Examples of Technology Transfer Data

Buyer	Seller	Contract Length (year)	Date	Technology	Contents	Fee
Samsung Electronics	Nippon Electronic (Japan)	10	02/24/1978	Color TV	Know-how Transfer, Licensing	Fixed \$800,000
LG Electronics	Hitachi (Japan)	9	04/01/1978	Color TV	Know-how Transfer, Licensing	Fixed \$100,000 Royalty 3%
Hyundai Motors	Kyukoto Engineering (Japan)	3	06/14/1979	Concrete mixer	Know-how Transfer	Royalty 5%

Table 2: Summary Statistics of South Korean Firms

	Ever-Adopted	Never-Adopted	Ever-Patented	Never-Patented	All
Emp.	1,230	311	1,783	355	635
Asset	185	18	310	23	70
Sales	204	29	316	33	96
Sales / Emp.	0.17	0.14	0.20	0.14	0.16
Patent (dummy)	0.05	0.00	0.13	N/A	0.02
Adopt (dummy)	0.14	N/A	0.12	0.02	0.03
# Firms	1,226	7,941	553	8,614	9,167
N	23,714	88,528	13,266	98,976	112,242

Notes. All nominal values are converted to 2015 US million dollars.

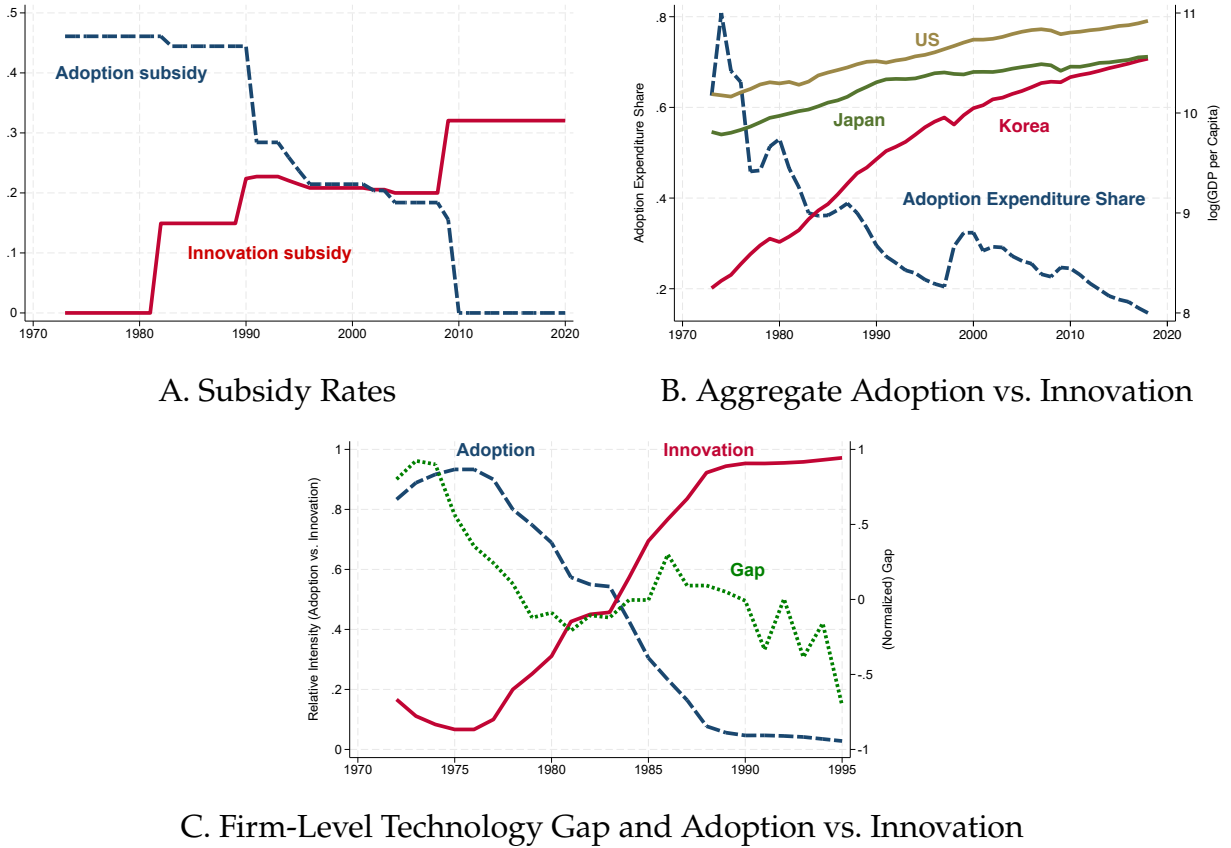
of the contracts involve know-how transfers (technical assistance, training, and blueprint sharing), while 5% are licensing-only agreements. These transactions occurred between independent firms and involved monetary compensation for technology. Table 1 illustrates the dataset structure. Fees took the form of a lump-sum payment, a royalty rate, or both, and were sizable. The average royalty rate was 3% over a 5-year contract, and fixed fees averaged 10% of annual sales. Japanese and US firms accounted for 50% and 27% of contracts.

To measure innovation by Korean firms, we use patent data from the Korean Intellectual Property Office (KIPO) and apply the cleaning procedure of [Lee et al. \(2020\)](#). The dataset includes the universe of patents registered in Korea since 1945. Because KIPO data lack citation information until 1990, we supplement them with data from the United States Patent and Trademark Office (USPTO), which includes all patent citations made by patents that are registered in the USPTO since 1975.

Korean firm-level data on employment, fixed assets, sales, and exports come from two sources. For 1970–1982, we use the digitized Annual Reports of Korean Companies, and for

fees account for only a small share of total adoption expenses ([Lee, 2022](#)), and we treat such informal channels as exogenous knowledge diffusion in the model.

Figure 1: State-Dependent Technology Policy and Adoption versus Innovation in Korea



Notes. Panel A plots adoption and innovation subsidy rates. Panel B plots real GDP per capita, alongside Korea's adoption expenditure share (adoption fees paid to foreign firms divided by total adoption and R&D expenditures). Panel C plots, for Korea's three largest firms, average shares of adoption contracts and patents (normalized by their sum) and average technology gaps (normalized by the median).

1983–1995, the KIS-VALUE. The combined dataset covers about 70% of total manufacturing gross output. Table 2 reports summary statistics by adoption and patenting status. Adopters and patenters tend to be larger than their counterparts. For foreign firms, we obtain outcomes from Compustat North America (US firms, 1970–1995) and Worldscope (non-US firms, 1980–1995), both covering publicly listed firms. We link Compustat to USPTO patent records using the concordance of [Bena et al. \(2017\)](#).

We obtain aggregate expenditures on adoption and R&D from Statistics Korea. When measuring aggregate adoption expenditure, we use total payments to foreign countries for the use of intellectual property. We obtain input-output (IO) tables from the Bank of Korea and trade data from Comtrade.

2.2 Background

In 1966, the Korean government first introduced subsidies for foreign technology adoption costs.⁵ The initial subsidy rate of 46% gradually fell to 28% by 1991 and further to 12% thereafter, where it remained until adoption subsidies were discontinued in 2010. Meanwhile, support for domestic innovation grew steadily.⁶ Beginning at 15% in 1982, the R&D subsidy rate rose to 22% by 1990 and to 32% by 2009, where it has since remained (Choe and Lee, 2012). Following Bloom et al. (2002) and Impullitti (2010), we translate statutory tax credits into effective subsidy rates using corporate tax rates (see Appendix A.2 for details).

Adoption and innovation subsidy rates diverged (Panel A of Figure 1), while South Korea’s real GDP per capita converged toward advanced economies (Panel B). Consistent with this policy shift, the aggregate adoption expenditure share declined, suggesting adoption was more central at earlier stages of development. Panel C shows the same shift at the firm level, with the transition tied directly to each firm’s technology gap, defined as the ratio of frontier to firm labor productivity:

$$\text{Gap}_{it} = \frac{z_{jt}^{\text{frontier}}}{z_{it}}, \quad (2.1)$$

where the frontier is the maximum productivity among US firms in the same sector-year jt .⁷ Korea’s three largest firms (Samsung Electronics, Hyundai Motors, and LG Electronics) shift toward innovation as they close their gaps, mirroring the aggregate pattern. This shift holds broadly across firms and is robust to alternative normalizations (Appendix Figure A.3).

3. STYLIZED FACTS

We present three stylized facts on adoption: advantage of backwardness, strategic pricing, and knowledge diffusion.

⁵Tax credits for adoption fees were provided under Article 24 of the Foreign Capital Inducement Act (1966), titled “On tax exemptions for expenses for the adoption of foreign technologies.” This policy is distinct from the HCI Drive (1973–1979), studied in recent work (e.g., Choi and Levchenko, 2025; Kim et al., 2021). Unlike the HCI Drive, which targeted specific heavy manufacturing subsectors, the policy we analyze was a broad, long-run subsidy applied across all manufacturing sectors.

⁶An R&D tax credit was introduced under Article 14 of the Tax Reduction and Exemption Regulation Act in 1982. At a conference that year, President Chun declared that “Korea needs to transition from the imitation stage of the 1970s to creativity and innovation in the 1980s.”

⁷Griffith et al. (2004), Gorodnichenko et al. (2010), Hashmi (2013), and Levchenko and Zhang (2016) use similar frontier measures. We use Compustat North America for its longer time coverage. Results are qualitatively similar using Worldscope, which begins in 1980 (Appendix Figure A.3).

Fact 1. Advantage of Backwardness

The advantage of backwardness implies that more technologically backward firms experience larger performance improvements following adoption. To test this prediction, we compare first-time adopters (treated) to firms that had never or not yet adopted (control), estimating heterogeneous effects by their technology gap measured 1 year before adoption. The control group includes firms that never adopt during the full sample period and firms that adopt only after the event window. The estimation sample is constructed by matching on observables over an event window spanning 4 years before and 7 years after adoption. Within sectors, we match first-time adopters to non-adopting firms on pre-event observables (log sales, employment, and technology gap) using Mahalanobis distance. Each treated firm is matched to at most 2 control firms, with replacement, so the same firm may serve as a control in multiple matches. The procedure yields 515 matched pairs involving 1,064 unique firms.

A remaining concern is that matching on observables may still be biased if firms select into adoption based on time-varying unobservable shocks. To address this, we supplement the baseline matching with a winners vs. losers design following [Choi and Shim \(forthcoming\)](#). Winners are firms that successfully adopted, and losers are firms whose negotiations failed because the foreign seller canceled the contract for reasons plausibly exogenous to the loser. These cancellation reasons are documented in the raw contract records (available through 1985), which firms were required to submit to the government upon cancellation. We exclude cancellations initiated by Korean firms to minimize endogeneity concerns. Because both groups revealed adoption intent, selection on unobservables is mitigated. While [Choi and Shim \(forthcoming\)](#) estimate the average effect of adoption, we extend the design to study heterogeneous effects by technology gap. Appendix B.1 provides further details.

For each loser, we match up to 2 winners in the same sector and region who adopted technology in the year of the cancellation and were most similar in pre-event size (measured by total assets). This design yields 41 matches with 91 unique firms.⁸ Another caveat is that, unlike the baseline matching, the sample is not restricted to first-time adopters, but the results remain informative as long as the advantage of backwardness holds for repeated adopters as well. We also restrict the sample to ensure treated firms are compared to a clean control group. Of the 41 matches, 18 involve losers that never subsequently adopted and 23 involve losers that eventually adopted after a delay. We construct the estimation sample using a rolling control group, dropping matches once delayed losers adopt, so that the coefficients

⁸Our matching variables follow [Choi and Shim \(forthcoming\)](#), but we obtain 41 matches rather than their 38, owing to our more aggregated regional classification. Because labor productivity is missing in some years, we define the large- and small-gap indicators using the average gap over the 4 pre-adoption years.

are identified solely from within-match comparisons.

Using the estimation samples from the two designs, we estimate

$$y_{imt} = \beta_L (\mathbb{1}[\text{Gap}_m^{\text{large}}] \times \mathbb{1}[\text{Adopt}_{imt}]) + \beta_S (\mathbb{1}[\text{Gap}_m^{\text{small}}] \times \mathbb{1}[\text{Adopt}_{imt}]) + \delta_{im} + \delta_{mt} + \varepsilon_{imt}, \quad (3.1)$$

where i indexes firms, m matches, and t years. $\mathbb{1}[\text{Adopt}_{imt}]$ is the post-treatment indicator that switches from 0 to 1 in years after the event year $t(m)$. The indicators $\mathbb{1}[\text{Gap}_m^{\text{large}}]$ and $\mathbb{1}[\text{Gap}_m^{\text{small}}]$ flag matches in which the treated firm's pre-adoption gap lies above or below the sample median, respectively. δ_{im} denotes firm-match fixed effects and δ_{mt} match-year fixed effects. ε_{imt} is the error term. Standard errors are two-way clustered at the match and firm levels.

β_L and β_S capture the heterogeneous post-adoption effects for large- and small-gap groups. Our stacked-by-event specification is robust to biases arising from heterogeneous treatment effects across cohorts, as emphasized in the recent staggered diff-in-diff literature (Roth et al., 2023). We also assess robustness to alternative gap measures.

To examine dynamic effects and pre-trends, we also estimate the following event study:

$$y_{imt} = \sum_{\tau=-4}^7 \beta_l^\tau \left(\mathbb{1}[\text{Gap}_m^{\text{large}}] \times \mathbb{1}[\text{Adopt}_{imt}^\tau] \right) + \sum_{\tau=-4}^7 \beta_s^\tau \left(\mathbb{1}[\text{Gap}_m^{\text{small}}] \times \mathbb{1}[\text{Adopt}_{imt}^\tau] \right) + \delta_{im} + \delta_{mt} + \varepsilon_{imt}, \quad (3.2)$$

where $\mathbb{1}[\text{Adopt}_{imt}^\tau]$ is an event-time indicator equal to 1 when $t = t(m) + \tau$ for treated firms.

For both research designs, the identifying assumption is that adoption timing is uncorrelated with time-varying unobservables (i.e., the parallel-trends assumption holds).⁹ Treated and control groups are well balanced not only on the variables used for matching but also on those not used for matching, and pre-event observables do not predict treatment status (see Appendix Tables B.1–B.2 for baseline and B.3–B.4 for winners vs. losers), which supports the identifying assumption. We assess the parallel-trends assumption by examining pre-event coefficients in the event study specifications and find no evidence of pre-trends.

Columns 1–4 of Table 3 report results for the baseline matching. Column 1 shows that, on average, adoption raises labor productivity by 9%. Column 2 reports heterogeneous effects. Firms that are more backward (larger gaps) exhibit larger gains, while the effect for

⁹Due to the inclusion of firm-match fixed effects, selection based on time-invariant unobservables does not bias our estimates, but time-varying unobservables may result in spurious correlations. For example, productivity shocks that make firms more likely to adopt also improve firm performance, leading to spurious correlations between adoption and performance.

Table 3: Advantage of Backwardness: Post-Adoption Effects by Technology Gap

Research Design	Baseline matching				Winners vs. losers			
Dep. Var.	Log labor prod.		Log sales		Log labor prod.		Log sales	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\mathbb{1}[\text{Adopt}_{imt}]$	0.09*** (0.03)		0.15*** (0.04)		0.62*** (0.18)		0.37* (0.19)	
$\times \mathbb{1}[\text{Gap}_m^{\text{large}}]$		0.19*** (0.05)		0.25*** (0.07)		0.87** (0.34)		0.82** (0.35)
$\times \mathbb{1}[\text{Gap}_m^{\text{small}}]$		0.02 (0.04)		0.07* (0.04)		0.41** (0.15)		0.01 (0.13)
$p\text{-val } (\beta_L = \beta_S)$		[0.01]		[0.02]		[0.22]		[0.04]
Dep. mean	11.29	11.29	17.45	17.45	11.13	11.13	18.03	18.03
N	11,448	11,448	11,448	11,448	820	820	820	820
Fixed effects	Firm $\times$ match, Match $\times$ year							
# Cl. (firm $\times$ match)	1,064 $\times$ 515				91 $\times$ 41			

Notes. Standard errors in parentheses are two-way clustered at the firm and match levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the OLS estimates of equation (3.1). p -values in the brackets are associated with the hypotheses that the estimated coefficients of more backward and the other groups are equal. In columns 1–2 and 5–6 (3–4 and 7–8), the dependent variables are log labor productivity (log sales).

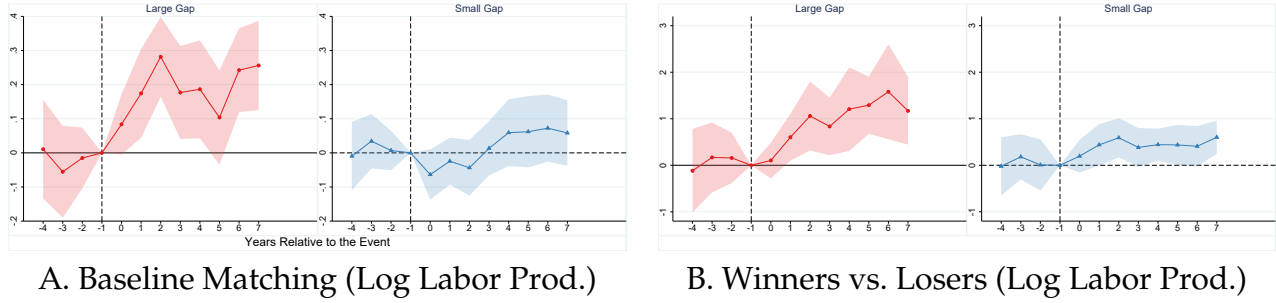
less-backward firms is smaller and imprecisely estimated. The null hypothesis that the two coefficients are equal is rejected at the 5% level. Columns 3–4 show a qualitatively similar pattern using log sales as the outcome, with more-backward firms expanding more after adoption. Columns 5–8 report results from the winners vs. losers design.¹⁰

The event study results also confirm these findings (Figure 2). Importantly, we find no pre-trends in either design. Adoption effects grow over the first 3 or 4 years after the event and then stabilize, and the large-gap (more backward) group exhibits substantially larger labor productivity gains throughout the post-event window.

The estimated average and dynamic effects from the winners vs. losers design are qualitatively consistent with the baseline matching but larger in magnitude. These larger effects likely reflect two factors. First, the cancellation records are available only through 1985, while the baseline matching covers post-1985 events. Because the advantage of backwardness has weakened as the gap has narrowed, post-1985 events show smaller effects. Restricting the baseline matching to pre-1985 events increases its magnitude by 50%. Second, the winners vs.

¹⁰ A potential concern with this design is finite-sample bias from highly leveraged observations (Young, 2019) and the small number of clusters (Cameron et al., 2008). Despite the smaller sample, the results are robust to randomization inference and bootstrap methods that address these concerns (see Appendix Table B.5).

Figure 2: Advantage of Backwardness: Post-Adoption Effects by Technology Gap



Notes. This figure presents the estimated β_l^τ and β_s^τ in equation (3.2) for the baseline matching and winners vs. losers research designs in Panels A and B. Shaded bands are 90% confidence intervals based on standard errors two-way clustered at the match and firm levels. In both panels, the dependent variable is log labor productivity.

losers sample is more skewed toward large and exporting firms, and the baseline matching based on observables may not completely remove selection bias.¹¹

These heterogeneous effects are robust to varying the number of matches. Because labor productivity may not capture sectoral differences in capital intensity, we also consider revenue TFP (TFPR) as an alternative performance measure, estimated as residuals from production functions. We similarly consider market shares and TFPR as alternative gap measures. Appendix Figure B.2 and Appendix Table B.6 report these results.

To examine whether the advantage of backwardness extends to innovation, we conduct the same matching-based analysis for firms around their first patent application. We use the same matching procedure and set of observables as for the adoption analysis. Treated and control groups are well balanced for the innovation analysis as well (Appendix Tables B.7–B.8). Table 4 reports results, and the event study shows no pre-trends (Appendix Figure B.3). For innovation, we find no evidence of the advantage of backwardness. This is consistent with the idea that innovation builds on a firm’s own technological capabilities, making the technology gap less relevant to the returns from own R&D. For both labor productivity and sales, the coefficients for the large- and small-gap groups are similar in magnitude, and we cannot reject the hypothesis that they are equal.¹² These results suggest that the advantage of backwardness is specific to adoption and does not extend to a firm’s own innovation.

¹¹ Although the winners vs. losers sample tends to be larger in firm size and more likely to export, its sectoral and regional composition closely resembles that of the full contract sample, consistent with cancellation episodes being plausibly random (Appendix Figure B.1).

¹² Turning to average effects, first-time patenting is associated with an increase in sales, but its effect on labor productivity is close to zero and statistically insignificant. The average productivity effect is sensitive to the measure used. Using TFPR yields statistically significant effects (Appendix Table B.9), but even for TFPR we cannot reject equality of the coefficients for the two gap groups. This sensitivity is well documented (e.g., Syverson, 2011).

Table 4: No Evidence of Advantage of Backwardness for Innovation

Dep. Var.	Log labor prod.		Log sales	
	(1)	(2)	(3)	(4)
$\mathbb{1}[\text{Innovate}_{imt}]$	0.02 (0.03)		0.17*** (0.04)	
$\times \mathbb{1}[\text{Gap}_m^{\text{large}}]$		0.02 (0.03)		0.16*** (0.05)
$\times \mathbb{1}[\text{Gap}_m^{\text{small}}]$		0.01 (0.05)		0.18*** (0.06)
$p\text{-val } (\beta_l = \beta_s)$		[0.92]		[0.78]
Dep. mean	11.70	11.70	18.26	18.26
N	8,674	8,674	8,674	8,674
Fixed effects	Firm $\times$ match, Match $\times$ year			
# Cl. (firm $\times$ match)	755 $\times$ 390			

Notes. Standard errors in parentheses are two-way clustered at the firm and match levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the OLS estimates of equation (3.1), with $\mathbb{1}[\text{Innovate}_{imt}]$ replacing $\mathbb{1}[\text{Adopt}_{imt}]$ as the post-event indicator. p -values in the brackets are associated with the hypothesis that the estimated coefficients of more backward and the other groups are equal. In columns 1–2 and 3–4, the dependent variables are log labor productivity and sales.

Fact 2. Strategic Pricing due to Competition

Next, we provide evidence on strategic pricing. Our hypothesis is that foreign sellers charge higher fees when buyers are neck-and-neck with them, because a buyer's productivity catch-up then poses a larger threat to the seller's future profits. This hypothesis is consistent with a negative correlation between adoption fees and the gap. To document this negative correlation, we run the following regression:

$$\log \text{Fee}_{ift} = \beta \log \text{Gap}_{it} + \mathbf{Quality}'_{ift} \gamma + \varrho \log \text{Assets}_{it} + \delta_{jct} + \varepsilon_{ift}. \quad (3.3)$$

Fee_{ift} is the adoption fee paid by Korean buyer i to foreign seller f for a contract in year t . $\log \text{Gap}_{it}$ is standardized. δ_{jct} denotes sector-country-year fixed effects. ε_{ift} is the error term. Standard errors are two-way clustered at the Korean buyer and foreign seller levels.

$\mathbf{Quality}_{ift}$ is a vector of characteristics capturing the quality and content of transferred technologies. Based on brief descriptions of the technologies transferred, we assign a 3-digit International Patent Classification (IPC) to each contract. To flexibly control for technology heterogeneity, we use a post-double-selection LASSO procedure following Belloni et al. (2014) to select covariates from country-IPC fixed effects, foreign sellers' technology levels

(measured by patent and citation stocks), and the full set of interactions between these variables (see Appendix B.3 for details). $\log \text{Assets}_{it}$ is log total assets, capturing firm size. We use log assets rather than sales as our size control because sales enter the construction of the gap, and assets better capture firm size as a stock. By including Quality_{ift} and $\log \text{Assets}_{it}$, we isolate variation in technology gaps from variation in technology quality and firm size.

We use fixed fees as our baseline outcome and report robustness checks using total fees (the sum of fixed fees and calculated royalty payments).¹³ We use fixed fees as the baseline because royalties require strong assumptions about the product composition of future sales and discount factors. Moreover, fixed fees account for the majority of payments. Of all contracts, 78% include a fixed-fee component: 42% specify fixed fees only, 36% specify both fixed fees and royalties, 17% specify royalties only, and 5% specify neither.

Table 5 reports the results. Column 1 shows a negative and highly significant correlation between technology gaps and fixed fees. Column 2 adds LASSO-selected quality controls and log assets. The coefficient attenuates by 52% but remains significant. A one standard deviation increase in the log gap is associated with a 23% reduction in fixed fees. The result is robust to using total fees as the outcome, to more granular fixed effects, and to alternative gap measures (Appendix Table B.10).¹⁴

An alternative explanation is that more advanced buyers with smaller gaps simply purchase higher-quality technologies and therefore pay higher fees. We view this as unlikely to drive the negative correlation for two reasons. First, the gap coefficient is stable after controlling for technology-quality proxies, and δ_{jct} absorbs technological trends at the sector-country-year level. Second, if more advanced Korean firms were systematically buying higher-quality technologies, we would expect those buyers to experience larger post-adoption performance improvements. However, Fact 1 shows larger gains for firms farther from the frontier, which is inconsistent with this alternative.

To further explore the mechanism, we examine heterogeneous effects based on sectoral competitive pressures. With strategic pricing, the gap–fee relationship should be stronger in sectors more exposed to competition. We consider three measures of competitive intensity. The first is import tariffs, where lower values imply greater foreign competition, so we classify

¹³The present-discounted royalty payment is calculated as $\sum_{\tau=1}^{T(c)} (1/1.3)^\tau \times \text{Royalty Rate}_c \times \frac{\text{sales}_{i,t(c)+\tau}}{\# \text{Product}_{it(c)}}$, where $T(c)$ denotes the duration of contract c , $t(c)$ denotes the year in which contract c was signed, and 1.3 reflects the average borrowing rate (Cole and Park, 1983). For a subset of firms and years, we digitize the products that firms produce and assign them to 4-digit SITC codes. For missing information, we impute the number of products using log assets, which are positively correlated with the number of products. Since sales per product are not available, we divide total sales by the imputed number. Our results remain robust using total fees as the outcome (Appendix Table B.10).

¹⁴Including more granular fixed effects based on 3-digit IPC classes reduces the number of observations, so we use the current set as the baseline.

Table 5: Adoption Fees Tend to be Higher with Smaller Gap

Dep. Var.	Log fixed fee									
	Average		Heterogeneous effect							
	effect		Tariff		Elast. subst. (BW)		Elast. subst. (S)		1–Lerner index	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
log Gap _{it}	−0.48*** (0.10)	−0.23** (0.10)								
×1 [High Sensitivity _j]			−0.66*** (0.11)	−0.38*** (0.11)	−0.71*** (0.14)	−0.40*** (0.13)	−1.11*** (0.23)	−1.11*** (0.19)	−0.71*** (0.16)	−0.47*** (0.15)
×1 [Low Sensitivity _j]			−0.29** (0.13)	−0.09 (0.12)	−0.28** (0.14)	−0.10 (0.13)	−0.38*** (0.11)	−0.11 (0.09)	−0.32*** (0.12)	−0.08 (0.11)
log Assets _{it}		0.21*** (0.03)		0.21*** (0.03)		0.21*** (0.03)		0.21*** (0.03)		0.21*** (0.03)
LASSO quality ctrl.		✓		✓		✓		✓		✓
<i>p</i> -val			[0.02]	[0.01]	[0.03]	[0.07]	[0.01]	[0.01]	[0.06]	[0.03]
Dep. mean	12.85	12.85	12.85	12.85	12.85	12.85	12.85	12.85	12.85	12.85
N	3,022	3,022	3,022	3,022	3,022	3,022	3,022	3,022	3,022	3,022
Fixed effects	Sector×country×year									
# Cl. (firm×seller)	558×2,213									

Notes. Standard errors in parentheses are two-way clustered at the Korean firm and foreign seller levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table reports OLS estimates of equation (3.3). log Gap_{it} is standardized. High (Low) Sensitivity denotes above- (below-)median competition sectors based on each measure. BW and S denote elasticities of substitution from Broda and Weinstein (2006) and Soderbery (2015). The Lerner index is computed following Aghion et al. (2005). *p*-val is the *p*-value for the test of equality between High and Low Sensitivity coefficients. Quality controls are selected via the post-double-selection LASSO of Belloni et al. (2014) and included in even-numbered columns.

sectors with below-median tariffs as high-competition sectors. The second is the elasticity of substitution, where higher values mean productivity gains more sharply erode sellers' market shares, so sectors with above-median elasticities are classified as high-competition sectors. We measure sectoral elasticities using estimates from Broda and Weinstein (2006) and Soderbery (2015), taking the median elasticity across narrowly defined products within each sector.¹⁵ Third, we use one minus the sectoral Lerner index following Aghion et al. (2005),¹⁶ which increases with competition, and we classify sectors with above-median values as high-competition sectors. Columns 3–10 report these heterogeneous effects. The coefficient for high-competition sectors is statistically larger in magnitude than that for low-competition sectors, which further supports strategic pricing.

¹⁵Broda and Weinstein (2006) estimate substitution elasticities from highly disaggregated trade and price data. Soderbery (2015) refines these estimates by addressing small-sample bias and weak-instrument concerns.

¹⁶For each US Compustat firm, we compute the Lerner index as operating profits net of financial costs divided by sales, where financial costs are proxied by 0.085 times assets as in Aghion et al. (2005). We then average this at the sector–year level and take one minus the resulting average.

Fact 3. Knowledge Diffusion from Adoption

We now turn to knowledge diffusion from adoption, a spillover that extends beyond direct adopters to Korean firms that never engaged in adoption. We use patent citations as a proxy for knowledge flows (Jaffe et al., 1993). Our design matches foreign sellers that sold technology for the first time (treated) to observationally similar foreign firms that had not yet sold or had never sold technology to Korean firms (control), and then traces whether treated sellers receive more citations from never-adopting Korean firms after the transfer. Consider two foreign firms, one that sold technology to a Korean firm and one that did not. If the transfer triggers diffusion beyond direct adopters and citations reflect idea flows, never-adopting firms should become more likely to cite the treated seller's patents than those of similar non-sellers after the transfer.

We construct the estimation sample by distance matching within firms' primary 3-digit IPC class, which is assigned based on their patent portfolio. For matching, we use cumulative patent stock, the number of new patents filed, and the number of new citations received (excluding self-citations), each measured one year prior to the event, along with their respective 3-year growth rates. For each treated seller, we select two control firms using Mahalanobis distance on these variables. This yields 205 matched sets with 565 unique firms. Treated and control groups are well balanced on pre-event observables, and pre-event observables do not predict treatment status (Appendix Tables B.11 and B.12).

Using the matched sample, we estimate the following specification:

$$y_{fmt} = \beta \mathbb{1}[\text{Seller}_{fmt}] + \delta_{fm} + \delta_{mt} + \varepsilon_{fmt}, \quad (3.4)$$

where f denotes foreign sellers, m match, and t year. $\mathbb{1}[\text{Seller}_{fmt}]$ equals one in years after the event year for the treated seller. δ_{fm} denote firm-match fixed effects and δ_{mt} match-year fixed effects. Standard errors are two-way clustered at the firm and match levels.

To examine dynamic effects and pre-trends, we also estimate an event study specification:

$$y_{fmt} = \sum_{\tau=-4}^7 \beta_{\tau} (\mathbb{1}[\text{Seller}_{fmt}^{\tau}]) + \delta_{fm} + \delta_{mt} + \varepsilon_{fmt}, \quad (3.5)$$

where $\mathbb{1}[\text{Seller}_{fmt}^{\tau}]$ denotes event-time indicators when $t = t(m) + \tau$ for treated sellers.

We consider two main outcomes. The first is an indicator for whether a seller receives citations from never-adopting Korean firms. We focus on the extensive margin of citations, because citation counts are frequently zero and our quantitative model is calibrated to the

Table 6: Knowledge Diffusion from Adoption: Average Effects

Panel A. Firm-Level Analysis					
Dep. Var.	Baseline			Placebo	
	Dum. citation (KR) (1)	Ihs. citation (KR) (2)	Tech. proximity (3)	Dum. citation (Other) (4)	Ihs. citation (Other) (5)
$\mathbb{1}[\text{Seller}_{f_{mt}}]$	0.04*** (0.01)	0.10*** (0.03)	0.02** (0.01)	0.01 (0.02)	0.06 (0.04)
Dep. mean	0.05	0.08	0.20	0.69	2.07
N	6,661	6,661	6,661	6,661	6,661
Fixed effects	Firm×match, Match×year				
# Cl. (firm×match)	565×205				

Panel B. Patent-Level Analysis				
Dep. Var.	Baseline		Placebo	
	Dum. citation (KR) (1)	Ihs. citation (KR) (2)	Dum. citation (Other) (3)	Ihs. citation (Other) (4)
$\mathbb{1}[\text{Seller}_{f_{mt}}]$	0.0004** (0.0002)	0.0009** (0.0005)	−0.0010 (0.0053)	0.0015 (0.0070)
Dep. mean	0.0007	0.0011	0.1236	0.1288
N	878,171	878,171	878,171	878,171
Fixed effects	Patent×firm×match, Match×year			
# Cl. (firm×match)	563×204			

Notes. Standard errors in parentheses are two-way clustered at the foreign firm and match levels. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Panels A and B report estimates of β from equations (3.4) and (3.7), respectively. In Panel A, the dependent variables are an indicator and the inverse hyperbolic sine of citations received from never-adopting Korean firms (columns 1–2), patent similarity (column 3), and the analogous citation measures from firms outside Korea as placebo outcomes (columns 4–5). In Panel B, the dependent variables are patent-level analogues of the citation indicator and inverse hyperbolic sine for never-adopting Korean firms (columns 1–2) and firms outside Korea (columns 3–4). Some matches and firms are dropped in Panel B due to the additional fixed effects.

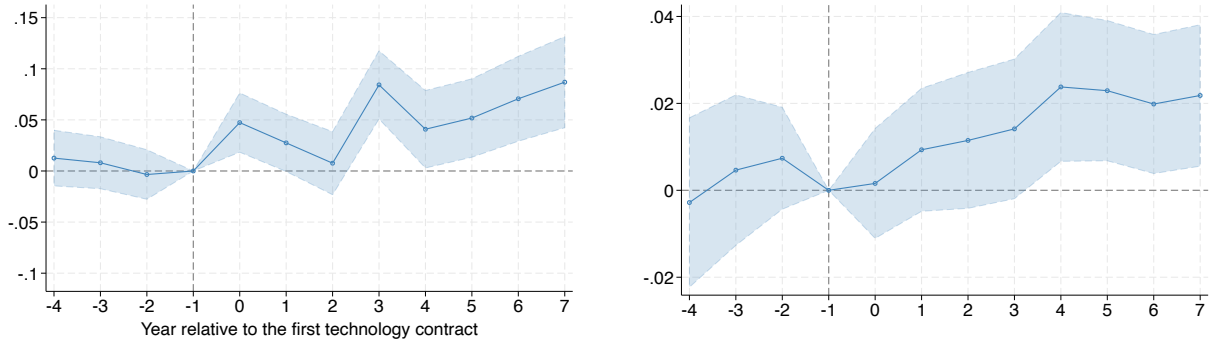
extensive margin. The second is technological proximity between foreign sellers and the set of never-adopting Korean firms. Following Bloom et al. (2013), we measure proximity as cosine similarity of patent stocks across 3-digit IPC classes:

$$\text{Tech. Proximity}_{f_{mt}} = \frac{F_{f_{mt}}^\top \cdot F_{\mathcal{K}_{mt}}}{\|F_{f_{mt}}\| \times \|F_{\mathcal{K}_{mt}}\|}, \quad (3.6)$$

where $F_{f_{mt}} = (p_{f_{1t}}, \dots, p_{f_{Kt}})'$ is the vector whose k -th element represents seller f 's patent stock in technological field k . $F_{\mathcal{K}_{mt}}$ is the analogous vector aggregated over the set of never-adopting Korean firms whose patenting activities overlap with seller f 's main fields. Under knowledge diffusion, we expect proximity to increase for treated sellers.

Panel A of Table 6 reports the average effect. In column 1, sellers are, on average, 4 percentage points more likely to receive citations from never-adopting Korean firms after the

Figure 3: Knowledge Diffusion from Adoption. Event Study



A. Citation Dummy

B. Technological Similarity

Notes. This figure plots event study coefficients from equation (3.5). Shaded bands are 90% confidence intervals based on standard errors two-way clustered at the foreign seller and match levels. The dependent variable in Panel A is an indicator for receiving citations from never-adopting Korean firms; in Panel B it is technological similarity between foreign sellers and never-adopting Korean firms within the same main 3-digit IPC class.

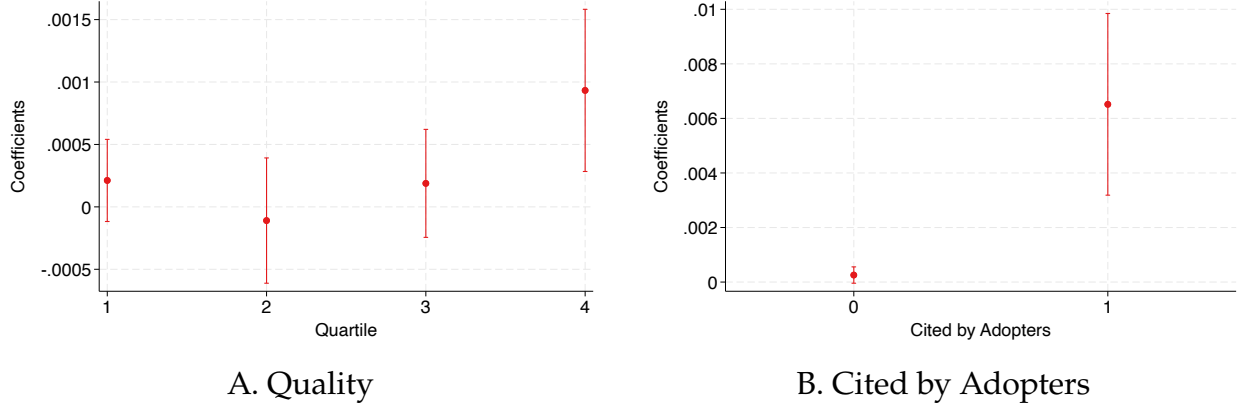
event. Column 2 uses the inverse hyperbolic sine of citations to capture both extensive and intensive margins, and the results remain qualitatively the same.¹⁷ In column 3, consistent with the citation increase, never-adopting Korean firms become more technologically similar to treated sellers. The event study in Figure 3 shows no pre-trends. Estimated effects grow in magnitude at longer post-event horizons, suggesting that knowledge diffusion occurs with lags.

If the parallel-trends assumption holds, the estimates have a causal interpretation. While the firm-match fixed effects absorb time-invariant unobservables, time-varying shocks that coincide with transfer timing can still violate the assumption. In that case, post-event increases in citations and technological proximity would reflect those unobserved shocks rather than knowledge diffusion, resulting in spurious correlations.

To address this concern, we implement two additional exercises. First, we run a placebo test using citations from firms outside Korea as an outcome. If unobservable shocks were driving our results, we would expect similar post-event citation increases from firms outside Korea, but columns 4–5 of Panel A in Table 6 show no such pattern. The placebo coefficients are small and statistically insignificant, and their 95% confidence intervals do not overlap with those for citations from never-adopting Korean firms. The placebo event study is likewise flat (Appendix Figure B.4).

¹⁷Log-like transformations with zeros can be sensitive to functional forms, as they are not unit-invariant (Chen and Roth, 2024). Our results are robust to alternative transformations (Appendix Table B.13).

Figure 4: Knowledge Diffusion. Patent-Level Analysis. Heterogeneous Effects by Patent Quality and Adopter-Citation Status



Notes. This figure reports patent-level estimates of β from equation (3.7). Error bars denote 90% confidence intervals based on standard errors two-way clustered at the firm and match levels. In both panels, the dependent variable is an indicator for receiving citations from never-adopting Korean firms. Panel A reports estimates separately for each within-firm patent quality quartile. Panel B reports heterogeneous effects by whether the patent was cited by adopters who engaged directly in a transfer contract.

Second, we estimate a patent-level specification restricted to patents filed before the event:

$$y_{pfmt} = \beta \mathbb{1}[\text{Seller}_{fmt}] + \delta_{pfm} + \delta_{mt} + \varepsilon_{pfmt}, \quad (3.7)$$

where p indexes patents and δ_{pfm} denotes patent-seller-match fixed effects.

The patent-level specification offers two advantages. First, by restricting attention to patents filed before the event, we avoid counting citations to patents created after the transfer, which reduces the risk that our results reflect contemporaneous unobservable shocks. Second, δ_{pfm} absorbs time-invariant, patent-specific heterogeneity at a finer level.

Panel B of Table 6 reports the patent-level results, which confirm the firm-level findings. Treated sellers' patents receive more citations after the event, while placebo results show no corresponding increase for firms outside Korea. Both the firm-level and patent-level effects are economically large relative to their respective means. The firm-level effect (0.04) is roughly 80% of the mean (0.05), and the patent-level effect (0.0004) is about 57% of its mean (0.0007). The results remain robust to adding more granular fixed effects that absorb finer technological trends (IPC by application year by country by calendar year; see Appendix Table B.14). The patent-level event study also shows no pre-trends (Appendix Figure B.5).

Next, we examine the nature of the knowledge being diffused through two exercises. First, we study heterogeneity by patent quality within seller firms. Diffusion could be limited to lower-quality, easily imitable components of transferred technologies, or it could extend to higher-quality technologies. We measure patent quality using pre-event forward citations

normalized by application year and 3-digit IPC class, rank patents into within-firm quality quartiles, and estimate equation (3.7) separately for each quartile.¹⁸ Panel A of Figure 4 shows a monotonic increase in the citation probability across quality quartiles. The effect is substantially larger for higher-quality patents, with the top quartile exhibiting the largest post-event citation response. The top-quartile effect is statistically distinguishable from those for lower quartiles at conventional significance levels. Second, we examine whether diffusion concentrates in patents cited by adopters who directly engaged in a transfer contract. If particular adopted technologies diffuse to non-adopters, we expect non-adopters to disproportionately cite the same patents as these direct adopters. Panel B confirms this pattern. These two results suggest that diffusion extends to high-quality, frontier technologies and operates through the specific technologies transferred under contracts.

4. SIMPLE MODEL

Having documented these three facts, we now develop a model to understand their implications for optimal technology policy. To build intuition, we begin with a static, closed-economy model that incorporates the advantage of backwardness and knowledge diffusion (Facts 1 and 3) and delivers closed-form optimal subsidy expressions. Although this model abstracts from strategic pricing (Fact 2), it cleanly isolates the core mechanism driving the adoption-to-innovation transition: the state-dependent trade-off between adoption and innovation subsidies is shaped by the advantage of backwardness and the degree of domestic diffusion. Fact 2 requires open-economy interactions and endogenous fee dynamics, which we incorporate into the quantitative model in the next section.

4.1 Competitive Equilibrium

Setup. An economy has a unit continuum of symmetric sectors, $j \in [0, 1]$, and a representative household with labor endowment L . Each sector has two firms with fixed entry, a leader h and a fringe firm $\tilde{h}$, each producing a unique variety. The wage is the numeraire.

¹⁸Because newer patents have had less time to accumulate citations and citation norms differ across IPC classes, we normalize each patent's forward citation count by the mean within its IPC class and application-year cell (Hall et al., 2005).

Household. A household maximizes its utility, $U(C) = \log C$, with nested Cobb-Douglas–CES preferences:

$$C = \exp \left(\int_0^1 \log C_j dj \right), \quad C_j = \left(y_{hj}^{\frac{\sigma-1}{\sigma}} + y_{\tilde{h}j}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

subject to $PC = L + \Pi + T$, where P is the price index, Π total profits, and T a lump-sum transfer.

Firm. Firms are monopolistically competitive and charge a constant markup $\frac{\sigma}{\sigma-1}$. Production is linear in labor:

$$y_{ij} = z_{ij} l_{ij}, \quad i \in \{h, \tilde{h}\}, \quad (4.1)$$

where z_{ij} denotes productivity. A firm's equilibrium profit is

$$\pi_{ij} = \frac{1}{\sigma} \frac{z_{ij}^{\sigma-1}}{z_{hj}^{\sigma-1} + z_{\tilde{h}j}^{\sigma-1}} E, \quad (4.2)$$

where $E = L + \Pi + T$ is aggregate expenditure.

A leader chooses the probabilities of successful innovation x and adoption a , incurring labor costs

$$l_{hj}^x = \frac{\alpha}{\gamma} x^\gamma, \quad l_{hj}^a = \frac{\alpha}{\gamma} a^\gamma, \quad (4.3)$$

with $\alpha > 0$ and $\gamma > 1$. Because a leader is ex-ante identical across sectors, it chooses common efforts x and a , while productivity differs ex-post as uncertainty resolves. Conditional on these choices, a leader's productivity evolves as

$$z_{hj} = \begin{cases} \lambda_x & \text{with probability } x \\ \lambda_a & \text{with probability } a \\ 1 & \text{with probability } 1 - x - a, \end{cases} \quad (4.4)$$

where $\lambda_a, \lambda_x > 1$ are the step-size improvements from adoption and innovation. If both fail, a leader's productivity remains at 1.

We allow $\lambda_a \neq \lambda_x$. When the economy lags far behind the global technology frontier, the advantage of backwardness implies that adoption delivers larger gains than innovation ($\lambda_a \geq \lambda_x$). Conversely, near the frontier, innovation can be more effective ($\lambda_x \geq \lambda_a$). The simple model takes these step sizes as given, while the quantitative model endogenizes them as a function of the technology gap later.

Although a fringe firm does not invest in adoption or innovation, its productivity $z_{\tilde{h}j}$ may nonetheless benefit from a leader's investment through knowledge diffusion:

$$z_{\tilde{h}j} = \begin{cases} z_{hj} & \text{with probability } \delta \\ 1 & \text{with probability } 1 - \delta. \end{cases} \quad (4.5)$$

Given subsidies for adoption and innovation (κ_a and κ_x), a leader solves

$$\max_{x,a} \mathbb{E}_{x,a}[\pi_{hj}] - (1 - \kappa_x) \frac{\alpha}{\gamma} x^\gamma - (1 - \kappa_a) \frac{\alpha}{\gamma} a^\gamma, \quad (4.6)$$

where the expected profit is

$$\mathbb{E}_{x,a}[\pi_{hj}] = (\delta + (1 - \delta)(1 - x - a)) \frac{1}{2} \frac{E}{\sigma} + (1 - \delta)a \frac{\lambda_a^{\sigma-1}}{1 + \lambda_a^{\sigma-1}} \frac{E}{\sigma} + (1 - \delta)x \frac{\lambda_x^{\sigma-1}}{1 + \lambda_x^{\sigma-1}} \frac{E}{\sigma}. \quad (4.7)$$

The first-order conditions for adoption and innovation are

$$(1 - \kappa_k) \alpha k^{\gamma-1} = (1 - \delta) \frac{\lambda_k^{\sigma-1} - 1}{2(1 + \lambda_k^{\sigma-1})} \frac{E}{\sigma}, \quad k \in \{a, x\}, \quad (4.8)$$

where the right-hand side is the private marginal benefit from adoption and innovation.

Equilibrium. In the competitive equilibrium, a household maximizes utility, firms charge prices and choose adoption and innovation efforts to maximize profits, and market clearing conditions hold. Goods market clearing requires $PC = p_{hj}y_{hj} + p_{\tilde{h}j}y_{\tilde{h}j}$ for all $j \in [0, 1]$. Labor market clearing is

$$L = \int_0^1 (l_{hj} + l_{\tilde{h}j} + l_{hj}^x + l_{hj}^a) dj. \quad (4.9)$$

The government runs a balanced budget: $T = - \int_0^1 (\kappa_x l_{hj}^x + \kappa_a l_{hj}^a) dj$.

4.2 Social Planner and Optimal Subsidy

The social planner maximizes household welfare subject to the technological and resource constraints:

$$\max_{\{l_{hj}, l_{\tilde{h}j}\}, x, a} \left(\int_0^1 \log \left(y_{hj}^{\frac{\sigma-1}{\sigma}} + y_{\tilde{h}j}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} dj \right), \quad (4.10)$$

$$\text{subject to equations (4.1), (4.4), (4.5), and (4.9).} \quad (4.11)$$

Although each sector faces uncertain adoption, innovation, and diffusion outcomes, the continuum structure makes aggregate consumption deterministic from the planner's perspective. The planner chooses x and a before uncertainty resolves and allocates labor ex-post according to realized productivities.

The planner's first-order conditions are

$$\alpha k^{\gamma-1} = \left(\delta \log \lambda_k^{\sigma-1} + (1-\delta) \log \frac{1 + \lambda_k^{\sigma-1}}{2} \right) \frac{E}{\sigma}, \quad k \in \{a, x\}, \quad (4.12)$$

where the right-hand side is the social marginal benefit. Comparing (4.8) and (4.12), the private return falls short of the social return, so firms underinvest in both activities.

In the competitive equilibrium, the optimal subsidy that implements the planner's allocation satisfies:

$$1 - \kappa_k = \frac{(1-\delta) \frac{\lambda_k^{\sigma-1}-1}{2(1+\lambda_k^{\sigma-1})}}{\delta \log \lambda_k^{\sigma-1} + (1-\delta) \log \frac{1+\lambda_k^{\sigma-1}}{2}} < 1, \quad k \in \{a, x\}. \quad (4.13)$$

$1 - \kappa_k$ measures the wedge between the private and social marginal returns to investment. When this wedge falls below 1, the social return exceeds the private return, calling for a subsidy $\kappa_k > 0$ to bridge the gap.

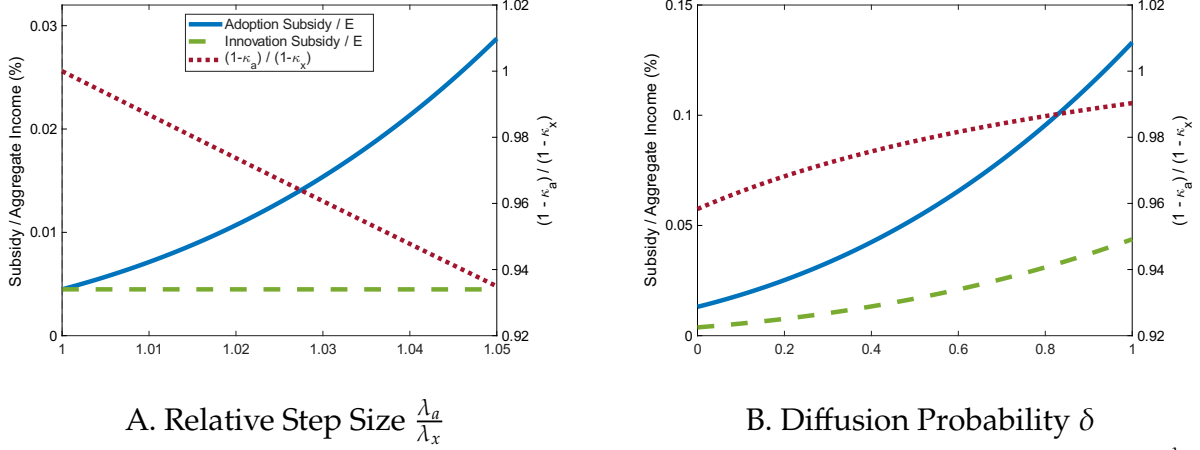
Even without diffusion ($\delta = 0$), firms underinvest relative to the social optimum. Under monopolistic competition, productivity gains translate partly into lower prices, so firms capture only a fraction of the total surplus they create (Jones and Williams, 2000). This surplus appropriability problem deepens with the step size λ_k , so larger steps require larger subsidies ($\frac{\partial(1-\kappa_k)}{\partial \lambda_k} < 0$). As a result, when adoption delivers larger productivity gains than innovation ($\lambda_a > \lambda_x$), the optimal policy favors adoption ($\frac{1-\kappa_a}{1-\kappa_x} < 1$).

Diffusion ($\delta > 0$) deepens the wedge through two channels. First, leaders do not internalize this positive externality to fringe firms, causing further underinvestment. Second, diffusion erodes leaders' post-success market shares, reducing the return to investment (Helpman, 1993).¹⁹ Both channels imply that higher diffusion calls for larger subsidies.

While higher diffusion raises both adoption and innovation subsidy rates, it also narrows the gap between them. The CES structure exhibits diminishing marginal returns within each sector. When the leader is already highly productive (as under adoption with large

¹⁹When leaders succeed, fringe firms catch up with probability δ , compressing leaders' market shares from $\lambda_k^{\sigma-1}/(1+\lambda_k^{\sigma-1})$ to $1/2$. Helpman (1993) identifies this channel in the context of intellectual property rights: when imitation (or diffusion) is possible, innovators anticipate that competitors will erode their market position after success, reducing the expected return to investment.

Figure 5: Optimal Subsidy Rates by Relative Step Size and Diffusion Parameter



Notes. Panel A (Panel B) plots optimal adoption and innovation subsidies against the relative step size $\frac{\lambda_a}{\lambda_x}$ (the diffusion probability δ). In Panel A, $\lambda_x = 1.068$ and $\delta = 0.044$. In Panel B, $\lambda_a = 1.10$ and $\lambda_x = 1.068$. Other parameters are $\sigma = 6$, $\alpha = 0.595$, and $\gamma = 1.5$.

λ_a), the sector is nearly satiated, and diffusion to the fringe yields a small marginal welfare gain. When the leader's advantage is modest (as under innovation with smaller λ_x), aggregate output is more responsive to the fringe's catch-up. Cobb-Douglas aggregation amplifies this asymmetry, since the planner values balanced productivity gains across sectors.²⁰ Consequently, higher diffusion raises the social return to innovation relatively more, compressing the optimal ratio of adoption to innovation subsidy rates ($\frac{\partial[(1-\kappa_a)/(1-\kappa_x)]}{\partial\delta} > 0$). Total subsidy expenditures may nonetheless diverge as adoption and innovation levels adjust endogenously.

Figure 5 illustrates these results, which are summarized in Proposition 1.

Proposition 1. *Let $\delta \in [0, 1)$, $\lambda_a > 1$, $\lambda_x > 1$, and $\lambda_a > \lambda_x$ without loss of generality.*

- (i) *The optimal subsidy in equation (4.13) implements the social planner's allocation.*
- (ii) *The wedge $1 - \kappa_k$ is decreasing in step size (i.e., $\frac{\partial(1-\kappa_k)}{\partial\lambda_k} < 0$), and the wedge ratio $\frac{1-\kappa_a}{1-\kappa_x} \in (0, 1)$ is decreasing in $\frac{\lambda_a}{\lambda_x}$ (i.e., $\frac{\partial[(1-\kappa_a)/(1-\kappa_x)]}{\partial(\lambda_a/\lambda_x)} < 0$).*
- (iii) *The wedge $1 - \kappa_k$ is decreasing in diffusion (i.e., $\frac{\partial(1-\kappa_k)}{\partial\delta} < 0$), but higher diffusion compresses the wedge ratio (i.e., $\frac{\partial[(1-\kappa_a)/(1-\kappa_x)]}{\partial\delta} > 0$).*

Proof. See Appendix C. □

The simple model implies that optimal subsidies favor the investment with the larger step size, and this bias is moderated by diffusion. While the model is static, it has direct

²⁰More generally, this asymmetry holds for any welfare function of the form $\int_0^1 u(C_j) dj$ when u is sufficiently concave.

implications for state-dependent technology policy. Fact 1 establishes that the adoption step size is large when firms lag far behind the frontier and declines as firms approach it. It is therefore optimal to subsidize adoption most intensively at the outset and gradually shift toward innovation as Korean firms close the gap. Fact 3 shows that adoption also generates domestic diffusion, which further shapes the optimal policy mix. The simple model, however, abstracts from open-economy forces, strategic pricing (Fact 2), and transitional dynamics, all of which are central to the quantitative analysis in the next section.

5. QUANTIFICATION

5.1 Quantitative Model

5.1.1 Setup

Time is continuous, $t \in [0, \infty)$. There are two countries, Home and Foreign, indexed by $C \in \{H, F\}$. Each country is endowed with time-varying labor L_{Ct} . There is a continuum of sectors $j \in [0, 1]$. Within each sector, firms produce a unique variety that is tradable across countries subject to iceberg costs $\tau \geq 1$. In each country, a representative household owns domestic firms and supplies labor inelastically, with labor internationally immobile. There is no international trade in assets, ruling out cross-country borrowing and lending.

For expositional convenience, we describe the model from the perspective of Home, and analogous conditions hold for Foreign.

5.1.2 Household

A representative household maximizes discounted utility subject to the budget constraint:

$$U_{Ht} = \int_t^\infty \exp(-\rho(s-t)) \log C_{Hs} ds,$$

$$\text{s.t. } r_{Ht}A_{Ht} + w_{Ht}L_{Ht} = P_{Ht}C_{Ht} + T_{Ht} + \dot{A}_{Ht}$$

where C_{Ht} and P_{Ht} denote consumption and its price index, $\rho > 0$ the discount rate, r_{Ht} the interest rate, w_{Ht} the wage, T_{Ht} a lump-sum transfer, and A_{Ht} asset holdings with time derivative $\dot{A}_{Ht}$.

Household preferences are Cobb-Douglas across sectors and CES within sectors:

$$C_{Ht} = \exp \left(\int_0^1 \log \left(\sum_{i \in \mathcal{I}_H} y_{ijt}^{\frac{\sigma-1}{\sigma}} + \sum_{i \in \mathcal{I}_F} (y_{ijt}^*)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} dj \right),$$

where $\sigma > 1$ is the elasticity of substitution and $\mathcal{I}_C$ is the set of operating firms in country C . y_{ijt} and y_{ijt}^* denote quantities demanded for varieties produced by Home and Foreign firms, with the asterisk indicating exported varieties.

5.1.3 Firm

In each country and sector, there are two firm types, a leader and a unit-mass continuum of homogeneous fringe firms. The set of firms in each country is given by $\mathcal{I}_H = \{h, \tilde{h}\}$ and $\mathcal{I}_F = \{f, \tilde{f}\}$, where h and f denote leaders and $\tilde{h}$ and $\tilde{f}$ fringe firms. Because fringe firms are identical, we treat them as a single representative firm.²¹

Firm i 's production function is

$$\mathcal{Y}_{ijt} = z_{ijt} l_{ijt},$$

where $\mathcal{Y}_{ijt}$ denotes total output, z_{ijt} productivity, and l_{ijt} labor inputs. Total output is the sum of quantities demanded in both markets gross of trade costs, $\mathcal{Y}_{ijt} = y_{ijt} + \tau y_{ijt}^*$.

Leaders compete on prices à la Bertrand, charging endogenous markups over their marginal costs (Atkeson and Burstein, 2008):

$$p_{hjt} = \mathcal{M}_{hjt} \frac{w_{Ht}}{z_{hjt}}, \quad p_{hjt}^* = \mathcal{M}_{hjt}^* \frac{\tau(1+t_t^F)w_{Ht}}{z_{hjt}},$$

where $\mathcal{M}_{hjt}$ and $\mathcal{M}_{hjt}^*$ are markups charged in Home and Foreign markets, increasing in market shares, and t_t^F is a Foreign import tariff. Fringe firms charge monopolistically competitive constant markups: $p_{\tilde{h}jt} = \frac{\sigma}{\sigma-1} \frac{w_{Ht}}{z_{\tilde{h}jt}}$ and $p_{\tilde{h}jt}^* = \frac{\sigma}{\sigma-1} \frac{\tau(1+t_t^F)w_{Ht}}{z_{\tilde{h}jt}}$.

Adoption and Innovation. To improve productivity, leaders can either innovate or adopt technologies from foreign leaders. For simplicity, adoption occurs only between Home and Foreign leaders and there is no adoption between domestic firms or from foreign fringe firms.

²¹This market structure with leaders and fringe firms is also used in Cavenaile et al. (2025).

A Home leader chooses a Poisson innovation intensity x_{hjt} at flow labor cost²²:

$$l_{hjt}^x = \alpha_{Hx} \frac{x_{hjt}^{\gamma_x}}{\gamma_x}, \quad (5.1)$$

where α_{Hx} governs the scale and γ_x the curvature. With intensity x_{hjt} , the leader's productivity jumps by one step $\lambda > 1$:

$$z_{hj,t+\Delta t} = \lambda z_{hjt}.$$

The leader may instead search for foreign technology by choosing an adoption intensity a_{hjt} at flow labor cost²³:

$$l_{hjt}^a = \alpha_{Ha} \frac{a_{hjt}^{\gamma_a}}{\gamma_a}, \quad (5.2)$$

where α_{Ha} governs the scale and γ_a the curvature.²⁴ With intensity a_{hjt} , Home and Foreign leaders enter Nash bargaining over a one-time adoption fee $\mathcal{F}_{hjt}$. If bargaining succeeds, the Home firm pays the fee and adopts the technology; otherwise, adoption does not occur and no fee is paid.

After successful adoption, productivity improves by

$$z_{hj,t+\Delta t} = \begin{cases} \min\{\lambda z_{hjt}, z_{fjt}\} & \text{with intensity } 1 - \eta \\ z_{fjt} & \text{with intensity } \eta \end{cases}, \quad 0 < \eta < 1. \quad (5.3)$$

With intensity η , a Home firm fully catches up to the Foreign seller's level z_{fjt} , whereas with intensity $1 - \eta$ it improves by one step, capped at the Foreign level. The parameter η governs the strength of the advantage of backwardness. A larger η raises the probability of full catch-up, so the expected gain from adoption increases more steeply with the technology gap.

Adoption and innovation are substitutes for productivity improvement, but they differ in three key respects. First, adoption cannot raise productivity above the Foreign level, whereas

²²Leaders' innovation and adoption have both defensive and expansionary motives, as in [Akcigit et al. \(forthcoming\)](#) (see Appendix Figure E.2). When neck-and-neck with a foreign rival, even a marginal productivity gain can shield the leader from competition at home or expand its export market share abroad.

²³Adoption labor costs capture that adoption requires researchers who investigate, learn, and implement foreign technologies.

²⁴The functional form for adoption labor costs and the Nash-bargaining determination of adoption fees are similar to [Santacreu \(2025a\)](#). [Santacreu \(2025a\)](#) assumes innovators only develop new varieties and collect royalties and adopters produce a unique variety after adopting blueprints from these innovators. However, we model adoption and innovation as substitutes, as we focus on the effectiveness of these technology investments depending on development stages.

innovation has no such bound. Second, adoption exhibits an advantage of backwardness and the expected gain from adoption increases with technology gap, while innovation does not. This is consistent with Fact 1, which shows that only adoption features gap-dependent heterogeneous effects. Finally, adoption requires a one-time fee.

Our specification nests several cases in the literature. When $\eta = 0$, adoption yields a one-step improvement independent of the gap, as in König et al. (2022). When $\eta = 1$, it yields immediate full catch-up, as in Perla and Tonetti (2014) and Benhabib et al. (2021). Our flexible specification helps the model match the micro facts. $\eta = 0$ rules out the advantage of backwardness, which is inconsistent with Fact 1. $\eta = 1$ implies that both adopters' gains and sellers' losses increase monotonically in the gap. Thus, adoption fees rise monotonically with the gap, contradicting Fact 2, as discussed below.

Unlike the simple model, where fringe firms are passive and benefit only through diffusion, fringe firms here can also actively innovate or adopt. Their costs take the same functional forms as leaders but differ in scale: $l_{hjt}^x = \frac{\tilde{\alpha}_{Hx}}{\gamma_x} x_{hjt}^{\gamma_x}$ and $l_{hjt}^a = \frac{\tilde{\alpha}_{Ha}}{\gamma_a} a_{hjt}^{\gamma_a}$. Upon successful innovation, the fringe firm jumps one step above the incumbent leader ($z_{\tilde{h}jt} = \lambda z_{hjt}$) and replaces it. Upon successful adoption, it pays an adoption fee $\mathcal{F}_{\tilde{h}jt}$ and productivity evolves as

$$z_{\tilde{h}j,t+\Delta t} = \begin{cases} \min\{\lambda z_{hjt}, z_{fjt}\} & \text{with intensity } 1 - \eta \\ z_{fjt} & \text{with intensity } \eta \end{cases} \quad (5.4)$$

with the fringe firm again replacing the incumbent leader.

Knowledge Diffusion. The model features three types of exogenous knowledge diffusion. First, motivated by Fact 3, we introduce domestic diffusion from leaders to fringe firms, as in the simple model. With Poisson intensity δ^D , a fringe firm can attain the domestic leader's productivity level without engaging in innovation or adoption. The assumption of full catch-up is also consistent with evidence that non-adopters tend to cite high-quality patents. The second is intertemporal diffusion, which is standard in quality-ladder models. When a fringe firm innovates or adopts, it builds on the incumbent leader's current productivity level (Grossman and Helpman, 1991). Third, there is international diffusion. With intensity $\delta^F > 0$, leaders in both countries gain access to the global frontier, attaining the same productivity level. This captures other channels of technology transfer outside formal adoption contracts (e.g., diffusion through trade or imitation). Moreover, this diffusion guarantees a non-degenerate stationary distribution of technology gaps (Aghion et al., 2001).²⁵

²⁵In this class of models, Foreign firms with higher productivity innovate at a faster rate than Home firms. If $\delta^F = 0$, Foreign firms remain permanently ahead, producing a degenerate distribution. A sufficiently large

5.1.4 Government Policy

The Home government subsidizes fractions κ_{Hat} and κ_{Hxt} of adoption and innovation costs for both leaders and fringe firms, financed by lump-sum taxes on households. It also levies an ad valorem import tariff $t_t^H \geq 0$, with revenues rebated to households as a lump sum.

5.1.5 Equilibrium

Appendix D.1 provides detailed expressions for value functions and the law of motion for the state distribution. We formally define the Markov perfect equilibrium and balanced growth path in Appendix D.2.

State Variable. In equilibrium, firms' strategies depend only on integer-valued technology gaps, which constitute the payoff-relevant state variable:

$$\frac{z_{fjt}}{z_{hjt}} = \lambda^{m_{jt}^F}, \quad \frac{z_{hjt}}{z_{\tilde{h}jt}} = \lambda^{m_{jt}^D}, \quad \frac{z_{fjt}}{z_{\tilde{f}jt}} = \lambda^{m_{jt}^{DF}}. \quad (5.5)$$

$m_{jt}^F > 0$ implies that the Foreign leader is m_{jt}^F steps ahead of the Home leader. Because fringe firms have lower productivity than leaders and only replace them upon catching up, $m_{jt}^D, m_{jt}^{DF} \geq 0$. By sector symmetry, $\mathbf{m}_{jt} = \{m_{jt}^F, m_{jt}^D, m_{jt}^{DF}\}$ fully characterizes the payoff-relevant state. We henceforth suppress sector and time subscripts where convenient.

Adoption Fee. Adoption fees, determined by Nash bargaining, internalize adopters' expected gains and foreign sellers' expected losses.²⁶

$$\begin{aligned} \mathcal{F}_{ht}(\mathbf{m}) &= \arg \max_{\mathcal{F}_{ht}(\mathbf{m})} \left(\Delta V_{ht}^a(\mathbf{m}) - (1 - \kappa_{Hat}) \mathcal{F}_{ht}(\mathbf{m}) \right)^\xi \times \left(\Delta V_{ft}^a(\mathbf{m}) + \mathcal{F}_{ht}(\mathbf{m}) \right)^{1-\xi} \\ &= \frac{(1 - \xi) \Delta V_{ht}^a(\mathbf{m}) - \xi (1 - \kappa_{Hat}) \Delta V_{ft}^a(\mathbf{m})}{1 - \kappa_{Hat}}. \end{aligned} \quad (5.6)$$

$0 \leq \xi \leq 1$ is the adopter's bargaining power. $\Delta V_{ht}^a(\mathbf{m})$ is the adopter's value gain from successful adoption, and $\Delta V_{ft}^a(\mathbf{m})$ is the seller's value loss.

Equation (5.6) implies that fees rise when sellers incur larger losses or when buyers obtain larger gains. Domestic diffusion also increases sellers' losses, because successful adoption

advantage of backwardness could prevent that outcome, but our calibrated η is not large enough to do so.

²⁶To focus on buyer-seller interaction, we impose two restrictions on bargaining. Contracts cannot be contingent on future behavior (e.g., no export bans), and sellers cannot commit not to resell the technology, which are consistent with the fact that only 1.3% of contracts restrict exports, and that there was limited commitment in practice. Also, sellers cannot credibly manipulate buyers' adoption rate by proposing a fee that deviates from $\mathcal{F}_{ht}(\mathbf{m})$, because such deviations are time-inconsistent. Conditional on an adoption event, it is optimal to charge $\mathcal{F}_{ht}(\mathbf{m})$, and buyers anticipate this.

raises domestic fringe productivity and intensifies future competition.

Adoption fees vary with the technology gap through two opposing forces. When the gap is large, a buyer can fully catch up to the Foreign frontier with intensity η , generating a large value gain and pushing fees up. When the gap is small, buyers and sellers are neck-and-neck, so even the buyer's one-step gain can substantially shift global market shares away from the seller, prompting sellers to charge higher fees as compensation.²⁷ The first force implies fees fall as the gap narrows, while the second implies they rise. When the latter dominates, fees rise as the gap narrows, consistent with Fact 2.

Market Clearing. Asset markets clear in each period: $A_{Ht} = \int_0^1 \sum_{i \in \mathcal{I}_H} V_{ijt} dj$. Goods markets clear as

$$\sum_{i \in \mathcal{I}_H} p_{ijt} y_{ijt} + \sum_{i \in \mathcal{I}_F} p_{ijt}^* y_{ijt}^* = P_{Ht} C_{Ht}, \quad \forall j \in [0, 1]$$

and labor markets clear as

$$L_{Ht} = \int_0^1 \left[\sum_{i \in \mathcal{I}_H} l_{ijt} + l_{ijt}^x + l_{ijt}^a \right] dj.$$

The balance of payments condition requires:

$$\int_0^1 \sum_{i \in \mathcal{I}_F} \left[\frac{1}{1 + t_t^H} p_{ijt}^* y_{ijt}^* - a_{ijt} \mathcal{F}_{ijt} \right] dj = \int_0^1 \sum_{i \in \mathcal{I}_H} \left[\frac{1}{1 + t_t^F} p_{ijt}^* y_{ijt}^* - a_{ijt} \mathcal{F}_{ijt} \right] dj.$$

The government budget is balanced each period. Total government spending on subsidies, inclusive of the deadweight cost of public funds θ , equals the sum of tax and tariff revenues:

$$T_{Ht} + \int_0^1 \left[\frac{t_t^H}{1 + t_t^H} \sum_{i \in \mathcal{I}_F} p_{ijt}^* y_{ijt}^* \right] dj = (1 + \theta) \int_0^1 \left[\sum_{i \in \mathcal{I}_H} \kappa_{Hat} \left(\mathcal{F}_{ijt} + w_{Ht} l_{ijt}^a \right) + \kappa_{Hxt} w_{Ht} l_{ijt}^x \right] dj.$$

A value of $\theta > 0$ means the government must raise $1 + \theta$ units of revenue to finance one unit of expenditure, so subsidies carry a fiscal cost beyond the direct transfer.

5.1.6 TFP Growth and Technology Gaps

The following proposition decomposes aggregate TFP growth into contributions from adoption, innovation, domestic diffusion, and international diffusion.

²⁷ Appendix Figure E.1 shows an example of a Home leader's profit function over gaps. Its slope is flatter when the Home firm is further behind but becomes steeper as the gap narrows toward zero.

Proposition 2. Define aggregate TFP as output per unit of production labor, $Z_{Ht} \equiv \frac{Y_{Ht}}{L_{Ht}^p}$. Its growth rate can be first-order approximated as

$$\frac{d \log Z_{Ht}}{dt} \approx \log \lambda \times \sum_{\mathbf{m}} \mu_t \times \left[\underbrace{\omega_{ht} \left\{ \underbrace{(x_{ht} + x_{\tilde{h}t})}_{\text{Innovation}} \right\}}_{\text{Adoption}} + \underbrace{\left(a_{ht} + a_{\tilde{h}t} \right) \mathbb{1}_{m^F > 0} (\eta m^F + (1 - \eta)) + \delta^F \max\{m^F, 0\}}_{\text{International diffusion}} \right] + \omega_{\tilde{h}t} \underbrace{\delta^D m^D}_{\text{Domestic diffusion}}, \quad (5.7)$$

where μ_t is the gap distribution (equation D.5) and ω_{it} are markup-adjusted revenue shares (equation D.6).

Proof. See Appendix D.3. □

This decomposition implies that technology policy should be state-dependent. It shows that when firms lag further behind (larger m^F), the expected step size from adoption, $\eta m^F + (1 - \eta)$, is larger, so adoption contributes more to TFP growth. The larger step size deepens the surplus appropriability problem, calling for larger subsidies, as shown in the simple model. Once the gap narrows, the expected step size from adoption declines and eventually vanishes, while innovation plays a growing role, implying that policy should shift toward innovation. As gaps narrow, the contribution of international diffusion shrinks while that of domestic diffusion grows.

5.1.7 Taking Stock

Trade Policy. In an open economy, Foreign sellers' incentives to transfer technology depend on Home import tariffs. Higher tariffs segment markets and reduce Foreign firms' export profits, weakening their outside option and making technology transfers to Home firms more attractive.²⁸ In the extreme case of an infinite tariff, exporters face no loss from selling technology, so adoption fees fall since sellers need not be compensated for any market-share loss.

Additional Market Failures. The quantitative model also features three sources of market failures identified in the simple model: surplus appropriability, positive spillovers from in-

²⁸This channel is empirically documented by [de Souza et al. \(2025\)](#), who find that larger tariff reductions decrease the number of technology transfers in a given sector in Brazil.

tertemporal and domestic diffusion, and anticipated profit erosion from diffusion.²⁹ These failures lead to underinvestment in both adoption and innovation. Beyond these three failures, the model features two additional market failures absent from the simple model. First, Home fringe firms invest with a business-stealing motive, displacing the incumbent leader to capture its rents. Because the social planner is indifferent about which firm produces a given variety, this motive could induce overinvestment (Aghion and Howitt, 1992). Second, oligopolistic market power causes leaders to produce below the social optimum.

Strategic Motive. In an open-economy setup, a strategic motive arises. Adoption subsidies can benefit Korea by shifting profits from Japan (Spencer and Brander, 1983; Hémous et al., 2025; Santacreu, 2025a; Milicevic et al., 2026). Foreign leaders always benefit from technology transfer, because adoption fees, set through bargaining, fully compensate them for the rents lost as Home firms close the gap. But these fees reflect only the leader’s private loss and not the harm to Foreign fringe firms, who also lose market share when Home firms close the gap. Because fees fall short of the full social cost of transfer to the Foreign economy, Home adoption subsidies can shift rents from Foreign fringe firms to Home firms, giving the Home planner a profit-shifting motive for intervention.

Discussion of Model Assumptions. We assume Home firms adopt only from Foreign leaders, not from other Home firms. This is consistent with the fact that adoption fees paid to domestic firms account for only 6.3% of total adoption expenditures (Lee, 2022). Domestic transfers are also less likely to be profitable in equilibrium, because Home firms face the same wages and trade costs, limiting the surplus available from such contracts.

The model features only two countries, with a single source country for adoption. Multiple foreign sellers would compete to compress fees, raising adoption rates and amplifying the quantitative importance of adoption subsidies, so our two-country abstraction likely understates adoption’s quantitative role. We also assume a simple Nash bargaining contract with a lump-sum adoption fee. Richer contracting environments could affect equilibrium adoption rates and welfare. We maintain this simple contracting rule to focus on the transition from adoption to innovation in Korea’s growth miracle.

5.2 Taking the Model to the Data

In this section, we describe our calibration procedure.

²⁹In the extended model, the last channel also operates dynamically. Diffused technologies enable fringe firms to replace or surpass leaders’ productivity levels over time, eroding leaders’ stream of future profits, the Schumpeterian appropriability problem (Aghion and Howitt, 2006).

5.2.1 Parameterization

For computational tractability, we impose additional structure on the model. The initial technology gap m_{j0}^F is assumed to be normally distributed across sectors, $m_{j0}^F \sim \mathcal{N}(\mathcal{D}, \mathcal{V})$, where larger $\mathcal{D}$ implies greater average backwardness. We cap the Foreign–Home leader gap at 30 and the domestic leader–fringe gap at 3, with caps chosen by incrementally raising each bound until results stabilize.³⁰ Foreign firms’ R&D and adoption cost scales are proportional to Home firms’ ($\alpha_{Fx} = \alpha_F \alpha_{Hx}$, $\alpha_{Fa} = \alpha_F \alpha_{Ha}$), and fringe firms’ scales are proportional to Home leaders’ by a factor $\tilde{\alpha}$ ($\tilde{\alpha}_{Hx} = \tilde{\alpha} \alpha_{Hx}$, $\tilde{\alpha}_{Ha} = \tilde{\alpha} \alpha_{Ha}$). We impose a common curvature for adoption and innovation costs, $\gamma_x = \gamma_a = \gamma$.

5.2.2 Calibration

We set Home to Korea and Foreign to Japan, which was Korea’s largest source of foreign technologies, taking 1973 as the initial year. We solve the transition path from the initial condition to the balanced growth path, with time-varying subsidies and tariffs under perfect foresight, and update parameters to minimize the distance between model and data moments. Appendix E.1 presents the algorithm. Table 7 reports a summary of the 22 calibrated parameters.

We take 8 parameters $\{L_{Ht}, L_{Ft}, \kappa_{Hat}, \kappa_{Hxt}, t_t^H, t_t^F, \mathcal{V}, \delta^D\}$ directly from the data. Labor endowments are human-capital-adjusted working hours from Lee and Lee (2016).³¹ Subsidy rates κ_{Hxt} and κ_{Hat} come from our dataset (Figure 1), and tariffs t_t^H and t_t^F are computed as import-weighted averages across sectors. We calibrate domestic diffusion intensity $\delta^D = 0.044$ to match the estimate in Table 6. We assume that when domestic diffusion occurs, fringe firms cite patents underlying the foreign technologies adopted by the domestic leader, so the probability of citing foreign patents rises with δ^D following adoption.

We externally calibrate 4 parameters $\{\rho, \gamma, \sigma, \theta\}$, setting the discount rate $\rho = 0.03$, the R&D cost curvature $\gamma = 1.5$ (Aghion et al., 2023), the elasticity of substitution $\sigma = 8$ consistent with the median estimate in Broda and Weinstein (2006), and the deadweight cost $\theta = 1$ following Feldstein (1999).³²

We estimate the remaining 10 parameters, $\Theta = \{\lambda, \eta, \alpha_a, \alpha_x, \tilde{\alpha}, \alpha_F, \tau, \xi, \delta^F, \mathcal{D}\}$, by targeting 10 data moments. We choose Θ to minimize the distance between model moments $M_i(\Theta)$

³⁰Results are robust to alternative values of the maximum gaps (Appendix Table E.1).

³¹Appendix Figure E.3 documents faster labor endowment growth in Korea than in Japan due to faster growth in population and human capital, with working hours slightly declining over time in both countries. We take this as exogenous in our model. We later conduct robustness checks using endogenous labor supply.

³²The nested aggregation structure follows Atkeson and Burstein (2008), whose calibrated CES elasticities of 1.01 (upper) and 10 (lower) are consistent with our Cobb-Douglas upper aggregator and our value of $\sigma = 8$.

and their data counterparts M_i^D :

$$\min_{\Theta} \sum_{i=1}^{10} \left(\frac{M_i^D - M_i(\Theta)}{\frac{1}{2}(M_i^D + M_i(\Theta))} \right)^2.$$

Although the parameters are estimated jointly, we describe the moments that are most informative for each parameter.

Using model-simulated data, we run the regression as in equation (3.2) and match the difference in adoption coefficients between groups with large and small pre-adoption technology gaps, obtained from the baseline matching.³³ This moment primarily identifies the full catch-up intensity η , which governs the advantage of backwardness. We calibrate the adopter bargaining parameter ξ to match the average ratio of total adoption fees to annual sales (27.6%), as greater adopter bargaining power implies lower adoption fees.

To pin down innovation and adoption labor cost scales α_x and α_a , we target the average manufacturing ratios of innovation R&D and adoption-fee expenditures to value-added in 1981–1990 (2.6% and 1.1%). We set the foreign R&D cost scale parameter α_F so that the long-run productivity ratio between Home and Foreign leaders equals 1. We target the mean exports-to-GDP ratio (0.26, 1970–1993) to calibrate iceberg trade costs τ . Korea’s post-2000 annual TFP growth (0.82%) identifies the step size λ , which governs the model’s balanced-growth rate. The 1973 real GDP per capita ratio between Korea and Japan (0.21) pins down the mean initial gap $\mathcal{D}$. The 2018 ratio (0.995) informs the international diffusion parameter δ^F , as higher δ^F accelerates convergence.

Table 7 reports estimation results and the targeted and non-targeted moments. The estimated full-catch-up probability η is 0.12. The step size λ is estimated at 1.068, implying a one-step productivity gain of 6.8%. We find $\alpha_a > \alpha_x$, indicating that adoption requires more labor input than innovation. The iceberg cost τ is estimated at 1.51, close to the estimate of 1.7 by [Anderson and Van Wincoop \(2004\)](#). The adopter’s bargaining power ξ is 0.845, implying adopters have stronger bargaining power than sellers. The mean initial gap $\mathcal{D}$ is 26.3, corresponding to Japanese firms being initially 466% more productive than Korean firms. Japan’s relative R&D cost $\alpha_F = 1.226$ suggests comparable R&D efficiencies between the two countries. Finally, the international diffusion rate δ^F is estimated at 0.029, near the lower end

³³We target the baseline matching instead of the winners vs. losers design for three reasons. First, the baseline sample is more representative, while the winners-losers sample covers only a few selected cancellation episodes through 1985. Second, the model does not feature exogenous variation from cancellations. Finally, even if selection bias remains after matching, our calibration explicitly accounts for it inside the model. Specifically, we simulate adopting firms, divide them into large- and small-gap groups by the median of m^F , and compute changes in log sales after adoption for each $\mathbf{m}$, analogous to matching in the regression. We then take a weighted average using $\mu(\mathbf{m})a(\mathbf{m})$ within each group to obtain the model-implied β_L and β_S .

Table 7: Calibration Summary

Parameter	Description	Value	Source/Moment	Model Fit	Data
Panel A. Directly From Data					
L_{Ht}, L_{Ft}	Labor endowment		Lee and Lee (2016)		
$\kappa_{Hat}, \kappa_{Hxt}$	Subsidy rate		Tax credit rate		
t_t^H, t_t^F	Import tariff rate		Average import tariff rate		
$\mathcal{V}$	Variance in initial gap	1.166	Data		
δ^D	Domestic diffusion	0.044	Fact 3 (Col. 1 Table 6)		
Panel B. Externally Calibrated					
ρ	Time preference	0.03	Literature		
σ	Elasticity of substitution	8	Broda and Weinstein (2006)		
γ	Adoption/innovation curvature	1.5	Aghion et al. (2023)		
θ	Deadweight cost of taxation	1	Feldstein (1999)		
Panel C. Jointly Calibrated through SMM					
λ	Step size	1.068	Long-run growth rate	0.009	0.008
η	Prob. catch up from adoption	0.120	Table 3 ($\beta_L - \beta_S$)	0.181	0.180
α_a	Adoption cost	0.924	Mfg. adoption/value-added	0.011	0.011
α_x	Innovation cost	0.620	Mfg. R&D/value-added	0.026	0.026
$\tilde{\alpha}$	Fringe relative cost	66.98	New firm patent share	0.056	0.055
α^F	Foreign relative cost	1.226	Long-run prod. ratio	0.999	1.000
τ	Iceberg trade cost	1.505	Mfg. export/value-added	0.258	0.259
ξ	Adopter bargaining power	0.845	Adoption fee/sale	0.276	0.276
δ^F	Across-country diffusion	0.029	2018 GDP ratio	0.995	0.995
$\mathcal{D}$	Initial prod. gap	26.302	1973 GDP ratio	0.210	0.210
Panel D. Untargeted Moments					
Δ Adoption expenditure share, 1973–1993				−0.76	−0.39
Corr. adoption fee and gap				−0.64	−0.23
Δ Firm concentration growth, 1973–1993 (%) (Choi et al., 2023)				38.56	63.49
Δ Export intensity change, 1973–1993 (p.p.)				29.35	10.16

of estimates in the literature.³⁴

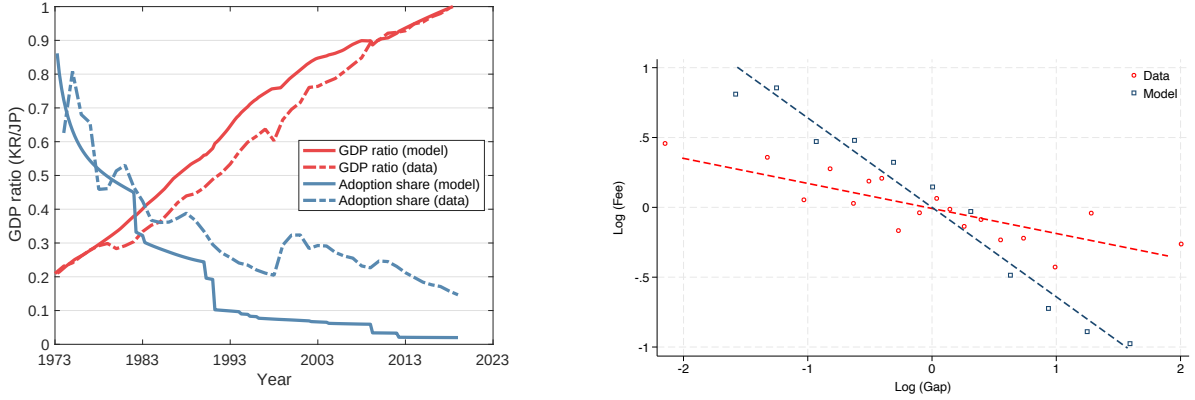
5.3 Model Validation

To validate the model, we present four untargeted moments (Panel D of Table 7 and Figure 6). The model reproduces the declining trend in the share of adoption expenditures over time. This pattern arises endogenously because firms increasingly prioritize innovation as technology gaps narrow.³⁵ The model also reproduces the negative correlation between adoption fees and technology gaps (Fact 2), driven endogenously by strategic pricing.

³⁴ Akcigit and Ates (2023) report values from 0.034 to 0.084 in a closed-economy setting, Cavenaile et al. (2023) estimate 0.016, and Sui (2023) report values between 0.01 and 0.09 in open-economy settings.

³⁵ The downward trend is not solely driven by the policy, as the model generates a qualitatively similar pattern even without subsidies (Appendix Figure E.5).

Figure 6: Untargeted Moments



A. Adoption Expenditure Share

B. Adoption Fee over Technology Gap

Notes. This figure illustrates untargeted moments from the data and the model. Panel A plots adoption fee expenditure as a share of adoption fee plus innovation cost. Panel B plots the log adoption fee against the log technology gap. Both the log fee and the log gap are standardized.

The model further captures two salient features of Korea's growth miracle (Appendix Figure E.2). First, it generates a marked rise in firm concentration, documented by Choi et al. (2023), who show that productivity gains of large firms were a main driver of this increase. In our model, adoption and innovation produce comparable top-end productivity gains and hence a similar rise in concentration.³⁶ At the same time, rising concentration increases static misallocation through higher markups, a pattern qualitatively consistent with plant-level evidence in Lee and Shin (2024).³⁷ Second, the model reproduces export-led growth, as firms that increase productivity through adoption expand exports, raising aggregate export intensity (Ventura, 1997).

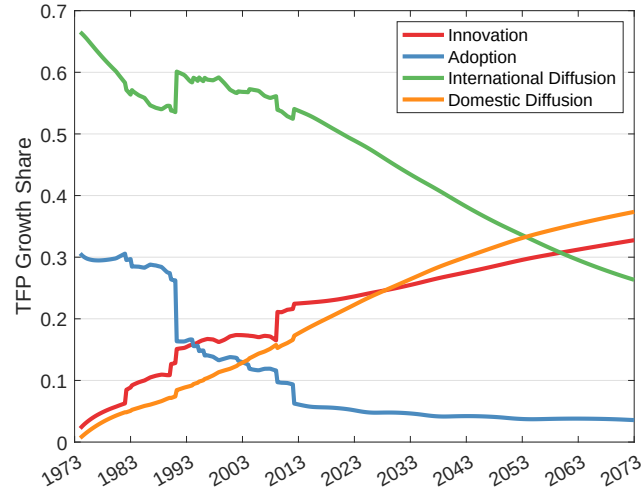
6. QUANTITATIVE RESULTS

We use our calibrated model to decompose the sources of Korea's TFP growth and perform counterfactual analysis.

³⁶In the model, we compute leaders' sales as a share of total gross output. In the data, the analog is the sales share of firms that have ever engaged in adoption or innovation, which we view as conceptually consistent with model leaders.

³⁷While the model captures this pattern qualitatively, the magnitude of misallocation, measured as dispersion in TFPR, is substantially smaller than the figures reported by Lee and Shin (2024), who use establishment-level data. This likely reflects two factors. First, markup dispersion is the only source of static misallocation in our framework, whereas other sources of distortion may show up in their data. Second, our model features two firms per sector, making direct quantitative comparison with establishment-level data infeasible.

Figure 7: Contributions to TFP Growth from Adoption, Innovation, and Diffusion over Time



Notes. This figure plots contributions to TFP growth by adoption, innovation, domestic diffusion, and international diffusion over time, based on equation (5.7) in Proposition 2.

6.1 TFP Growth Decomposition

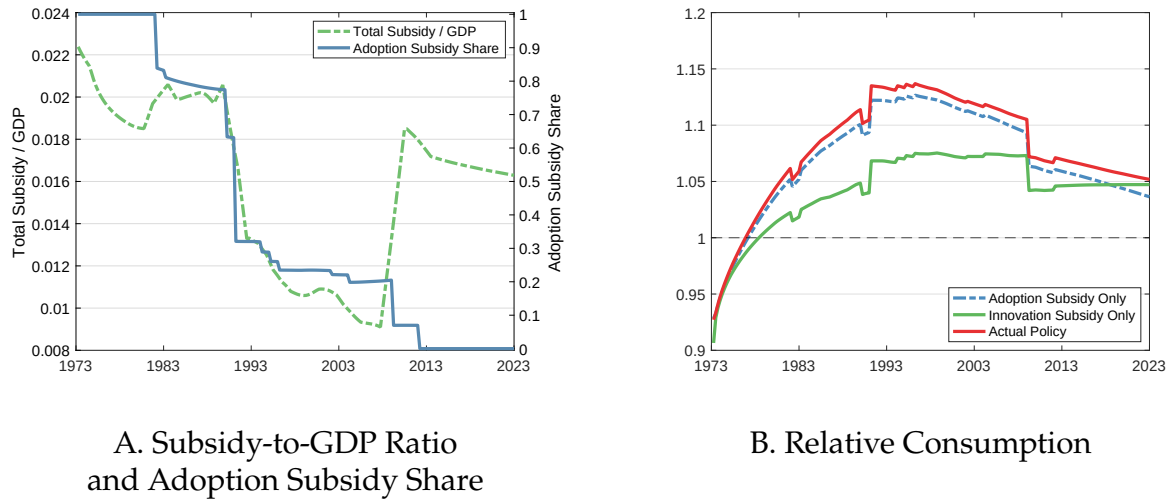
Figure 7 reports the decomposition from Proposition 2. In the early stage, when technology gaps are large, adoption and international diffusion dominate TFP growth. In 1973, adoption alone accounts for 31% of TFP growth and international diffusion for 67%. Both shares fall steadily as the economy approaches the frontier, with adoption declining to 5% and international diffusion to 49% by 2023. By contrast, innovation contributes little early on but rises gradually from 2% to 24%, and domestic diffusion grows from 1% to 22% as firms improve their productivity.³⁸ Domestic diffusion comes mainly from adoption early on, and from innovation later. These patterns show that Korea's catch-up was initially driven by adoption and international diffusion, gradually shifting toward innovation and domestic diffusion as firms approached the frontier.

6.2 Policy Evaluation

Actual Policy. We evaluate Korea's actual state-dependent policy against a no-subsidy counterfactual, measuring welfare as consumption-equivalent variation. We then consider two alternatives, one reallocating the entire budget to adoption subsidies and the other to innovation subsidies, holding the subsidy-to-GDP ratio fixed as in the actual policy. This

³⁸This result is consistent with Santacreu (2015), who, using a multi-country framework, finds that adoption drives more TFP growth in developing countries while innovation dominates in advanced economies.

Figure 8: Government Subsidy Share and Consumption Paths under Counterfactuals



Notes. Panel A plots subsidy-to-GDP ratio and the adoption subsidy share (adoption subsidy expenditure divided by total subsidy expenditure) under the actual policy. Panel B plots real consumption in each scenario as a ratio to consumption in the no-subsidy scenario. In all counterfactuals, the subsidy-to-GDP ratio is held fixed at the level of the actual policy.

Table 8: Welfare Effects of Actual, Adoption-Only, and Innovation-Only Policies

	Actual policy	Adoption only	Innovation only
Δ Welfare (%)	6.22	4.62	3.63

Notes. Welfare gains are measured relative to the no-subsidy scenario.

design evaluates the effectiveness of each policy under a fixed budget. Panel A of Figure 8 shows the model-implied subsidy-to-GDP ratio and its breakdown between adoption and innovation over time. Average total subsidy expenditure is 1.6% of GDP over 1973–2023. The adoption subsidy share begins at 100% in 1973 and declines gradually to zero by 2011, tracing the shift in policy focus from adoption to innovation.

Panel B reports real consumption relative to the no-subsidy scenario. The adoption-only policy produces faster early-stage growth, lifting consumption above the no-subsidy level within 4 years, but gains peak around 1993 and decline thereafter, because adoption subsidies do not raise long-run growth. The innovation-only policy delivers smaller gains until 2019 but higher consumption thereafter, as early innovation subsidies divert resources away from adoption. The actual policy combines the short-run benefits of adoption with the long-run gains of innovation, outperforming both single-instrument policies. Table 8 shows welfare gains of 6.22% under the actual policy, versus 4.62% for the adoption-only and 3.63% for the innovation-only policy.³⁹

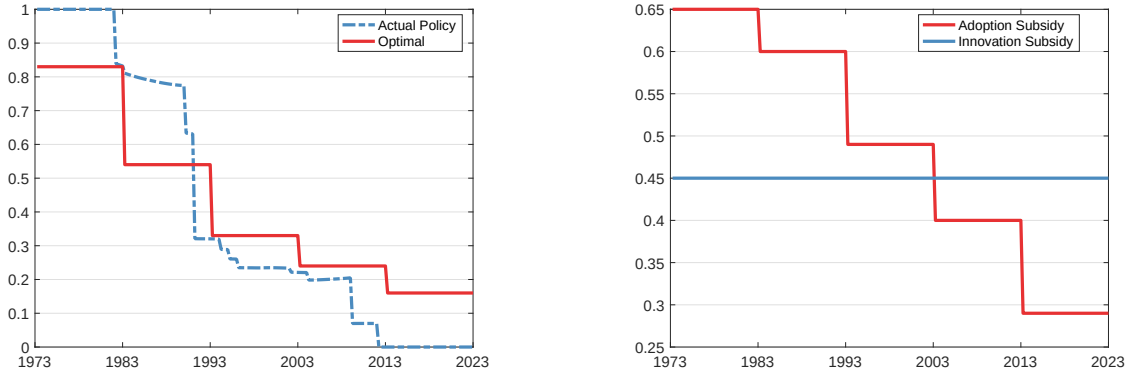
³⁹de Souza (2021) shows that innovation subsidies in Brazil led firms to substitute productive but skill-biased

Table 9: Welfare Effects of Optimal Technology Policies

	Optimal share (constrained)	Optimal rate (unconstrained)
Δ Welfare (%)	6.51	10.96

Notes. Welfare gains are measured relative to the no-subsidy counterfactual. Column 1 reports welfare gains from the constrained optimal policy, where the subsidy-to-GDP ratio is fixed as in the actual policy. Column 2 reports welfare gains from the unconstrained optimal policy.

Figure 9: Optimal Subsidy Policy: Constrained and Unconstrained



A. Optimal Share (Fixed Budget)

B. Optimal Rate

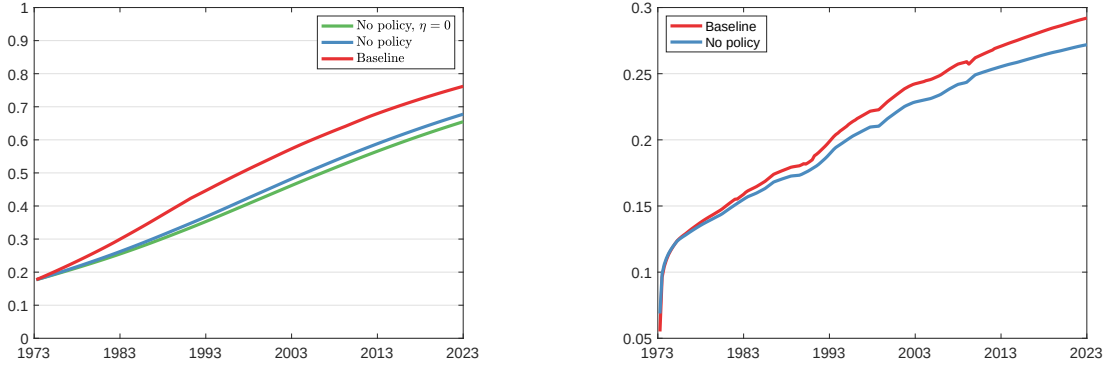
Notes. Panel A plots optimal adoption subsidy shares, holding the subsidy-to-GDP ratio fixed at the level of the actual policy. Panel B plots the optimal adoption and innovation subsidy rates over time.

Optimal Policy. To explore how far the actual policy is from the optimum, we compute constrained and unconstrained optimal policies. First, we impose the actual policy's fiscal constraint and allow the planner to reallocate the fixed subsidy-to-GDP ratio between adoption and innovation every 10 years over a 50-year horizon. Restricting reallocation to every 10 years captures realistic limits on policy adjustment and reduces the computational burden, while the fixed budget share reflects the limited fiscal space that developing-country governments typically face.⁴⁰ Panel A of Figure 9 shows the constrained optimal adoption share over time. The optimal path broadly matches the historical decline through 1993, but it retains a positive adoption share in the long run while the actual adoption subsidy falls to zero by 2010. Relative to the no-subsidy scenario, the constrained optimal policy raises welfare by 6.51% (Table 9), modestly above the 6.22% gain from the actual policy.

foreign technologies for less productive domestic innovation. Korea's college-educated share surpassed that of the US by 2003, suggesting the distributional consequences are less severe than in Brazil.

⁴⁰This reflects the fact that many developing-country governments face limited fiscal space and cannot easily raise spending because borrowing costs are high due to financial frictions or sovereign default risk (e.g., Buera and Shin, 2013). Indeed, many governments and policy institutes target government spending-to-GDP ratios in their fiscal plans (e.g., IMF World Economic Outlook).

Figure 10: Korea's Productivity Catch-Up and Global Profit Share



A. Productivity Ratio of Korea to Japan

B. Korea's Global Profit Share

Notes. Panel A plots Korea's average productivity relative to Japan. Panel B plots Korea's profit share, defined as its profits divided by the combined profits of Korea and Japan.

Second, we remove the budget constraint and allow the planner to choose subsidy rates freely (Panel B). The unconstrained optimal adoption subsidy rate declines over time while the optimal innovation subsidy rate remains flat. The unconstrained optimum raises welfare by 10.96%, but the average subsidy-to-GDP ratio is 5% over 1973–2023, 213% higher than the 1.6% under the actual policy. The time path of the optimal policy is consistent with the simple model's prediction that subsidies should target technology investment with the larger step size. The simple model also predicts that domestic diffusion compresses the subsidy gap between adoption and innovation. Consistent with this prediction, shutting down domestic diffusion ($\delta^D = 0$) and fixing adoption fees to match the average adoption rate in the data delays the optimal switch toward innovation (Appendix Figure E.6).

Catching Up and Profit Shifting. In an open-economy setup, a strategic motive for profit shifting arises. Although Japan's leaders always gain from technology transfers in present discounted value terms, they do not internalize losses to Japan's fringe firms, and Korea's catch-up shifts profits from Japan to Korea by eroding fringe-firm market shares. Panel A of Figure 10 shows that under the actual policy, Korea converged toward Japan's productivity more rapidly than under the no-subsidy scenario. Korea's productivity was 18% of Japan's in 1973 and reached approximately 76% by 2023, compared with 68% under the no-subsidy scenario. The advantage of backwardness alone cannot account for this rapid catch-up. Setting $\eta = 0$ under the no-subsidy scenario, Korea still reaches 65% of Japan's productivity level, only 3 percentage points below the 68% achieved with the advantage of backwardness. This small difference indicates that the advantage of backwardness plays little role in the

Table 10: Welfare Effects of Optimal Tariff

	Optimal tariff & actual policy	Jointly optimal tariff & share
Δ Welfare (%)	6.38	6.60

Notes. Welfare gains are measured relative to the no-subsidy scenario.

catch-up and the policy itself is the key driver.

Panel B of Figure 10 shows that the actual policy shifted profits toward Korea. Under the actual policy, Korea's global profit share increased from 6% in 1973 to 29% in 2023, compared with 27% under the no-subsidy scenario. To isolate these profit-shifting effects, we consider an alternative scenario in which Japan's global profit share is held constant through a lump-sum transfer from Korea.⁴¹ In this scenario, Korea's welfare gain from the policy is 5.41%, implying that roughly 13% of the gain arises from the profit-shifting channel.

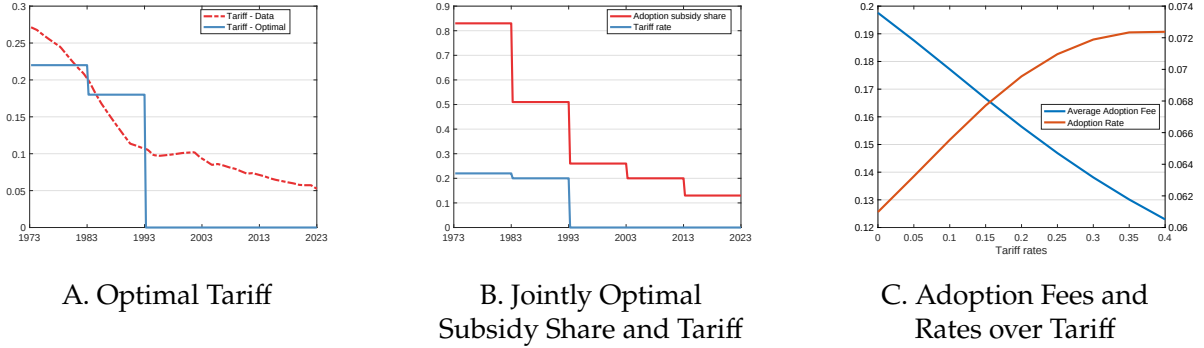
Trade Policy. Korea's import tariffs affect adoption rates by changing Japanese sellers' outside option and thus their willingness to sell. To explore the role of tariffs, we compute the optimal tariff path given actual subsidy rates. Starting near 22%, it declines to zero over time (Panel A of Figure 11), yielding additional welfare gains of 0.16 percentage points over the actual policy (Table 10). Under jointly optimal tariffs and subsidy shares, holding the subsidy-to-GDP ratio fixed (Panel B of Figure 11), the additional welfare gains rise to 0.38 percentage points. In both cases, the optimal tariff is positive initially but declines to zero over time and tends to be lower than the actual tariff. Higher tariffs initially lower adoption fees and raise adoption rates (Panel C of Figure 11), making early tariffs optimal. In later stages, however, tariffs weaken innovation incentives by reducing competition, outweighing the positive terms-of-trade gains and driving the optimal tariff to zero.⁴²

Endogenous vs. Fixed Fee. These tariff results hinge on adoption fees being endogenously determined. Higher tariffs reduce sellers' outside option, push fees down, and raise adoption. If fees are instead fixed, this channel shuts off. We confirm this by re-solving the model with technology transferred at a fixed adoption-fee-to-sales ratio, independent of the gap. The optimal policy then sets tariffs to zero throughout and delays the switch from adoption to innovation (Appendix Figure E.7).

⁴¹Specifically, we first compute Japan's profit share in the no-subsidy case, $\frac{\Pi_{Ft}^c}{\Pi_{Ht}^c + \Pi_{Ft}^c}$. We then construct counterfactual Japanese profits by applying this share to total profits under the actual policy, and require Korea to compensate Japan for the difference between actual and counterfactual profits through a lump-sum transfer. Appendix Figure E.8 shows consumption paths under both scenarios. Japan loses 0.25% under the actual policy but gains 0.03% under this alternative scenario, compared with the no-policy case.

⁴²This finding is consistent with Akcigit et al. (forthcoming), who also find the optimal tariff to be zero in a step-by-step growth framework because tariffs dampen firms' innovation efforts.

Figure 11: Optimal Tariff



Notes. Panel A plots the actual and optimal tariff paths. Panel B plots the jointly optimal subsidy share and tariff. Panel C reports average adoption fees and adoption rates across tariff levels, conditional on the actual subsidy schedule. For each tariff level, we solve the model's transition and compute gap-weighted averages using the equilibrium distribution of gaps.

Foreign Ban on Technology Transfer. We also consider a counterfactual in which the Japanese government bans all technology transfers and the Korean government reallocates the full subsidy budget to innovation. In this case, Korea's welfare falls by 6.2% while Japan gains by 1.7%, due to Korea's slower catch-up and a shift of profits toward Japan (Appendix Figure E.9).⁴³ The welfare loss for Korea also depends on the timing of the ban. When the gap is large, the benefits from adoption are greater, so an early ban is more costly. As the gap narrows, these benefits diminish and the welfare loss shrinks (Appendix Figure E.10).

Robustness. We conduct a battery of robustness checks, varying the discount rate, elasticity of substitution, deadweight costs, R&D cost convexity, iceberg trade costs, exogenous diffusion, bargaining power, and maximum gap, and find the results robust throughout. We also extend the model to allow for endogenous labor supply (detailed in Appendix D.4). With endogenous labor supply, subsidies generate larger welfare gains as higher adoption and innovation raise real wages and hours worked. All results are reported in Appendix Table E.1.

7. CONCLUSION

We study the transition from adoption-led to innovation-led growth. Using novel historical microdata, we document three stylized facts about technology adoption: the advantage of backwardness, strategic pricing, and knowledge diffusion. We build a two-country endogenous growth model consistent with these facts. The relative returns to adoption versus

⁴³With multiple countries providing alternative adoption sources, the effect of the ban may be mitigated.

innovation change with the technology gap, calling for a state-dependent policy that favors adoption in the initial phase and gradually shifts toward innovation. Using the calibrated model, we show that Korea's state-dependent policy accelerated its growth and improved its welfare.

Three directions stand out for future work. Empirically, we document the advantage of backwardness and knowledge diffusion, yet the micro-level mechanisms behind these patterns remain incompletely understood. On the modeling side, a multi-country extension or endogenous foreign responses would generate richer buyer-seller competition and additional policy trade-offs. Finally, we abstract from political-economy frictions that can impede the shift from adoption-led to innovation-led strategies. Modeling distributional consequences and transition costs would help explain why this transition may be politically difficult.

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ONLINE APPENDIX

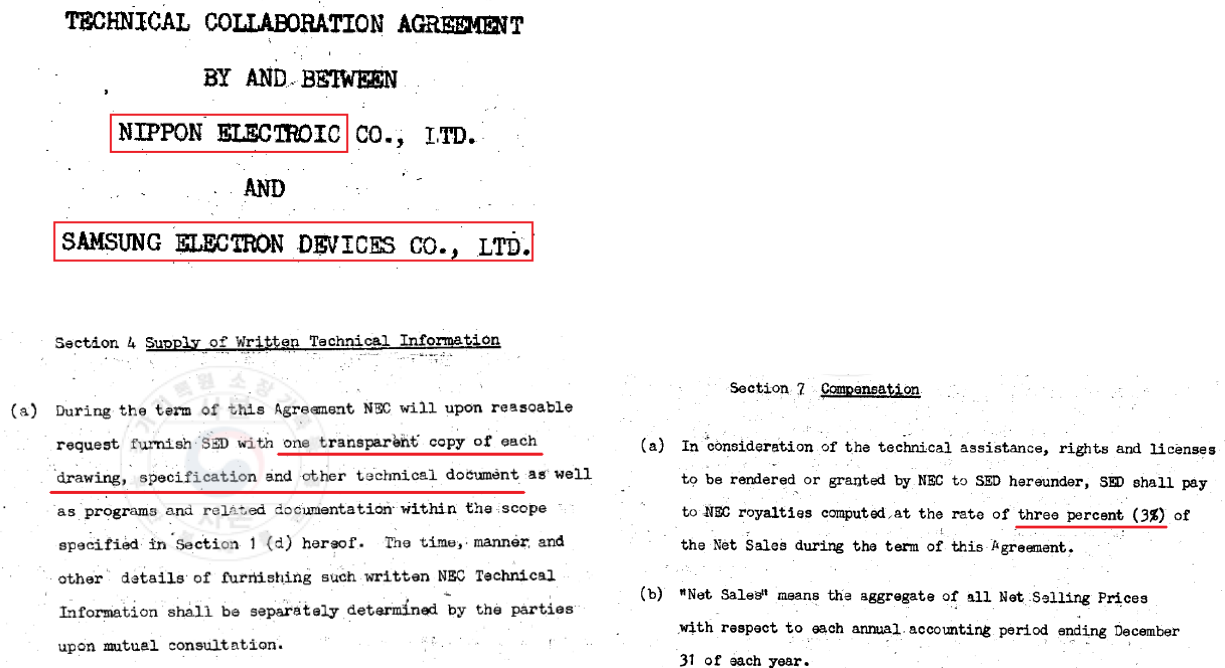
(NOT FOR PUBLICATION)

A. APPENDIX: DATA AND BACKGROUND

A.1 Data Construction

Technology Transfer Data. During the period 1962–1993, Korean firms were mandated under the Foreign Capital Inducement Act to report all foreign currency transactions, including foreign technology transfer contracts. Under this Act, firms reported details of technology transfer contracts to the Economic Planning Board, which directed Korea’s economic policies. Records of the Economic Planning Board, held in Korea’s national archives, include the universe of technology transfer contracts between Korean and foreign firms. We collect and digitize these contracts. Figure A.1 shows a sample contract. Table A.1 reports the distribution of these contracts by country and sector. Table A.2 reports descriptive statistics on adoption fees.

Figure A.1: Example of Adoption Contract



Korea Industrial Technology Association (1995) classifies contracts into four categories: sharing information, technical guidance, patent licensing, and trademark licensing. We con-

sider the first two as know-how transfers and the third and fourth as licensing. Know-how transfer includes sharing blueprints, design specifications, production details, and training of Korean employees. 95% of the contracts involve know-how transfer: 51% involve only know-how transfer, 44% involve both know-how and licensing, and 5% involve only licensing. We exclude intra-multinational transfers (subsidiary–headquarters contracts), which account for about 3% of the sample.

Table A.1: Top 10 Industries and Source Countries among Technology Transfer Contracts

Country	Share (%)	Sector	Share (%)
Japan	49.88	Machinery	26.66
United States	26.29	Electronics	24.89
Germany (West)	5.56	Chemical manufacturing	16.09
France	4.07	Chemical fiber	4.97
United Kingdom	3.69	Metal	4.93
Italy	1.75	Food	3.08
Switzerland	1.60	Shipbuilding	2.70
Netherlands	1.36	Non-metallic products	2.66
Canada	0.94	Pharmaceutical	2.45
Sweden	0.70	Construction	1.81
Others	4.16	Others	9.76

Table A.2: Descriptive Statistics. Adoption Fees

	Fixed Fee	Royalty Fee	Total Fee	Fixed / Total
	(2015 USD \$1M)			Ratio (%)
	(1)	(2)	(3)	(4)
Mean	12.48	4.16	16.64	56.88
Median	0.22	0.03	0.90	94.86
SD	256.23	25.13	257.39	45.67
N	4,303	4,303	4,303	4,303

Notes. All nominal values are converted to 2015 US million dollars.

Figure A.2: Snapshot of Annual Reports of Korean Companies

一信石材工藝 (株) (Il Shin Stoneworks Co., Ltd.)				一信石材工藝 (株) (Il Shin Stoneworks Co., Ltd.)			
〔設立年月日〕 1971.2.26 〔事業者登録番號〕 101-81-08982				〔startyear〕 1971.2.26 〔Business ID〕 101-81-08982			
〔營業項目〕 大理石工製品加工業				〔sector〕 marble processing			
〔任 員〕 (代)劉孝永				〔CEO〕 (代)劉孝永			
〔本 社〕 〒110 月全路區實業洞45-1 (75) 8335-7				〔address〕 〒110 月全路區實業洞45-1 (75) 8335-7			
〔工 場〕 〒520 全北全州市八福洞1街270-4 (4) 1731-3				〔address2〕 〒520 全北全州市八福洞1街270-4 (4) 1731-3			
〔Cable〕 "ILSTONE" Seoul 〔従業員〕 217名				〔Cable〕 "ILSTONE" Seoul 〔Employees〕 217名			
〔主 製 品〕 大理石花瓶, 彫刻品, 花崗岩原石				〔products〕 marble vase, sculpture, granite stone			
〔生産能力〕 大理石花瓶 15,000個, 花崗岩 10,000才				〔production cap〕 marble vase 15,000 granite 10,000才			
〔購入原材料年間消費量〕 大理石原石 10,000個 (300m ³)				〔material used〕 raw marble 10,000個 (300m ³)			
〔貿易業許可番號〕 750374				〔trade ID〕 750374			
〔輸 出〕 大理石花瓶, 大理石彫刻花瓶, 花崗岩原石, 純形彫刻品				〔export〕 marble vase, sculpture, granite stone			
〔輸 入〕 大理石原石, 主要 石加工機械				〔import〕 raw marble, processing machinery			
〔去 來 銀 行〕 第一銀行 通商支店				〔bank〕 第一銀行 通商支店			
〔経 営 實 績〕 (決算期 12月末 現在) (單位: 千円)				〔 (決算期 12月末 現在) (單位: 千円)〕			
資 産・資 本				year			
1977年 1978年 1979年				1977年 1978年 1979年			
総 資 産				total asset			
410,737 681,743 735,121				410,737 681,743 735,121			
流動資産				current assets			
142,313 326,734 373,651				142,313 326,734 373,651			
固定資産				fixed asset			
225,344 278,096 344,960				225,344 278,096 344,960			
繰延資産				deferred asset			
43,078 76,293 16,504				43,078 76,293 16,504			
流動負債				current debt			
123,721 270,994 195,225				123,721 270,994 195,225			
固定負債				fixed debt			
59,133 116,258 161,935				59,133 116,258 161,935			
資 本 金				capital			
160,000 200,000 200,000				160,000 200,000 200,000			
剰 余 金				surplus			
67,882 94,430 177,961				67,882 94,430 177,961			
費 出 損 益				sale			
賣 出 額				456,036 680,454 —			
456,036 680,454 —				profit			
純 利 益				48,361 58,598 128,480			
48,361 58,598 128,480							

Korean Firm Outcome Data. For 1970–1982, we use firm outcome information from the Annual Reports of Korean Companies, published by the Korea Productivity Center. Figure A.2 shows a sample page, which we digitize. This dataset is merged with KIS-VALUE, which provides information starting in 1980 for firms with assets above 3 billion Korean won (2.65 million dollars in 2015). We use firm-level sales, fixed assets, total assets, employment, and industry codes. Table A.3 presents the industry classification used, based on ISIC Rev 3.1 codes. For firms with missing employment or sales, we impute values using total assets, which are non-missing throughout the sample.

Foreign Firm Outcome Data. Our foreign firm outcome data come from Compustat. We follow Imrohoroğlu and Tüzel (2014) for data cleaning. We use PPEGT to measure foreign firms' capital. We drop observations with missing values of employment, fixed assets, and sales. To convert NAICS codes to ISIC Rev 3.1, we use the crosswalk provided by the United Nations Statistics Division.

Aggregate and Sectoral Data. For 1973–1982, sectoral tariffs are digitized from Luedde-Neurath (1986). From 1983 to 1988, Korea's import tariff data are sourced from the Korea Customs Service, and Japan's from the average import tariff of G7 countries calculated by Yi (2003). From 1989 onward, tariff rates are obtained from the World Development Indicators. Aggregate R&D expenditure data are obtained from The Ministry of Science and Technology (1990) and Statistics Korea.

Merging the Datasets. The firm outcome data and the technology transfer dataset are combined by matching on firm names. Next, this merged dataset is combined with data

from the Korean patent office, using both business IDs and firm names. The KIS-VALUE dataset provides Business IDs, which facilitate the merge with the patent office data. For foreign firms, we use the names listed in the technology transfer dataset to merge with USPTO data. We match firms in the adoption data with the USPTO data using company names. We first run fuzzy matching using the Python package `fuzzymatcher`, after stripping common suffixes such as “co,” “ltd,” and “inc.” We require a minimum similarity score of 0.35. For remaining unmatched firms, we manually assign USPTO IDs using PatentsView data. Finally, foreign firms are matched with Compustat using the Compustat–USPTO link constructed by [Bena et al. \(2017\)](#). If two assignees share the same gvkey, we merge them and treat them as one firm.

Table A.3: Sector Classification

Aggregated Industry	Industry
Petrochemicals	Coke oven products (231), Refined petroleum products (232)
Chemicals, rubber, and plastic products	Basic chemicals (241), Other chemical products (242) Man-made fibres (243, excl. pharmaceuticals and medicine chemicals, 2423) Rubber products (251), Plastic products (252)
Pharmaceuticals	Pharmaceuticals and medicine chemicals (2423)
Electronics	Office, accounting, and computing machinery (30) Electrical machinery and apparatus n.e.c. (31) Radio, television, and communication equipment and apparatus (32) Medical, precision, and optical instruments, watches, and clocks (33)
Metals	Basic metals (27), Fabricated metals (28)
Machinery and transportation equipment	Machinery and equipment n.e.c. (29) Motor vehicles, trailers, and semi-trailers (34) Other transport equipment (35)
Food	Food products and beverages (15), Tobacco products (16)
Textiles, apparel, and leather	Textiles (17), Apparel (18) Leather, luggage, handbags, saddlery, harness, and footwear (19)
Manufacturing n.e.c.	Manufacturing n.e.c. (369)
Wood	Wood and wood products, cork (20), Paper and paper products (21) Publishing and printing (22), Furniture (361)
Other nonmetallic mineral products	Glass and glass products (261), Non-metallic mineral products n.e.c. (269)

Notes. Numbers in parentheses denote ISIC Rev. 3.1 codes.

A.2 Construction of Subsidy Rates from Tax Credit Rates

From 1966 to 1982, the government granted a corporate tax exemption covering 100% of adoption fees for the first 5 years following contract signing and 50% thereafter. From 1983 to 1990, the exemption was limited to the first 5 years. Given an average contract duration of

5.7 years, the effective tax credit rates are:

$$\begin{aligned} 1966\text{--}1982: 31\% &\approx 100 \times \text{corporate tax rates}_t \times \frac{1 \times 5 + 0.5 \times 0.7}{5.7} \\ 1983\text{--}1990: 30\% &\approx 100 \times \text{corporate tax rates}_t \times \frac{1 \times 5}{5.7}, \end{aligned}$$

where the final term is the share of adoption expenditures eligible for tax exemption over the average contract life. We multiply by the corporate tax rate because the policy operates through exemptions from corporate income tax liabilities. We obtain both corporate and R&D tax credit rates from [Choe and Lee \(2012\)](#); corporate tax rates decreased from 33% in 1970 to 22% in 2010.

In 1991, the policy was further restricted to contracts involving only advanced technologies. In the contract data for 1991–1993, we observe whether each contract received tax credits and find that 42% did. We accordingly set the eligible fraction at 42% for this period. For subsequent changes in exemption rules, we apply the same formula using prevailing corporate tax rates. The government discontinued tax credits for adoption fees in 2010.

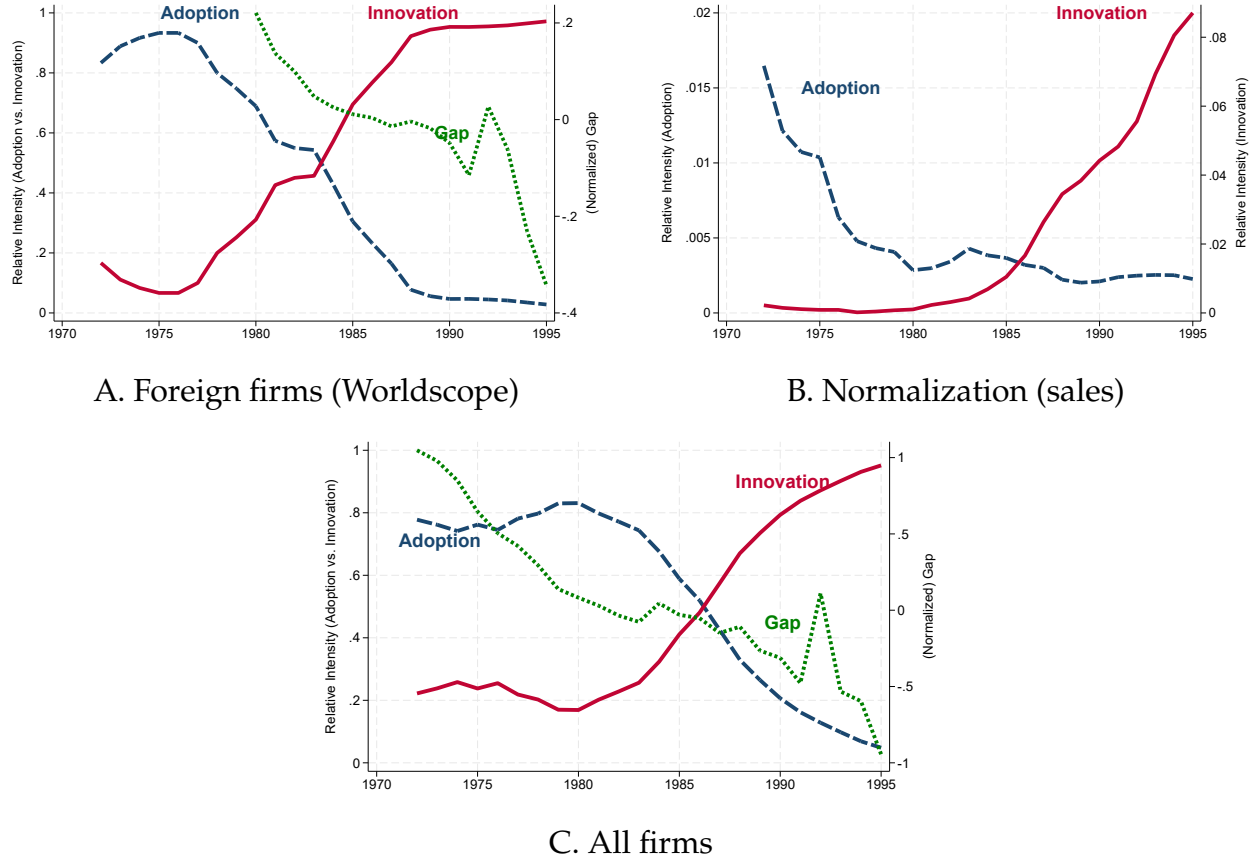
Following [Bloom et al. \(2002\)](#), we translate statutory tax credits for adoption and innovation into effective subsidy rates using corporate tax rates as follows:

$$\text{Adoption (or innovation) subsidy rate}_t = \frac{\text{Adoption (or innovation) tax credit rate}_t}{1 - \text{Corporate tax rate}_t}. \quad (\text{A.1})$$

This expression converts tax credits into equivalent proportional reductions in the unit cost of adoption or innovation.

To derive equation (A.1), note that tax credits reduce a firm's tax liability dollar-for-dollar for eligible adoption or R&D expenditures. Let v be the before-tax marginal value of innovation or adoption, τ^c the corporate tax rate, A_d the depreciation allowance, and A_c the tax credit rate. Firms equate marginal benefits and costs according to $(1 - \tau^c)v = (1 - A_d - A_c)$, where the left-hand side is the after-tax marginal benefit and the right-hand side is the marginal cost net of depreciation allowances and tax credits. Following [Impullitti \(2010\)](#), we assume full expensing ($A_d = \tau^c$), which gives $v = 1 - \frac{A_c}{1 - \tau^c}$. The term $\frac{A_c}{1 - \tau^c}$ therefore represents the effective subsidy rate faced by firms.

Figure A.3: Robustness. Adoption vs. Innovation



Notes. This figure presents robustness checks for the patterns shown in Panel C of Figure 1. In Panel A, we construct the technology gap using Worldscope data, which covers publicly listed firms from the US and other economies. Given the limited sample size in the early years, we define the world productivity frontier as the sectoral median to mitigate the influence of outliers. In Panel B, we normalize adoption and innovation outcomes by the average sales of the three frontier firms rather than the sum of contracts and patents. In Panel C, we replicate the figure using all firms in the dataset rather than limiting the sample to the three frontier firms. The gaps are normalized by the median across the sample period.

B. APPENDIX: STYLIZED FACTS

B.1 Details on Winners vs. Losers Research Design

This section provides additional details on the winners vs. losers identification strategy. We refer readers to [Choi and Shim \(forthcoming\)](#) for further details. In the winners vs. losers design, we compare firms that successfully adopted technology (winners) with firms that initially signed contracts but ultimately failed to adopt or were delayed in adopting due to external factors (losers). Winners serve as the treated group and losers as the control group.

Losers are firms whose government-approved contracts were ultimately canceled by the foreign party for reasons plausibly exogenous to the Korean firm. Examples of such

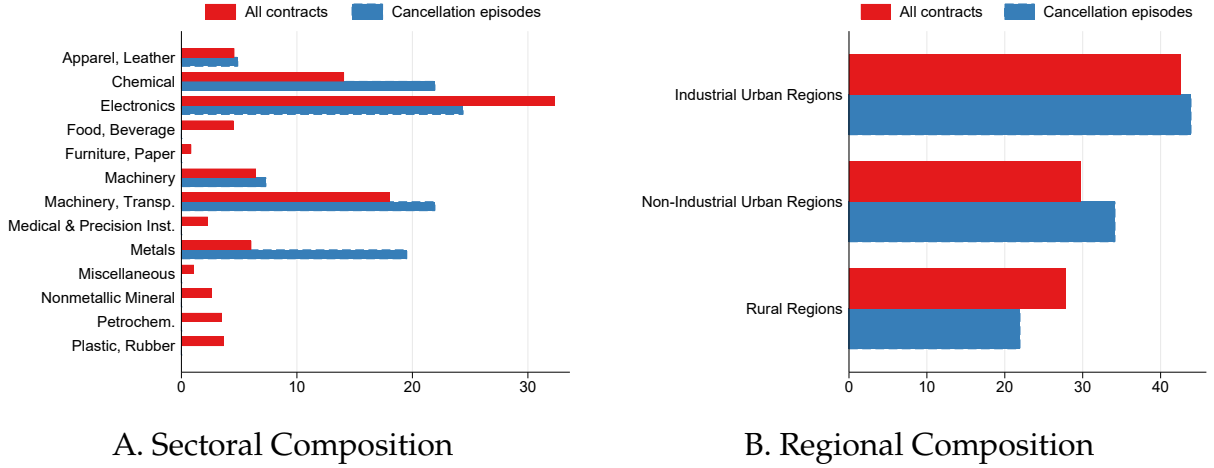
cancellations include foreign firms' bankruptcy, changes in management teams, or sudden requests to modify contractual terms after signing. We exclude cancellations initiated by Korean firms, such as those driven by sudden cash flow issues, to minimize concerns about endogeneity. When contracts were canceled after government approval, Korean firms were required to report the reasons to the government. We manually collected these cancellation cases from the contract cancellation documents submitted to the government under the Foreign Capital Inducement Act.

Losers fall into two types. Delayed adopters are firms that eventually adopted technology but experienced a delay due to the cancellations. After the initial cancellations, these firms did not re-sign with the foreign firms that canceled but signed contracts with other firms a few years later. Never adopters are firms that did not adopt technology at any point following the cancellations. Therefore, the cancellations create exogenous variation in adoption timing and status.

Each loser is matched with up to 2 winners who signed contracts in the same year as the loser's canceled contract. The matching procedure involves two steps. First, we match exactly on region-sector to absorb common shocks within region-sector cells, such as market size, local labor market conditions, and the presence of industrial complexes. Second, within each region-sector cell, we select winners most similar to the loser based on the Mahalanobis distance, with replacement, using log assets. This yields 41 matches among 91 unique firms, comprising 23 delayed adopters and 18 never adopters. Relative to the 38 matches in [Choi and Shim \(forthcoming\)](#), the higher count reflects our use of a more aggregated regional classification. The composition of matched firms may also differ slightly, as we use a more disaggregated sector classification.

Figure [B.1](#) reports the sectoral and regional compositions of the cancellation episodes and all contracts. They closely mirror the distribution of all contracts, consistent with these cancellation episodes being plausibly exogenous.

Figure B.1: Sectoral and Regional Composition of Cancellation Episodes



Notes. This figure presents the distributions of cancellation episodes and all contracts across sectors (Panel A) and regions (Panel B). In Panel B, industrial urban regions are the 8 regions targeted by the HCI Drive, where industrial complexes were constructed between 1973 and 1979. Non-industrial urban regions refer to Seoul, Incheon, Daegu, Daejeon, and Gwangju. Rural regions comprise all remaining areas.

Using this design, [Choi and Shim \(forthcoming\)](#) only estimate average effects using the following specification:

$$y_{imt} = \sum_{\tau=-4}^7 \beta^{\tau} \mathbb{1}[\text{Adopt}_{imt}^{\tau}] + \delta_{im} + \delta_{mt} + \varepsilon_{imt},$$

whereas this paper allows for heterogeneous effects depending on technology gaps (equation 3.2).

B.2 Production Function Estimation

To calculate Korean firms' TFP, we estimate production functions using two approaches. The first applies the control function approach of [Olley and Pakes \(1996\)](#) and [Akerberg et al. \(2015\)](#), with investment as a proxy variable. For the period 1970–1982, when the data are drawn from the Annual Reports of Korean Companies, value-added information is not directly available. We therefore construct value added by multiplying firm-level revenue by the industry-year value-added share from the input-output tables. Investment is computed as $\text{Capital}_{i,t+1} - 0.94 \times \text{Capital}_{it}$, assuming an annual depreciation rate of $1 - 0.94 = 0.06$, and deflated using the investment price deflator. We estimate sector-specific production functions with year fixed effects using moment conditions based on the timing assumption that labor is a flexible input while capital is predetermined from the previous period. TFP is the residual from this estimation.

The second approach follows [Choi et al. \(2023\)](#), which we briefly summarize here. In the KIS-VALUE data covering the post-1983 period, information on material inputs is available. Under the Cobb–Douglas assumption, the coefficient on material inputs is identified using the share of material expenditures in revenue. This approach addresses the identification challenges with control function methods for gross output production functions, as pointed out by [Gandhi et al. \(2020\)](#). After estimating the material input coefficient, we estimate a gross-output production function under the timing assumption that only labor is a flexible input while capital is predetermined. To extend these estimates to the 1970–1982 period, when material input data are not available, we convert the estimated gross output production function into a value-added production function. Specifically, we compute the value-added share as the sum of the labor and capital coefficients relative to total estimated returns to scale and rescale them accordingly.

In the Compustat data, we follow [İmrohoroglu and Tüzel \(2014\)](#). We construct value added as sales minus the cost of goods sold (COGS) plus the wage bill, where the wage index comes from the Social Security Administration. We measure investment using capital expenditure (CAPX) and capital using fixed assets (PPEGT), both deflated by the price index for non-residential private fixed investment from NIPA.

B.3 LASSO Selection of Controls

Given limited guidance on which variables to include for technology heterogeneity and a large number of potential controls, we select technology-quality controls using the approach of [Belloni et al. \(2014\)](#). We consider the following specification:

$$\log \text{Fee}_{ift} = \beta \log \text{Gap}_{it} + \sum_{k,c} (\text{IPC}_k \times D_c \times \mathbf{Z}_{ft})' \gamma_{kc} + \varrho \log \text{Assets}_{it} + \delta_{jct} + \varepsilon_{ift},$$

where IPC_k is an indicator that equals 1 for 3-digit IPC class k , D_c is a country indicator for country c , γ_{kc} is a 5×1 vector of coefficients, and $\mathbf{Z}_{ft}$ is a 5×1 vector of variables related to foreign seller f 's technology levels:

$$\mathbf{Z}_{ft} = \begin{bmatrix} 1 \\ \mathbb{1}[\text{Cum. Citation}_{ft} > 0] \\ \text{Ihs}(\text{Cum. Citation}_{ft}) \\ \mathbb{1}[\text{Cum. Patent}_{ft} > 0] \\ \text{Ihs}(\text{Cum. Patent}_{ft}) \end{bmatrix}.$$

Here, $\mathbb{1}[\text{Cum. Citation}_{ft} > 0]$ is an indicator that equals 1 if foreign seller f has positive cumulative citations, and $\text{Ihs}(\text{Cum. Citation}_{ft})$ is the inverse hyperbolic sine of cumulative citations. $\mathbb{1}[\text{Cum. Patent}_{ft} > 0]$ and $\text{Ihs}(\text{Cum. Patent}_{ft})$ are defined analogously for cumulative patents. We include dummies and the inverse hyperbolic sine transformation to capture extensive and intensive margins more flexibly. Given that a firm's productivity in the model reflects its cumulative number of past innovations, the cumulative patent stock is consistent with our theoretical framework. Cumulative forward citations additionally capture quality differences across patents, since firms with higher-quality patents are more likely to receive citations. In total, there are 585 unique combinations of 3-digit IPC classes and foreign sellers' countries; with 5 variables in $\mathbf{Z}_{ft}$, this yields $2,925 (= 585 \times 5)$ parameters.

The methodology follows the post-double-selection procedure, which proceeds in two steps. First, we run a LASSO regression of the outcome variable ($\log \text{Fee}_{ift}$) on the full candidate pool to select controls that predict the fee. Second, we run a separate LASSO regression of the treatment variable ($\log \text{Gap}_{it}$) on the same candidate pool to select controls that predict the gap. We then take the union of the variables selected in both steps and include them as controls in the OLS estimation of equation (3.3). In both steps, we use the theoretically motivated penalty parameter of [Belloni et al. \(2014\)](#). We partial out the fixed effects δ_{jct} and do not penalize $\log \text{Assets}_{it}$, so these variables are always included.

B.4 Additional Figures and Tables

Table B.1: Advantage of Backwardness. Covariate Balance. Baseline Matching

	Treated				Control				(Col. 1 - Col. 5)	
	Mean	Med.	SD	Obs.	Mean	Med.	SD	Obs.	<i>t</i> -stat.	<i>p</i> -val
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log sales	16.95	17.01	1.61	800	16.97	16.99	1.59	1,540	0.20	[0.65]
Log emp.	5.96	5.86	1.22	800	5.96	5.89	1.18	1,540	0.00	[0.98]
Log sales per emp.	10.98	10.93	1.03	800	11.00	11.00	1.05	1,540	0.43	[0.63]
Log fixed assets	15.79	15.81	1.83	800	15.76	15.81	1.78	1,537	0.42	[0.52]
Log fixed assets per emp.	9.83	9.81	1.24	800	9.79	9.80	1.17	1,537	0.66	[0.42]
Log assets	16.83	16.90	1.60	800	16.80	16.80	1.55	1,540	0.80	[0.37]
Export dummy	0.39	0	0.49	800	0.41	0	0.49	1,540	0.25	[0.61]
Ihs. exports	6.65	0	8.38	800	6.71	0	8.25	1,540	0.03	[0.87]

Notes. This table reports covariate balance between treated and control firms in the baseline matched sample from 4 to 2 years prior to the adoption event. Column 9 reports *t*-statistics for the difference in means, with *p*-values in column 10. All monetary values are in 2015 US dollars.

Table B.2: Fact 1. Advantage of Backwardness. Balance Test. Baseline Matching

Dep. Var.	$\mathbb{1}[\text{Adopt}_{im}]$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log sales	-0.00 (0.01)						-0.05* (0.03)
Log emp.		-0.00 (0.01)					-0.01 (0.01)
Log fixed assets			0.00 (0.01)				-0.01 (0.04)
Log assets				0.00 (0.01)			0.05 (0.04)
Export dummy					-0.01 (0.01)		-0.13** (0.06)
Ihs. exports						-0.00 (0.01)	0.13** (0.06)
F- <i>p</i>							[0.26]
N	2,340	2,340	2,337	2,340	2,340	2,340	2,337

Notes. Standard errors, two-way clustered at the match and firm levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the balance test for the baseline matching from 4 to 2 years prior to the adoption event. We regress a treatment dummy on pre-event observables: log sales, log employment, log fixed assets, log assets, export dummy, and the inverse hyperbolic sine of exports. F-*p* is the *p*-value from a joint *F*-test that all coefficients are zero. The smaller sample in columns 3 and 7 reflects missing observations for fixed assets.

Table B.3: Advantage of Backwardness. Covariate Balance. Winners vs. Losers

	Winners				Losers				(Col. 1 - Col. 5)	
	Mean	Med.	SD	Obs.	Mean	Med.	SD	Obs.	<i>t</i> -stat.	<i>p</i> -val
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log sales	17.69	17.87	1.72	140	18.01	18.04	1.76	84	0.84	[0.37]
Log emp.	6.82	6.80	1.33	140	6.85	7.05	1.54	84	0.01	[0.94]
Log sales per emp.	10.86	10.76	1.11	140	11.16	10.97	1.47	84	1.17	[0.29]
Log fixed assets	16.92	17.00	2.00	140	17.16	17.53	2.11	84	0.31	[0.58]
Log fixed assets per emp.	10.10	9.92	1.34	140	10.31	10.14	1.63	84	0.50	[0.48]
Log assets	17.86	17.91	1.79	140	18.08	18.21	1.86	84	0.34	[0.56]
Export dummy	0.56	1	0.50	140	0.55	1	0.50	84	0.01	[0.92]
Ihs. exports	9.64	14.77	8.78	140	9.41	15.08	8.73	84	0.02	[0.89]

Notes. This table reports covariate balance between winners and losers in the winners vs. losers matched sample from 4 to 2 years prior to the adoption event. Column 9 reports *t*-statistics for the difference in means, with *p*-values in column 10. All monetary values are in 2015 US dollars.

Table B.4: Advantage of Backwardness. Balance Test. Winners vs. Losers

Dep. Var.	$\mathbb{1}[\text{Adopt}_{im}]$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log sales	−0.02 (0.03)						−0.07 (0.06)
Log emp.		0.00 (0.04)					0.03 (0.05)
Log fixed assets			−0.01 (0.02)				0.01 (0.08)
Log assets				−0.01 (0.03)			0.02 (0.12)
Export dummy					−0.01 (0.09)		−0.25 (0.54)
Ihs. exports						−0.00 (0.00)	0.01 (0.03)
F- <i>p</i>							[0.89]
N	152	152	152	152	152	152	152

Notes. Standard errors, two-way clustered at the match and firm levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the balance test for the winners vs. losers research design from 4 to 2 years prior to the adoption event. We regress a treatment dummy on pre-event observables: log sales, log employment, log fixed assets, log assets, export dummy, and the inverse hyperbolic sine of exports. F- p is the p -value from a joint F -test that all coefficients are zero.

Table B.5: Robustness. Advantage of Backwardness. Alternative Inference

Research Design	Baseline matching		Winners vs. losers	
	Large gap (1)	Small gap (2)	Large gap (3)	Small gap (4)
Coefficient	0.19	0.02	0.87	0.41
Baseline	[0.00]	[0.58]	[0.01]	[0.01]
Randomization inference (Young, 2019)	[0.00]	[0.57]	[0.00]	[0.01]
Wild bootstrap (Cameron et al., 2008)	[0.00]	[0.79]	[0.02]	[0.01]
Jackknife (MacKinnon et al., 2023)	[0.00]	[0.78]	[0.02]	[0.01]
Clustering (Match)	[0.00]	[0.59]	[0.02]	[0.01]
Clustering (Firm)	[0.00]	[0.57]	[0.01]	[0.01]

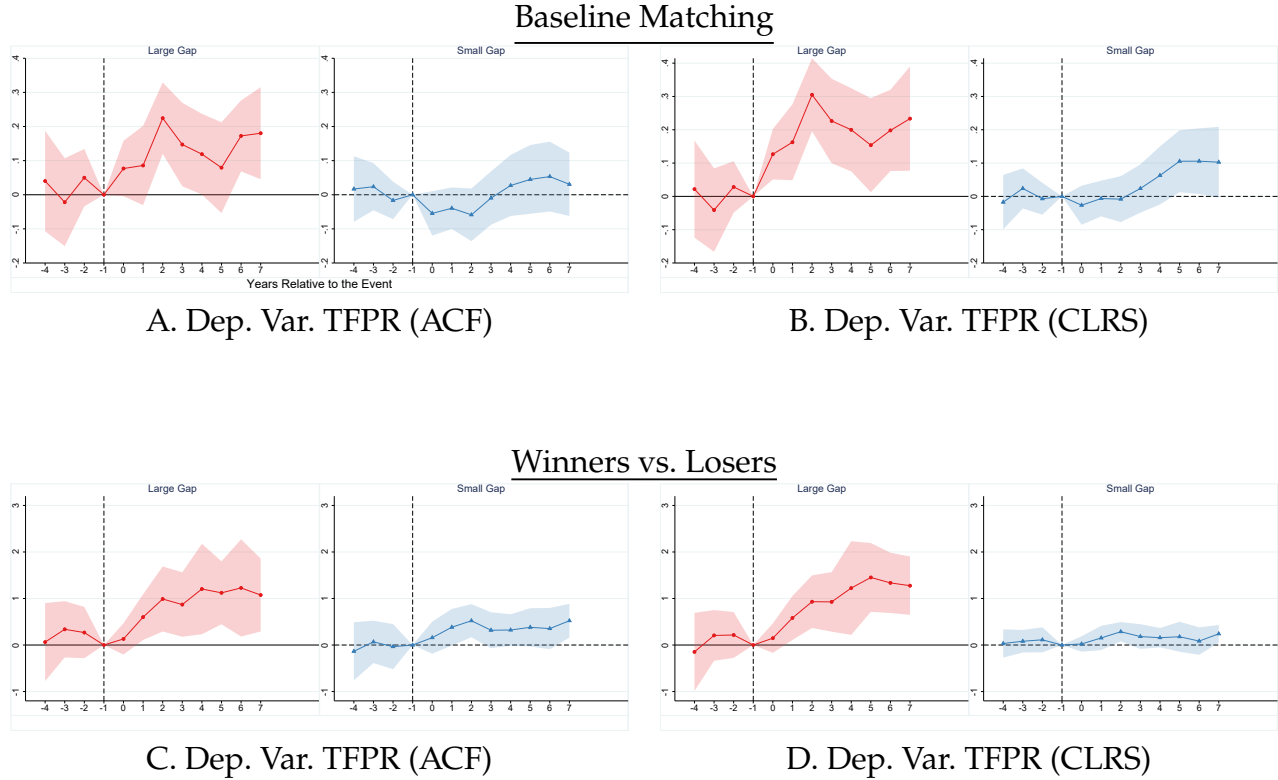
Notes. This table reports robustness checks of the estimated β in equation (3.1) using alternative inference procedures for both the baseline matching and winners vs. losers research designs. We consider randomization inference (Young, 2019), which is robust to finite-sample bias from highly leveraged observations, as well as wild-bootstrap (Cameron et al., 2008) and jackknife (MacKinnon et al., 2023) standard errors, both of which are more reliable with few clusters. We also report results under alternative clustering schemes at the match and firm levels.

Table B.6: Robustness. Advantage of Backwardness. Alternative Measures of Gap and Performance, and the Number of Matches

Robustness	Alt. gap			Alt. performance		Alt. # match	
	Market share	TFPR (ACF)	TFPR (CLRS)	Labor productivity			
	Log labor productivity			TFPR (ACF)	TFPR (CLRS)	Log labor productivity	
	#=2			#=1	#=3	#=3	
(1) (2) (3) (4) (5) (6) (7)							
Panel A. Baseline Matching							
$\mathbb{1}[\text{Gap}_m^{\text{large}}] \times \mathbb{1}[\text{Adopt}_{imt}]$	0.18*** (0.05)	0.18*** (0.05)	0.19*** (0.05)	0.11** (0.04)	0.18*** (0.05)	0.17*** (0.06)	0.19*** (0.05)
$\mathbb{1}[\text{Gap}_m^{\text{small}}] \times \mathbb{1}[\text{Adopt}_{imt}]$	0.04 (0.04)	0.03 (0.04)	0.03 (0.03)	0.00 (0.03)	0.04 (0.03)	-0.00 (0.04)	0.03 (0.03)
p-val	[0.03]	[0.01]	[0.01]	[0.05]	[0.03]	[0.01]	[0.00]
Dep. mean	11.29	11.29	11.29	6.36	11.39	11.31	11.27
# Cl. (firm×match)	1,064×515	1,064×515	1,064×515	1,062×515	1,062×515	816×500	1,248×517
N	11,448	11,448	11,448	11,391	11,391	7,032	15,762
Panel B. Winners vs. Losers							
$\mathbb{1}[\text{Gap}_m^{\text{large}}] \times \mathbb{1}[\text{Adopt}_{imt}]$	0.81** (0.32)	0.87** (0.34)	0.40** (0.18)	0.76** (0.33)	0.82** (0.33)	0.78** (0.36)	0.74** (0.29)
$\mathbb{1}[\text{Gap}_m^{\text{small}}] \times \mathbb{1}[\text{Adopt}_{imt}]$	0.42*** (0.15)	0.40** (0.15)	0.86*** (0.32)	0.40** (0.15)	0.15 (0.11)	0.40** (0.17)	0.36** (0.16)
p-val	[0.28]	[0.21]	[0.22]	[0.33]	[0.06]	[0.35]	[0.26]
Dep. mean	11.13	11.13	11.13	6.57	12.09	11.16	11.13
# Cl. (firm×match)	91×41	91×41	91×41	91×41	91×41	72×40	101×41
N	820	820	820	820	820	570	963
Fixed effects							
Firm×match, Match×year							

Notes. Standard errors, two-way clustered at the match and firm levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports robustness checks using alternative measures of technology gaps and firm performance, as well as alternative numbers of matches. Panels A and B present results for the baseline matching and winners vs. losers designs, respectively. In columns 1–3, firms are classified into more-backward and less-backward groups using alternative gap measures based on within-sector domestic market shares, TFPR estimated following [Akerberg et al. \(2015\)](#), and TFPR estimated following [Choi et al. \(2023\)](#). Columns 4–7 classify firms into two groups based on labor productivity. In columns 4–5, the dependent variables are TFPR estimated following [Akerberg et al. \(2015\)](#) and [Choi et al. \(2023\)](#). In columns 1–3 and 6–7, the dependent variable is log labor productivity. Columns 6–7 use 1 and 3 matches, respectively, while columns 1–5 use 2 matches.

Figure B.2: Robustness. Alternative Performance Measures. Advantage of Backwardness. Event Study



Notes. This figure presents the estimated β_l^T and β_s^T in equation (3.2) for the baseline matching and winners vs. losers designs in Panels A–B and C–D, respectively. Shaded bands are 90% confidence intervals based on standard errors, two-way clustered at the match and firm levels. The dependent variables are TFPR estimated following [Akerberg et al. \(2015\)](#) (Panels A and C) and [Choi et al. \(2023\)](#) (Panels B and D).

Table B.7: No Evidence of Advantage of Backwardness for Innovation. Covariate Balance

	Treated				Control				(Col. 1 - Col. 5)	
	Mean	Med.	SD	Obs.	Mean	Med.	SD	Obs.	<i>t</i> -stat.	<i>p</i> -val
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log sales	18.11	18.16	1.43	681	18.02	18.09	1.31	1,332	3.97	[0.05]
Log emp.	6.49	6.43	1.22	681	6.43	6.36	1.13	1,332	2.65	[0.10]
Log sales per emp.	11.62	11.58	0.93	681	11.60	11.54	0.86	1,332	0.60	[0.44]
Log fixed assets	17.13	17.14	1.65	679	17.05	17.11	1.56	1,328	2.80	[0.09]
Log fixed assets per emp.	10.65	10.68	1.12	679	10.62	10.65	1.12	1,328	0.34	[0.56]
Log assets	18.02	17.98	1.46	681	17.95	17.98	1.37	1,332	2.86	[0.09]
Export dummy	0.42	0.00	0.49	681	0.39	0	0.49	1,332	1.40	[0.26]
Ihs. exports	7.30	0	8.74	681	6.76	0.00	8.64	1,332	1.29	[0.24]

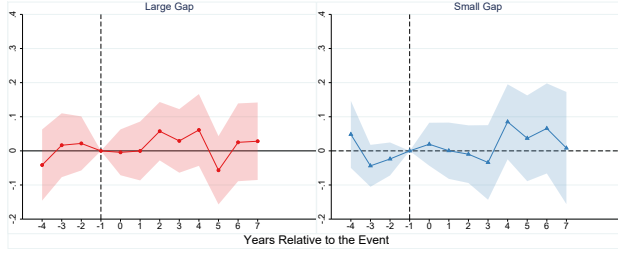
Notes. This table reports covariate balance between treated and control firms in the baseline matched sample from 4 to 2 years prior to the event. Column 9 reports *t*-statistics for the difference in means, with *p*-values in column 10. All monetary values are in 2015 US dollars.

Table B.8: No Evidence of Advantage of Backwardness for Innovation. Balance Test

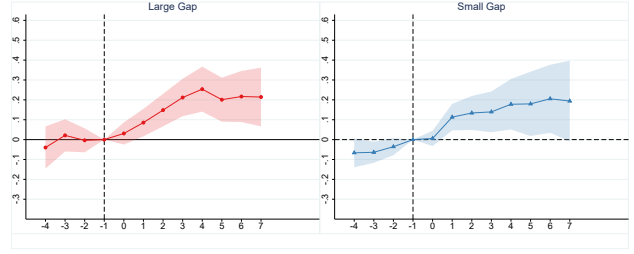
Dep. Var.	$\mathbb{1}[\text{Innovate}_{im}]$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log sales	0.01* (0.01)						0.02 (0.03)
Log emp.		0.01 (0.01)					0.00 (0.01)
Log fixed assets			0.01 (0.01)				0.02 (0.04)
Log assets				0.01 (0.01)			-0.03 (0.06)
Export dummy					0.01 (0.01)		0.06 (0.08)
Ihs. exports						0.01 (0.01)	-0.05 (0.08)
F- p							[0.45]
N	2,013	2,013	2,007	2,013	2,013	2,013	2,007

Notes. Standard errors, two-way clustered at the match and firm levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the balance test for the baseline matching from 4 to 2 years prior to the innovation event. We regress a treatment dummy on pre-event observables: log sales, log employment, log fixed assets, log assets, export dummy, and the inverse hyperbolic sine of exports. F- p is the p -value from a joint F -test that all coefficients are zero. The smaller sample in columns 3 and 7 reflects missing observations for fixed assets.

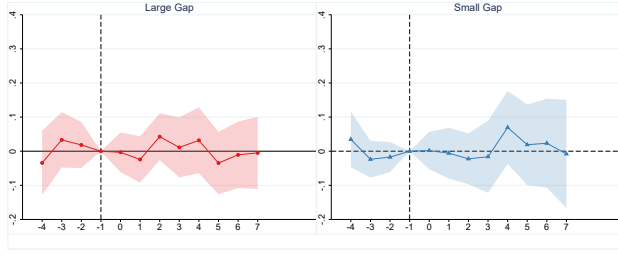
Figure B.3: No Evidence of Advantage of Backwardness for Innovation. Event Study



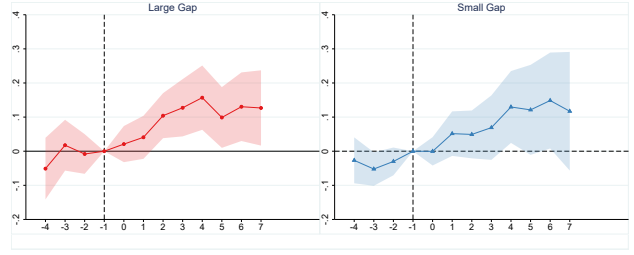
A. Log Labor Productivity



B. Log Sales



C. TFPR (ACF)



D. TFPR (CLRS)

Notes. This figure presents the estimated β_l^τ and β_s^τ in equation (3.2) for the baseline matching specification using first-time patents. Shaded bands are 90% confidence intervals based on standard errors, two-way clustered at the match and firm levels. The dependent variables are log labor productivity (Panel A), log sales (Panel B), TFPR estimated following [Akerberg et al. \(2015\)](#) (Panel C), and TFPR estimated following [Choi et al. \(2023\)](#) (Panel D).

Table B.9: Robustness. Alternative Productivity Outcomes. No Evidence of Advantage of Backwardness for Innovation

Dep. Var.	TFPR			
	ACF		CLRS	
	(1)	(2)	(3)	(4)
$\mathbb{1}[\text{Innovate}_{imt}]$	0.00 (0.03)		0.10*** (0.03)	
$\times \mathbb{1}[\text{Gap}_m^{\text{large}}]$		0.00 (0.03)		0.10*** (0.03)
$\times \mathbb{1}[\text{Gap}_m^{\text{small}}]$		0.00 (0.04)		0.10** (0.05)
$p\text{-val } (\beta_l = \beta_s)$		[0.99]		[0.99]
Dep. mean	11.70	11.70	18.26	18.26
N	8,649	8,649	8,649	8,649
Fixed effects	Firm $\times$ match, Match $\times$ year			
# Cl. (firm $\times$ match)	754 $\times$ 390			

Notes. Standard errors, two-way clustered at the firm and match levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports OLS estimates of equation (3.1). Bracketed p -values test equality of the large gap and small gap coefficients. The dependent variables are TFPR estimated following [Akerberg et al. \(2015\)](#) in columns 1–2 and [Choi et al. \(2023\)](#) in columns 3–4.

Table B.10: Robustness. Alternative Fees, Fixed Effects and Gap Measures. Adoption Fees Tend to be Higher with Smaller Gap

Robustness Dep. Var. Gap	Alt. fee		Alt. fixed effects				Log fixed fee		Alt. gap	
	Log total fee									
			Labor productivity				Market share		TFPR (ACF)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
logGap _{it}	-0.55*** (0.08)	-0.41** (0.16)	-0.49*** (0.16)	-0.32*** (0.06)	-0.48*** (0.09)	-0.28*** (0.06)	-0.37*** (0.06)	-0.42*** (0.06)	-0.26* (0.14)	-0.60*** (0.11)
Dep. mean	13.75	12.93	12.85	12.88	12.89	12.89	12.89	12.82	12.85	12.85
N	3,584	1,646	2,108	3,351	3,400	3,434	3,434	3,175	3,018	3,018
Country×sector×year FE	✓		✓					✓	✓	✓
IPC×country×sector×year FE		✓								
IPC×country×year FE			✓					✓	✓	✓
Country×year FE				✓						
Sector×year FE					✓					
Sector FE						✓				
Year FE						✓				
# Cl. (Firm×seller)	693×2,605	329×1,177	431×1,542	617×2,483	613×2,523	626×2,554	626×2,554	601×2,339	557×2,209	557×2,209

Notes. Standard errors, two-way clustered at the Korean firm and foreign seller levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports OLS estimates of equation (3.3). logGap_{it} is standardized. Column 1 uses total fees (fixed fees plus calculated royalties) as the outcome. Columns 2–7 report robustness to alternative fixed effects. Column 2 includes IPC×country×sector×year fixed effects. Column 3 includes IPC×country×year and country×sector×year fixed effects. Columns 4–7 include country×year, sector×year, sector-and-year, and year-only fixed effects, respectively. Columns 8–10 use alternative gap measures based on within-sector market shares, TFPR estimated following Akerberg et al. (2015), and TFPR estimated following Choi et al. (2023), respectively. In columns 1–7, the gap is based on labor productivity.

Table B.11: Knowledge Diffusion. Covariate Balance

	Sellers (Treated)				Non-sellers (Control)				(Col. 1 - Col. 5)	
	Mean	Med.	SD	Obs.	Mean	Med.	SD	Obs.	<i>t</i> -stat.	<i>p</i> -val
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dum. yearly new patents	0.62	1	0.49	410	0.61	1	0.49	810	0.03	[0.87]
lhs. yearly new patents	1.75	1.44	1.91	410	1.58	0.88	1.70	810	2.43	[0.12]
lhs. cum. patent stock	3.09	2.89	2.37	410	2.91	2.78	2.18	810	1.64	[0.20]
Dum. yearly new citations	0.60	1	0.49	410	0.60	1	0.49	810	0.05	[0.93]
lhs. yearly new citations	2.07	1.44	2.26	410	1.91	1.44	2.01	810	1.20	[0.21]
Dum. yearly new citations by Korean firms	0.01	0	0.12	410	0.01	0	0.08	810	1.60	[0.20]
lhs. yearly new citations by Korean firms	0.02	0	0.13	410	0.01	0	0.09	810	1.48	[0.22]
lhs. cum. citations	2.85	2.31	2.77	410	2.71	2.49	2.55	810	0.74	[0.39]
lhs. cum. citations (adj. by app. year)	3.04	2.81	2.37	410	2.88	2.78	2.19	810	1.35	[0.25]

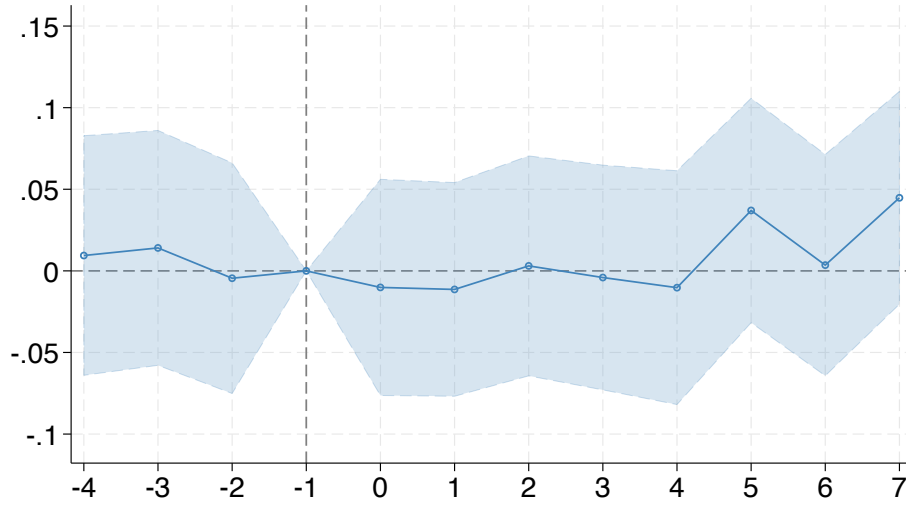
Notes. This table reports covariate balance between treated and control firms from 4 to 2 years prior to the adoption event. We report indicators and inverse hyperbolic sine transformations of key pre-event characteristics, including new patents filed, cumulative patent stock, new citations received, citations from never-adopting Korean firms, and cumulative citations (reported both in levels and adjusted by 3-digit IPC class and application year). Column 9 reports *t*-statistics for the difference in means, with *p*-values in column 10.

Table B.12: Knowledge Diffusion. Balance Test

Dep.	$\mathbb{1}[\text{Seller}_{fmt}]$									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dum. yearly new patents	0.00 (0.01)									-0.02 (0.02)
lhs. yearly new patents		0.02 (0.01)								0.01 (0.04)
lhs. cum. patent stock			0.02 (0.01)							0.04 (0.07)
Dum. yearly new citations				0.00 (0.01)						-0.02 (0.02)
lhs. yearly new citations					0.02 (0.01)					0.07 (0.06)
Dum. yearly new citations by Korean firms						0.02 (0.01)				0.03 (0.06)
lhs. yearly new citations by Korean firms							0.02 (0.01)			-0.02 (0.06)
lhs. cum. citations								0.01 (0.01)		-0.08 (0.06)
lhs. cum. citations (adj. by app. year)									0.02 (0.01)	0.00 (0.05)
F- <i>p</i>										[0.51]
N	1,220	1,220	1,220	1,220	1,220	1,220	1,220	1,220	1,220	1,220

Notes. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports the balance test from 4 to 2 years prior to the event. We regress a technology-seller dummy on pre-event observables, including new patents filed, cumulative patent stock, new citations received, citations from never-adopting Korean firms, and cumulative citations (reported both in levels and adjusted by 3-digit IPC class and application year). *F-p* is the *p*-value from a joint *F*-test that all coefficients are zero.

Figure B.4: Knowledge Diffusion. Event Study. Placebo



Notes. This figure presents the estimated event study coefficients from equation (3.5). Shaded bands are 90% confidence intervals based on standard errors, two-way clustered at the foreign seller and match levels. The dependent variable is an indicator for receiving citations from firms outside Korea.

Table B.13: Robustness. Knowledge Diffusion. Alternative Functional Forms and the Number of Matches

Robustness	Alt. functional forms				Alt. # matches	
	Log(1+y)	Level	Norm. level	Percentile	#=1	#=3
					Dummy	
Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}[\text{Seller}_{f_{mt}}]$	0.08*** (0.02)	0.36*** (0.14)	18.72*** (6.97)	0.18*** (0.05)	0.03*** (0.01)	0.04*** (0.01)
N	6,661	6,661	6,661	6,661	4,694	9,753
Fixed effects	Firm×match, Match×year					
# Cl. (firm×match)	566×204				378×198	566×204

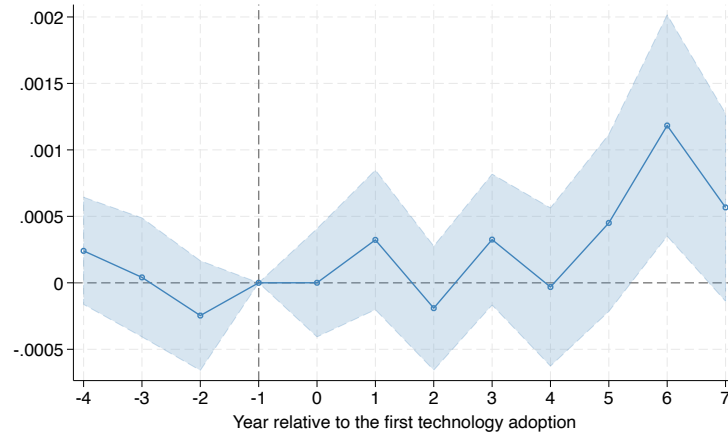
Notes. Standard errors, two-way clustered at the foreign firm and match levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports estimates of β from equation (3.4). Columns 1–4 report robustness to alternative transformations of citations, as suggested by [Chen and Roth \(2024\)](#). The dependent variables are $\log(1 + \text{citations})$, citation counts in levels, citation counts normalized by the pre-event mean, and citation percentiles, respectively. The estimated coefficient for the percentile outcome can be interpreted as the change in rank. In columns 5 and 6, which use 1 and 3 control firms per seller, respectively, the dependent variable is an indicator for receiving at least one citation from never-adopting Korean firms.

Table B.14: Robustness. Additional Fixed Effects. Knowledge Diffusion. Patent-Level Analysis

Dep. Var.	Dum. citation (KR) (1)	Ihs. citation (KR) (2)
$\mathbb{1}[\text{Seller}_{fmt}]$	0.0003* (0.0002)	0.0007* (0.0004)
Dep. mean	0.0007	0.0011
N	874,633	874,633
Fixed effects	Patent×firm×match, Match×year IPC×application-year×country×calendar-year	
# Cl. (firm×match)	549×201	

Notes. Standard errors, two-way clustered at the foreign firm and match levels, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This table reports estimates of β from equations (3.4) and (3.7). In columns 1 and 2, the dependent variables are an indicator and the inverse hyperbolic sine of citations from never-adopting Korean firms.

Figure B.5: Knowledge Diffusion. Patent-Level Analysis. Event Study



Citation Dummy

Notes. This figure presents the estimated β_τ from the event study specification: $y_{pfmt} = \sum_{\tau=-4}^7 \beta_\tau \mathbb{1}[\text{Seller}_{fmt}^\tau] + \delta_{pfm} + \delta_{mt} + \varepsilon_{pfmt}$. Shaded bands are 90% confidence intervals based on standard errors, two-way clustered at the foreign seller and match levels. The dependent variable is an indicator for whether a patent is cited by never-adopting Korean firms.

C. APPENDIX: SIMPLE MODEL

C.1 Social Planner Problem

Given productivities $(z_{hj}, z_{\tilde{h}j})$ and production labor $\tilde{L}$, the planner allocates labor within each sector to maximize C_j . With CES preferences and linear production $y_{ij} = z_{ij}l_{ij}$, the optimal allocation sets $l_{ij} \propto z_{ij}^{\sigma-1}$. Substituting into the CES aggregator yields

$$\log C_j = \frac{1}{\sigma-1} \log \left(z_{hj}^{\sigma-1} + z_{\tilde{h}j}^{\sigma-1} \right) + \log \tilde{L}.$$

Since there is a unit continuum of ex-ante identical sectors, the law of large numbers gives

$$W = \mathbb{E} \left[\frac{1}{\sigma-1} \log \left(z_h^{\sigma-1} + z_{\tilde{h}}^{\sigma-1} \right) \right] + \log \tilde{L},$$

where the expectation is over the joint distribution of $(z_h, z_{\tilde{h}})$ induced by equations (4.4)–(4.5).

The first expectation term can be rewritten as follows:

$$\mathbb{E} \left[\frac{1}{\sigma-1} \log \left(z_h^{\sigma-1} + z_{\tilde{h}}^{\sigma-1} \right) \right] = \frac{\log 2}{\sigma-1} + xT_{1x} + aT_{1a},$$

where

$$T_{1k} \equiv \frac{\delta}{\sigma-1} \log \lambda_k^{\sigma-1} + \frac{1-\delta}{\sigma-1} \log \frac{1 + \lambda_k^{\sigma-1}}{2}, \quad k \in a, x. \quad (\text{C.1})$$

The planner's objective is therefore

$$W = \text{const} + xT_{1x} + aT_{1a} + \log \tilde{L}, \quad \tilde{L} = L - \frac{\alpha}{\gamma} x^\gamma - \frac{\alpha}{\gamma} a^\gamma. \quad (\text{C.2})$$

First-Order Conditions. Differentiating (C.2) with respect to $k \in \{a, x\}$ gives the planner's first-order condition:

$$T_{1k} = \frac{\alpha k^{\gamma-1}}{\tilde{L}}, \quad k \in \{a, x\}. \quad (\text{C.3})$$

Market Clearing. Under monopolistic competition with constant markup $\frac{\sigma}{\sigma-1}$, each firm prices at $p_{ij} = \frac{\sigma}{\sigma-1} \frac{1}{z_{ij}}$. Total expenditure in sector j equals μ times the production labor employed there. Integrating over sectors, aggregate expenditure satisfies $E = \frac{\sigma}{\sigma-1} \tilde{L}$.

C.2 Proof of Proposition 1

Define $f(u) = \frac{u-1}{2(1+u)}$ and $g(u, \delta) = \delta \log u + (1 - \delta) \log \frac{1+u}{2}$ for $u > 1$, $\delta \in [0, 1)$. Setting $u_k = \lambda_k^{\sigma-1}$, note that $(\sigma - 1)T_{1k} = g(u_k, \delta)$.

Part (i). Substituting the social planner's first-order conditions (equation C.3) into the first-order conditions of firms in the competitive equilibrium, $(1 - \kappa_k)\alpha k^{\gamma-1} = (1 - \delta)\frac{u_k-1}{2(1+u_k)}\frac{E}{\sigma}$, and using $\frac{E}{\sigma} = \frac{\mu \tilde{L}}{\sigma} = \frac{\tilde{L}}{\sigma-1}$, we obtain that

$$1 - \kappa_k = \frac{(1 - \delta) f(u_k) \frac{\tilde{L}}{\sigma-1}}{\alpha k^{\gamma-1}} = \frac{(1 - \delta) f(u_k) \frac{\tilde{L}}{\sigma-1}}{T_{1k} \tilde{L}} = \frac{(1 - \delta) f(u_k)}{(\sigma - 1) T_{1k}} = \frac{(1 - \delta) f(u_k)}{g(u_k, \delta)},$$

which is equivalent to equation (4.13).

Hence, the competitive equilibrium with subsidy (4.13) satisfies the same first-order conditions as the social planner at the same allocation. Thus, market clearing conditions are also satisfied. $\square$

Part (ii). We show $\frac{\partial(1-\kappa_k)}{\partial \lambda_k} < 0$. Since $u_k = \lambda_k^{\sigma-1}$ is strictly increasing in λ_k , it suffices to show $\frac{\partial R}{\partial u} < 0$ where $R(u, \delta) \equiv \frac{(1-\delta)f(u)}{g(u, \delta)}$. This is equivalent to $\frac{f'_u}{f} < \frac{g'_u}{g}$ (i.e., $\frac{\partial \log R}{\partial u} < 0$).

Computing the derivatives:

$$\frac{f'(u)}{f(u)} = \frac{2}{(u-1)(1+u)}, \quad \frac{g_u(u, \delta)}{g(u, \delta)} = \frac{\delta/u + (1-\delta)/(1+u)}{\delta \log u + (1-\delta) \log \frac{1+u}{2}}.$$

The inequality $f'_u/f < g'_u/g$ is equivalent to

$$2u \left[\delta \log u + (1 - \delta) \log \frac{1+u}{2} \right] < (u-1)(u+\delta), \quad u > 1. \quad (\text{C.4})$$

Define $h(u, \delta) = (u-1)(u+\delta) - 2u \left[\delta \log u + (1 - \delta) \log \frac{1+u}{2} \right]$. Note $h(1, \delta) = 0$. Differentiating,

$$h_u(u, \delta) = (2u + \delta - 1) - 2g(u, \delta) - 2\delta - \frac{2(1-\delta)u}{1+u}.$$

One can verify that $h_u(1, \delta) = 0$ holds and that a second differentiation yields $h_{uu}(u, \delta) > 0$ for all $u > 1$, giving $h_u(u, \delta) > 0$ and hence $h(u, \delta) > 0$ for $u > 1$. This confirms (C.4) and thus $\frac{\partial(1-\kappa_k)}{\partial \lambda_k} < 0$.

To show that the ratio $\frac{1-\kappa_a}{1-\kappa_x}$ is decreasing in $\frac{\lambda_a}{\lambda_x}$, write the ratio as

$$\frac{1 - \kappa_a}{1 - \kappa_x} = \frac{f(u_a)/g(u_a, \delta)}{f(u_x)/g(u_x, \delta)}.$$

Fix λ_x . Since $\frac{f(u)}{g(u, \delta)}$ is strictly decreasing in u , and $u_a = \lambda_a^{\sigma-1}$ is increasing in λ_a , the ratio $\frac{1-\kappa_a}{1-\kappa_x}$ is strictly decreasing in λ_a , hence in $\frac{\lambda_a}{\lambda_x}$. $\square$

Part (iii). By an analogous argument (differentiating R with respect to δ), $\frac{\partial(1-\kappa_k)}{\partial\delta} < 0$, using that $\frac{\partial g}{\partial\delta} = \log \frac{2u}{1+u} > 0$ while $1 - \delta$ decreases in δ . Next, we show that the ratio $\frac{1-\kappa_a}{1-\kappa_x}$ is increasing in δ . Differentiating with respect to δ , the sign of $\frac{\partial[(1-\kappa_a)/(1-\kappa_x)]}{\partial\delta}$ equals the sign of $\frac{\partial \log[f(u_a)/g(u_a)]}{\partial\delta} - \frac{\partial \log[f(u_x)/g(u_x)]}{\partial\delta}$. Since $\frac{\partial \log R}{\partial\delta} = -\frac{1}{1-\delta} - \frac{\partial g/\partial\delta}{g}$ and $\frac{\partial g/\partial\delta}{g} = \frac{\log(2u/(1+u))}{g(u, \delta)}$ is decreasing in u (the numerator is bounded above by $\log 2$ while the denominator $g(u, \delta) \rightarrow \infty$), the expression $\frac{\partial \log R}{\partial\delta}$ is larger (less negative) at $u_a > u_x$, so the ratio increases in δ . $\square$

D. APPENDIX: QUANTITATIVE MODEL

D.1 Additional Expressions

In this section, we provide detailed expressions for the Euler equation, prices, firms' value functions, their optimal adoption and innovation rates, and the law of motion for technology gap distributions. For expositional convenience, we describe the model from the perspective of Home leaders, and analogous conditions hold for Foreign leaders and fringe firms.

Euler Equation. From household utility maximization, we can obtain the following Euler equation:

$$\frac{\dot{C}_{Ht}}{C_{Ht}} = \left(r_{Ht} - \frac{\dot{P}_{Ht}}{P_{Ht}} \right) - \rho.$$

Price. The price index is

$$P_{Ht} = \exp \left(\int_0^1 \log \left(\sum_{i \in \mathcal{I}_H} p_{ijt}^{1-\sigma} + \sum_{i \in \mathcal{I}_F} (p_{ijt}^*)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} dj \right),$$

where p_{ijt} and p_{ijt}^* are prices charged by Home and Foreign firms.

Their optimally charged prices are

$$p_{hjt} = \underbrace{\frac{1 - \frac{\sigma-1}{\sigma} s_{hjt}}{\frac{\sigma-1}{\sigma} (1 - s_{hjt})}}_{= \mathcal{M}_{hjt}} \frac{w_{Ht}}{z_{hjt}}, \quad p_{hjt}^* = \underbrace{\frac{1 - \frac{\sigma-1}{\sigma} s_{hjt}^*}{\frac{\sigma-1}{\sigma} (1 - s_{hjt}^*)}}_{= \mathcal{M}_{hjt}^*} \frac{\tau(1 + t_t^F) w_{Ht}}{z_{hjt}},$$

where s_{hjt} and s_{hjt}^* are their market shares in Home and Foreign:

$$s_{hjt} = \frac{p_{hjt} y_{hjt}}{\sum_{i \in \mathcal{I}_H} p_{ijt} y_{ijt} + \sum_{i \in \mathcal{I}_F} p_{ijt}^* y_{ijt}^*}, \quad s_{hjt}^* = \frac{p_{hjt}^* y_{hjt}^*}{\sum_{i \in \mathcal{I}_F} p_{ijt} y_{ijt} + \sum_{i \in \mathcal{I}_H} p_{ijt}^* y_{ijt}^*}.$$

Fringe firms charge $p_{\tilde{h}jt} = \frac{\sigma}{\sigma-1} \frac{w_{Ht}}{z_{\tilde{h}jt}}$ and $p_{\tilde{h}jt}^* = \frac{\sigma}{\sigma-1} \frac{\tau(1+t_t^F)w_{Ht}}{z_{\tilde{h}jt}}$.

Profits. Leaders' total operating profits are the sum of those earned from Home and Foreign, $\Pi_{hjt} = \pi_{hjt} + \pi_{hjt}^*$:

$$\pi_{hjt} = \frac{s_{hjt}}{\sigma - (\sigma - 1)s_{hjt}} P_{Ht} Y_{Ht}, \quad \pi_{hjt}^* = \frac{s_{hjt}^*}{\sigma - (\sigma - 1)s_{hjt}^*} P_{Ft} Y_{Ft},$$

where s_{hjt} and s_{hjt}^* are their market shares in Home and Foreign. Fringe firms' operating profits are

$$\pi_{\tilde{h}jt} = \frac{1}{\sigma} s_{\tilde{h}jt} P_{Ht} Y_{Ht}, \quad \pi_{\tilde{h}jt}^* = \frac{1}{\sigma} s_{\tilde{h}jt}^* P_{Ft} Y_{Ft}.$$

Value Function. A Home leader's value function can be expressed as follows:

$$\begin{aligned} & r_{Ht} V_{ht}(\mathbf{m}) - \dot{V}_{ht}(\mathbf{m}) \\ &= \max_{x_{ht}(\mathbf{m}), a_{ht}(\mathbf{m})} \left\{ \Pi_{ht}(\mathbf{m}) - (1 - \kappa_{Hxt}) l_{ht}^x(\mathbf{m}) w_{Ht} - (1 - \kappa_{Hat}) l_{ht}^a(\mathbf{m}) w_{Ht} \right. \\ & \quad + x_{ht}(\mathbf{m}) \Delta V_{ht}^x(\mathbf{m}) + a_{ht}(\mathbf{m}) [\Delta V_{ht}^a(\mathbf{m}) - (1 - \kappa_{Hat}) \mathcal{F}_{ht}(\mathbf{m})] \\ & \quad + a_{ft} \mathcal{F}_{ft}(\mathbf{m}) + a_{\tilde{f}t} \mathcal{F}_{\tilde{f}t}(\mathbf{m}) \\ & \quad - (x_{\tilde{h}t}(\mathbf{m}) + a_{\tilde{h}t}(\mathbf{m})) [0 - V_{ht}(\mathbf{m})] \\ & \quad \left. + \sum_{\mathbf{m}'} \mathbb{T}_{ht}(\mathbf{m}'; \mathbf{m}) [V_{ht}(\mathbf{m}') - V_{ht}(\mathbf{m})] \right\}, \end{aligned} \tag{D.1}$$

where $\Delta V_{ht}^a(\mathbf{m})$ and $\Delta V_{ht}^x(\mathbf{m})$ are changes in values after successful innovation and adoption:

$$\begin{aligned} \Delta V_{ht}^a(\mathbf{m}) &\equiv \mathbb{1}_{m^F > 0} \left(\eta V_{ht}(0, m^D + m^F, m^{DF}) + (1 - \eta) V_{ht}(m^F - 1, m^D + 1, m^{DF}) - V_{ht}(\mathbf{m}) \right), \\ \Delta V_{ht}^x(\mathbf{m}) &\equiv V_{ht}(m^F - 1, m^D + 1, m^{DF}) - V_{ht}(\mathbf{m}), \end{aligned} \tag{D.2}$$

where an indicator $\mathbb{1}_{m^F > 0}$ reflects that adoption raises productivity only when a Home leader is below a foreign seller.

The first line on the right-hand side of equation (D.1) denotes a Home leader's operating

profits net of innovation and adoption labor costs. The second line captures the net value gains from its own innovation and adoption, net of adoption fees. The third line reflects adoption fees received when a Foreign leader or fringe firm adopts from a Home leader. The fourth line represents the expected loss from displacement by successful fringe innovation or adoption. The final line captures value changes due to transitions in states, driven by foreign firms' innovation and adoption and by exogenous diffusion, governed by transition probabilities $\mathbb{T}_{ht}(\mathbf{m}'; \mathbf{m})$.

$$\mathbb{T}_{ht}(\mathbf{m}'; \mathbf{m}) = \begin{cases} x_{ft}(\mathbf{m}) + x_{\tilde{f}t}(\mathbf{m}) + (1 - \eta)(a_{ft}(\mathbf{m}) + a_{\tilde{f}t}(\mathbf{m})) & \text{if } \mathbf{m}' = \{m^F + 1, m^D, m^{DF} + 1\} \\ \eta(a_{ft}(\mathbf{m}) + a_{\tilde{f}t}(\mathbf{m})) + \delta^F & \text{if } \mathbf{m}' = \{0, m^D, m^{DF} - m^F\} \\ \delta^F & \text{if } \mathbf{m}' = \{0, m^D + m^F, m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, 0, m^{DF}\} \\ \delta^D & \text{if } \mathbf{m}' = \{m^F, m^D, 0\} \\ 0 & \text{otherwise.} \end{cases} \quad (\text{D.3})$$

Optimal Innovation and Adoption Rates. A Home leader's optimal innovation and adoption rates are

$$x_{ht}(\mathbf{m}) = \left(\frac{\Delta V_{ht}^x(\mathbf{m})}{(1 - \kappa_{Hxt})\alpha_{Hx}w_{Ht}} \right)^{\frac{1}{\gamma_x - 1}}, \quad a_{ht}(\mathbf{m}) = \left(\frac{\Delta V_{ht}^a(\mathbf{m}) - (1 - \kappa_{Hat})\mathcal{F}_{ht}(\mathbf{m})}{(1 - \kappa_{Hat})\alpha_{Ha}w_{Ht}} \right)^{\frac{1}{\gamma_a - 1}}, \quad (\text{D.4})$$

where $\Delta V_{ht}^a(\mathbf{m})$ and $\Delta V_{ht}^x(\mathbf{m})$ are value gains from successful adoption and innovation, defined in equation (D.2).

Law of Motion for Technology Gap Distributions. Let $\mu_t(\mathbf{m})$ denote a distribution of technology gaps in period t . A state variable $\mathbf{m}$ evolves according to its law of motion, $\dot{\mu}_t(\mathbf{m})$. Along the balanced growth path, $\dot{\mu}_t(\mathbf{m}) = 0$. Let $\mathbf{m}_{-n} = (m^F + n, m^D - n, m^{DF})$ and

$\mathbf{m}_{+n} = (m^F - n, m^D, m^{DF} - n)$. The law of motion is expressed as follows:

$$\begin{aligned}
\dot{\mu}_t(\mathbf{m}) = & \left(x_{ht}(\mathbf{m}_{-1}) + x_{\tilde{h}t}(\mathbf{m}_{-1}) + (1 - \eta) (a_{ht}(\mathbf{m}_{-1}) + a_{\tilde{h}t}(\mathbf{m}_{-1})) \right) \mu_t(\mathbf{m}_{-1}) \\
& + \left(x_{ft}(\mathbf{m}_{+1}) + x_{\tilde{f}t}(\mathbf{m}_{+1}) + (1 - \eta) (a_{ft}(\mathbf{m}_{+1}) + a_{\tilde{f}t}(\mathbf{m}_{+1})) \right) \mu_t(\mathbf{m}_{+1}) \\
& + \sum_{n=1}^{n=\bar{m}} \left(\eta (a_{ht}(\mathbf{m}_{-n}) + a_{\tilde{h}t}(\mathbf{m}_{-n})) + \delta^F \right) \mu_t(\mathbf{m}_{-n}) \\
& + \sum_{n=1}^{n=\bar{m}} \left(\eta (a_{ft}(\mathbf{m}_{+n}) + a_{\tilde{f}t}(\mathbf{m}_{+n})) + \delta^F \right) \mu_t(\mathbf{m}_{+n}) \\
& + \sum_{n=1}^{n=\bar{m}-m^D} \delta^D \mu_t(n, m^D + n, m^{DF}) + \sum_{n=1}^{n=\bar{m}-m^D} \delta^D \mu_t(n, m^D, m^{DF} + n) \\
& - \left(\sum_{i \in \mathcal{I}_H \cup \mathcal{I}_F} (a_{it}(\mathbf{m}) + x_{it}(\mathbf{m})) + \delta^F + 2\delta^D \right) \mu_t(\mathbf{m}).
\end{aligned} \tag{D.5}$$

The first line captures the flow of sectors whose state becomes $\mathbf{m}$ in period t because Home firms improve productivity by one step through innovation or adoption. The second line reflects transitions into $\mathbf{m}$ generated by innovation or adoption by Foreign firms. The third line captures the case in which Home firms fully catch up with the Foreign leader through adoption or through exogenous diffusion, while the fourth line describes analogous transitions generated by Foreign firms. The fifth line captures movements into $\mathbf{m}$ through domestic diffusion. The final line represents sectors that leave state $\mathbf{m}$ because of innovation, adoption, or diffusion, moving to other values of the state variables.

D.2 Equilibrium

In this section, we formally define a Markov perfect equilibrium and balanced growth path:

Definition 1. A Markov perfect equilibrium consists of $\{r_{Ct}, w_{Ct}, p_{ijt}, p_{ijt}^*, x_{ijt}, a_{ijt}, \mathcal{F}_{ijt}, T_{Ct}, C_{Ct}, Y_{Ct}, A_{Ct}, \mu_{\mathbf{m}t}\}$ such that:

- (Static equilibrium) A representative household maximizes utility; firms maximize profits; goods, labor, and asset markets clear; the balance of payments holds; and government budgets are balanced in each country and time.
- (Dynamic equilibrium) x_{ijt} and a_{ijt} solve a firm's dynamic problem; $\mathcal{F}_{ijt}$ solves Nash bargaining between buyers and sellers; and $\mu_{\mathbf{m}t}$ is consistent with x_{ijt} and a_{ijt} .

Definition 2. A balanced growth path is the equilibrium defined in Definition 1, in which $\{w_{Ct}, V_{ijt}, \mathcal{F}_{ijt}, T_{Ct}, C_{Ct}, Y_{Ct}, A_{Ct}\}$ grow at a constant rate g , and r_{Ct} and μ_t are constant over time.

D.3 Proof of Proposition 2

Plugging $L_{Ht}^p = \int_0^1 \sum_{i \in I_H} l_{ijt} dj$ into the definition of the aggregate TFP,

$$Z_{Ht} = \frac{Y_{Ht}}{L_{Ht}^p} = \frac{Y_{Ht}}{\int_0^1 \sum_{i \in I_H} \frac{y_{ijt} + \tau y_{ijt}^*}{z_{ijt}} dj} = \left(\int_0^1 \sum_{i \in I_H} \frac{y_{ijt} + \tau y_{ijt}^*}{Y_{Ht}} \frac{1}{z_{ijt}} dj \right)^{-1}.$$

Then, we can approximate $d \log Z_{Ht}$ to first order as

$$\frac{d \log Z_{Ht}}{dt} \approx \int_0^1 \sum_{i \in I_H} \frac{\partial \log Z_{Ht}}{\partial \log z_{ijt}} d \log z_{ijt},$$

where $\frac{\partial \log Z_{Ht}}{\partial \log z_{ijt}}$ can be represented as

$$\omega_{ijt} = \frac{\partial \log Z_{Ht}}{\partial \log z_{ijt}} = \frac{\frac{y_{ijt} + \tau y_{ijt}^*}{Y_{Ht}} \frac{1}{z_{ijt}}}{\left(\int_0^1 \sum_{i \in I_H} \frac{y_{ijt} + \tau y_{ijt}^*}{Y_{Ht}} \frac{1}{z_{ijt}} dj \right)}.$$

The numerator of ω_{ijt} can be expressed as

$$\frac{y_{ijt} + \tau y_{ijt}^*}{Y_{Ht}} \frac{1}{z_{ijt}} = \frac{\frac{p_{ijt} y_{ijt}}{P_{Ht} Y_{Ht}}}{\frac{p_{ijt} z_{ijt}}{w_{Ht}} w_{Ht}} + \frac{\frac{p_{ijt}^* y_{ijt}^*}{P_{Ft} Y_{Ft}} \frac{P_{Ft} Y_{Ft}}{P_{Ht} Y_{Ht}}}{\frac{p_{ijt}^* z_{ijt}}{\tau(1+t_f^F) w_{Ht}} w_{Ht}} = \frac{P_{Ht} Y_{Ht} + P_{Ft} Y_{Ft}}{P_{Ht} Y_{Ht}} \left(\frac{S_{Ht} s_{ijt}}{\mathcal{M}_{ijt} w_{Ht}} + \frac{(1 - S_{Ht}) s_{ijt}^*}{\mathcal{M}_{ijt}^* w_{Ht}} \right),$$

where the first equality uses $P_{Ht} = 1$ and the definitions of revenue shares and markups, and the second equality uses $S_{Ht} = \frac{P_{Ht} Y_{Ht}}{P_{Ht} Y_{Ht} + P_{Ft} Y_{Ft}}$.

Because markups and market shares are functions of $\mathbf{m}$, ω_{ijt} is also a function of $\mathbf{m}$. Therefore, ω_{ijt} can be expressed as

$$\omega_{ijt} = \omega_{it}(\mathbf{m}) = \frac{\left(\frac{S_{Ht} s_{it}(\mathbf{m})}{\mathcal{M}_{it}(\mathbf{m})} + \frac{(1 - S_{Ht}) s_{it}^*(\mathbf{m})}{\mathcal{M}_{it}^*(\mathbf{m})} \right)}{\left(\sum_{\mathbf{m}} \mu_t(\mathbf{m}) \sum_{i \in I_H} \frac{S_{Ht} s_{it}(\mathbf{m})}{\mathcal{M}_{it}(\mathbf{m})} + \frac{(1 - S_{Ht}) s_{it}^*(\mathbf{m})}{\mathcal{M}_{it}^*(\mathbf{m})} \right)}, \quad (\text{D.6})$$

where the common terms w_{Ht} and $\frac{P_{Ht} Y_{Ht} + P_{Ft} Y_{Ft}}{P_{Ht} Y_{Ht}}$ are canceled out in the numerator and the denominator.

$d \log Z_{Ht}$ can be approximated as follows:

$$\begin{aligned}
\frac{d \log Z_{Ht}}{dt} &\approx \int_0^1 \sum_{i \in \mathcal{I}_H} \omega_{ijt} d \log z_{ijt} \\
&= \sum_{\mathbf{m}} \int_{\mathbf{m}_j = \mathbf{m}} \left(\omega_{ht}(\mathbf{m}_j) d \log z_{hjt}(\mathbf{m}_j) + \omega_{\tilde{h}t}(\mathbf{m}_j) d \log z_{\tilde{h}jt}(\mathbf{m}_j) \right) d\mathbf{j} \quad (\text{D.7}) \\
&= \sum_{\mathbf{m}} \mu_t(\mathbf{m}) \left(\omega_{ht}(\mathbf{m}) d \log z_{hjt}(\mathbf{m}) + \omega_{\tilde{h}t}(\mathbf{m}) d \log z_{\tilde{h}jt}(\mathbf{m}) \right)
\end{aligned}$$

Next, we decompose $d \log z_{ijt}$. As the innovation and adoption rates are functions of $\mathbf{m}$, $d \log z_{ijt}$ is also a function of $\mathbf{m}$. Leader firms raise their productivity z_{hjt} through innovation, adoption, or exogenous international diffusion, which arrive at rates $x_{ht}(\mathbf{m})$, $a_{ht}(\mathbf{m})$, and δ^F . Fringe firms can also innovate and adopt; because any successful innovation or adoption by a fringe firm replaces the incumbent leader at rates $x_{\tilde{h}t}(\mathbf{m})$ and $a_{\tilde{h}t}(\mathbf{m})$, these activities likewise increase z_{hjt} . Therefore, $d \log z_{hjt}$ can be written as a function of the innovation rate ($x_{ht}(\mathbf{m})$, $x_{\tilde{h}t}(\mathbf{m})$), adoption rates ($a_{ht}(\mathbf{m})$, $a_{\tilde{h}t}(\mathbf{m})$), and the arrival rate of international diffusion δ^F as follows:

$$\begin{aligned}
d \log z_{hjt}(\mathbf{m}) &= \left[(x_{ht}(\mathbf{m}) + x_{\tilde{h}t}(\mathbf{m})) \right. \\
&\quad \left. + (a_{ht}(\mathbf{m}) + a_{\tilde{h}t}(\mathbf{m})) \mathbb{1}_{m^F > 0} ((1 - \eta) + \eta m^F) + \delta^F \max\{m^F, 0\} \right] \log \lambda + o(\Delta t), \quad (\text{D.8})
\end{aligned}$$

where the first term indicates the step size increase from innovation, the second from adoption, and the last one from exogenous international diffusion. $o(\Delta t)$ is the second-order term, representing the probability of firms experiencing more than one event. $\lim_{\Delta t \rightarrow 0} o(\Delta t)/\Delta t = 0$ holds in continuous time.

In addition, the productivity of fringe firms $z_{\tilde{h}jt}$ increases through exogenous domestic diffusion, which arrives at rate δ^D . $d \log z_{\tilde{h}jt}$ can be represented as a function of the arrival rate of domestic diffusion δ^D , as:

$$d \log z_{\tilde{h}jt}(\mathbf{m}) = \delta^D m^D \log \lambda + o(\Delta t), \quad (\text{D.9})$$

Combining equations (D.6), (D.7), (D.8), and (D.9), we obtain the expression in equation (5.7) in Proposition 2. $\square$

D.4 Endogenous Labor Supply

For robustness, we extend the baseline model to incorporate endogenous labor supply. Households have the following separable preference:

$$U_{Ht} = \int_t^\infty e^{-\rho(s-t)} \left(\log C_{Hs} - \psi_{Hs} \frac{h_{Hs}^{1+\iota}}{1+\iota} \right) ds,$$

where h_{Hs} denotes hours worked. Total labor supply is given by $L_{Ht} = h_{Ht} \bar{L}_{Ht}$, where $\bar{L}_{Ht}$ is the exogenously given human-capital-adjusted population of the Home country. Under this specification, the Euler equation remains unchanged, while optimal hours worked satisfy

$$h_{Ht} = \left(\frac{w_{Ht} \bar{L}_{Ht}}{\psi_{Ht} P_{Ht} C_{Ht}} \right)^{\frac{1}{\iota}}.$$

We set $\iota = 2$, following [Chetty et al. \(2011\)](#). For each discretized period, we calibrate ψ_{Ht} to match average hours worked in the data, as shown in Figure E.3. In the counterfactual exercises, we hold ψ_{Ht} fixed and allow hours worked to adjust according to the household problem. The welfare gains from adoption and innovation subsidies are larger than under exogenous labor supply, as reported in Panel J of Table E.1.

E. APPENDIX: QUANTIFICATION

E.1 Computational Algorithm

Balanced Growth Path. On the balanced growth path, wage and consumption in each country grow at the same rate g , while the distribution of gaps $\mu_t(\mathbf{m})$, innovation rate $x_{it}(\mathbf{m})$, adoption rate $a_{it}(\mathbf{m})$, and the relative price P_{Ht}/P_{Ft} stay the same. It is useful to divide equation (D.1) by $P_{Ht}Y_{Ht}$ and define $\tilde{V}_{it} = \frac{V_{it}}{P_{Ht}Y_{Ht}}$, $\tilde{w}_{Ht} = \frac{w_{Ht}}{P_{Ht}Y_{Ht}}$, and $\tilde{\mathcal{F}}_{ijt} = \frac{\mathcal{F}_{ijt}}{P_{Ht}Y_{Ht}}$ as normalized value function, wage, and adoption fee, respectively. Also, define the aggregate output share in each country as $S_{Ht} = \frac{P_{Ht}Y_{Ht}}{P_{Ht}Y_{Ht} + P_{Ft}Y_{Ft}}$ and represent the profit function as follows:

$$\begin{aligned} \Pi_{it}(\mathbf{m}) &= \tilde{\pi}_{it}(\mathbf{m}) \times P_{Ht}Y_{Ht} + \tilde{\pi}_{it}^*(\mathbf{m}) \times P_{Ft}Y_{Ft} \\ \tilde{\Pi}_{it} &= \frac{\Pi_{it}(\mathbf{m})}{P_{Ht}Y_{Ht}} = \tilde{\pi}_{it}(\mathbf{m}) + \tilde{\pi}_{it}^*(\mathbf{m}) \times \frac{1 - S_{Ht}}{S_{Ht}}, \end{aligned}$$

where $\pi_{it}(\mathbf{m})$ and $\pi_{it}^*(\mathbf{m})$ are normalized profits in domestic and foreign countries. Then, the normalized value function for $i \in \mathcal{I}_H$ can be expressed as follows:

$$\begin{aligned}
& (r_{Ht} - g_t) \tilde{V}_{ht}(\mathbf{m}) \\
&= \max_{x_{ht}(\mathbf{m}), a_{ht}(\mathbf{m})} \left\{ \tilde{\Pi}_{ht}(\mathbf{m}) - (1 - \kappa_{Hxt}) l_{ht}^x(\mathbf{m}) \tilde{w}_{Ht} - (1 - \kappa_{Hat}) l_{ht}^a(\mathbf{m}) \tilde{w}_{Ht} \right. \\
&\quad + x_{ht}(\mathbf{m}) \Delta \tilde{V}_{ht}^x(\mathbf{m}) + a_{ht}(\mathbf{m}) [\Delta \tilde{V}_{ht}^a(\mathbf{m}) - (1 - \kappa_{Hat}) \tilde{\mathcal{F}}_{ht}(\mathbf{m})] \\
&\quad + a_{ft} \tilde{\mathcal{F}}_{ft}(\mathbf{m}) + a_{\tilde{f}t} \tilde{\mathcal{F}}_{\tilde{f}t}(\mathbf{m}) \\
&\quad - (x_{\tilde{h}t}(\mathbf{m}) + a_{\tilde{h}t}(\mathbf{m})) [0 - \tilde{V}_{ht}(\mathbf{m})] \\
&\quad \left. + \sum_{\mathbf{m}'} \mathbb{T}_{ht}(\mathbf{m}'; \mathbf{m}) [\tilde{V}_{ht}(\mathbf{m}') - \tilde{V}_{ht}(\mathbf{m})] \right\}, \tag{E.1}
\end{aligned}$$

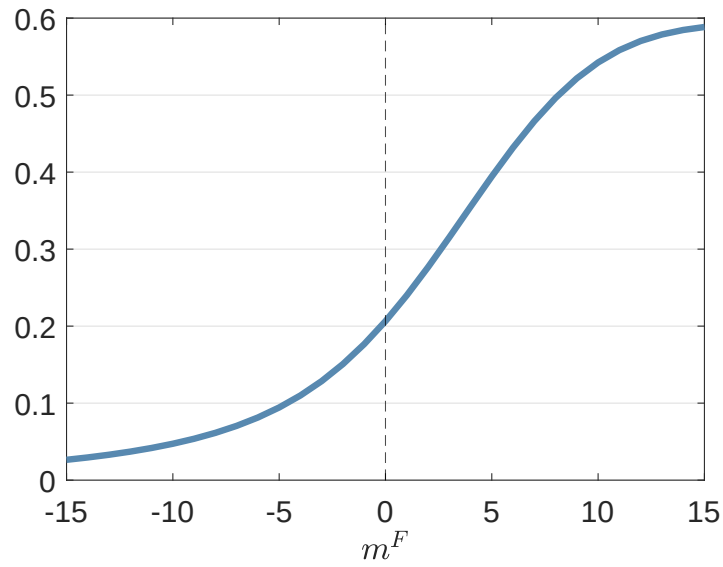
Note that from the household Euler equation, we know $r_{Ht} - g_t = \rho$ for any t . We solve the balanced growth path in two steps. First, we set an initial guess of $\{\tilde{w}_H, \tilde{w}_F, S_H\}$. Then, we form a guess for the value function for each $\mathbf{m}$, and iterate until it converges using equation (E.1). After the normalized value functions converge, we verify the labor market clearing conditions for each country and the trade balance conditions. We update these three variables until the labor market clears in each country and trade is balanced.

Transitional Dynamics. We solve the model's transition as follows.

1. We discretize the continuous-time model, setting $\Delta t = 2^{-5}$.
2. We solve the balanced growth path. We assume the economy converges to it by period T .
3. We form an initial guess for $\mathbb{X}_t^0 = \{\tilde{w}_{Ht}, \tilde{w}_{Ft}, S_{Ht}\}_{t=0}^{t=T}$.
4. Given the guess, we solve value functions, innovation rates, and adoption rates backward from period T to period 0.
5. Given the innovation and adoption decisions, we solve the distribution of technology gaps $\{\mu_t(\mathbf{m}_h)\}_{t=0}^{t=T}$ forward from period 0 to period T . $\mu_{H\mathbf{m}0}$ is given as the initial condition. We also compute the implied $\tilde{\mathbb{X}}_t^1 = \{\tilde{w}_{Ht}, \tilde{w}_{Ft}, S_{Ht}\}_{t=0}^{t=T}$ using $\{\mu_t(\mathbf{m}_h)\}_{t=0}^{t=T}$.
6. We compute the distance $\|\mathbb{X}_t^0 - \tilde{\mathbb{X}}_t^1\|$ between the guess and the implied value. We use the Euclidean norm.
7. We update the guess according to $\mathbb{X}_t^{i+1} = (1 - \Delta)\mathbb{X}_t^i + \Delta\tilde{\mathbb{X}}_t^{i+1}$, until $\|\mathbb{X}_t^0 - \tilde{\mathbb{X}}_t^1\| < \varepsilon$, where $0 < \Delta < 1$ is the damping parameter.
8. Once we find the equilibrium $\mathbb{X}$, we simulate the transition using the distribution $\mu_{H\mathbf{m}t}$ and compute Y_{Ht} .

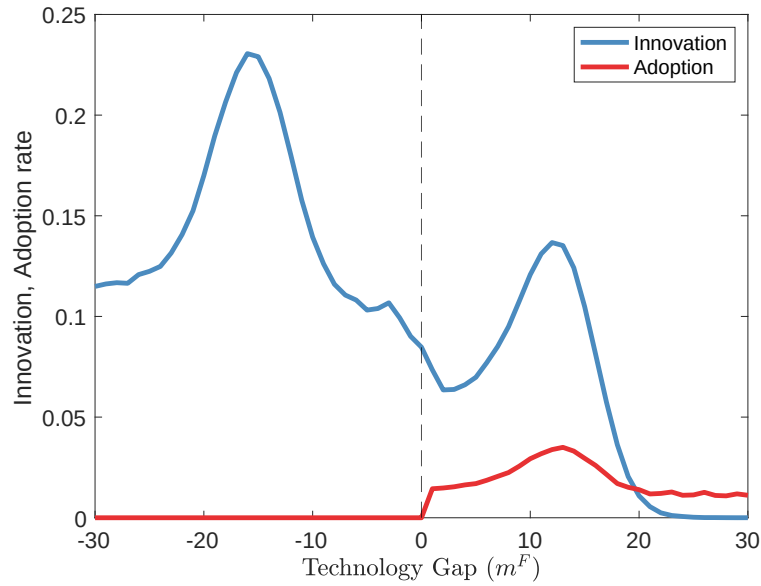
E.2 Additional Figures and Tables

Figure E.1: Profit Function over Technology Gaps



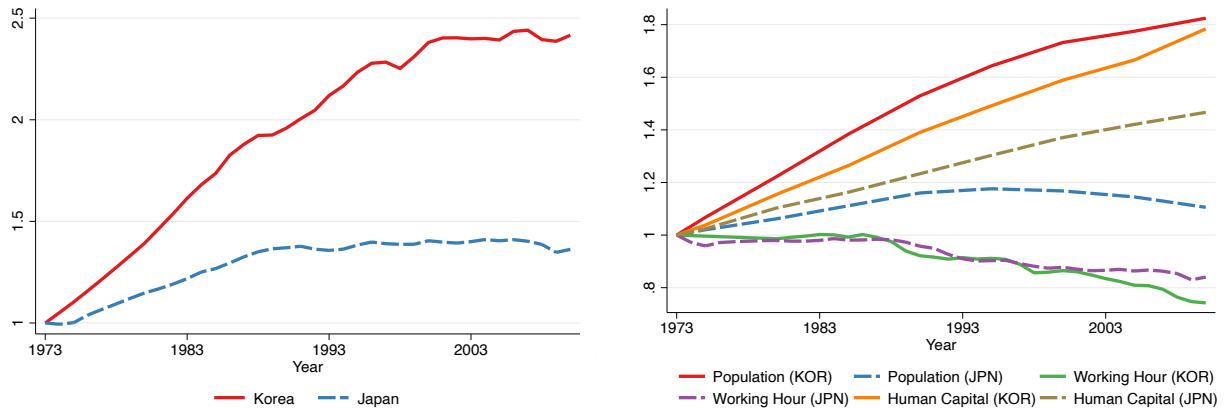
Notes. This figure plots the profits of a Home leader on the y-axis and the technology gap relative to a Foreign leader on the x-axis, holding all other variables fixed.

Figure E.2: Innovation Rates across Gaps: Defensive and Expansionary Motives



Notes. This figure plots innovation and adoption rates across m^F at the balanced growth path. The left and right peaks capture expansionary and defensive R&D efforts, respectively. For each m^F , we take the average of rates over m^D and m^{DF} , weighted by $\mu(\mathbf{m})$. Adoption rates are zero for $m^F \leq 0$.

Figure E.3: Labor Endowment in Korea and Japan over Time

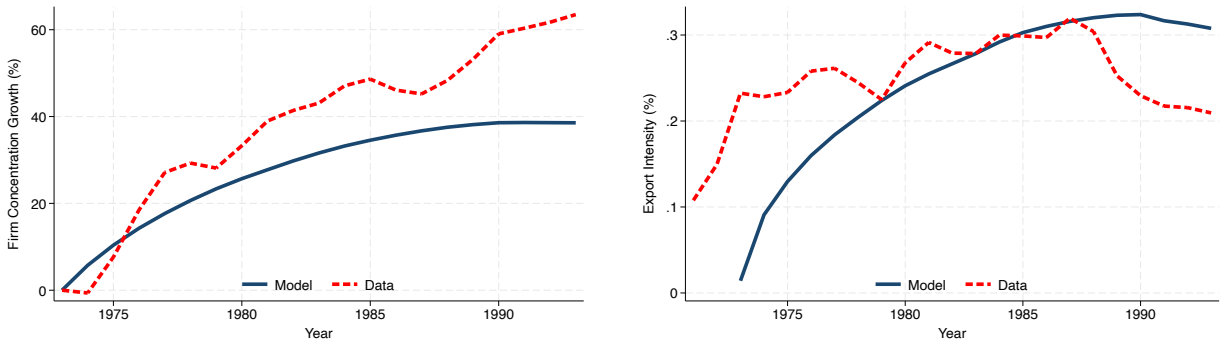


A. Labor Endowment

Notes. Panel A plots the total labor endowment in Korea and Japan over time, and Panel B separately presents population, the average working hours per person, and human capital measures. Population and human capital measures are from [Lee and Lee \(2016\)](#) and average working hours per person is from FRED. All variables are normalized to one in the initial year.

B. By Components

Figure E.4: Untargeted Moments: Firm Concentration Growth and Export Expansion

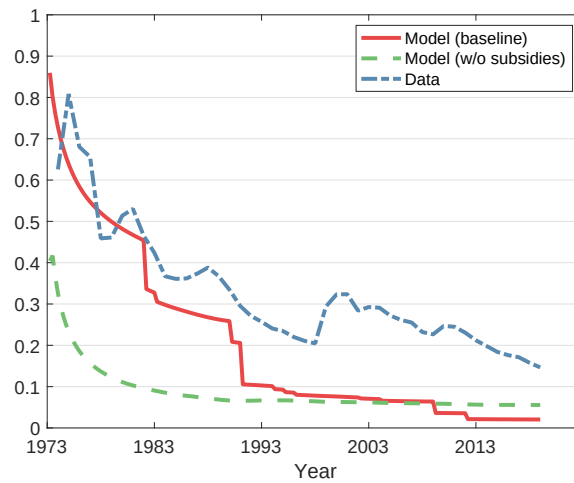


A. Firm Concentration Growth

B. Export-Led Growth

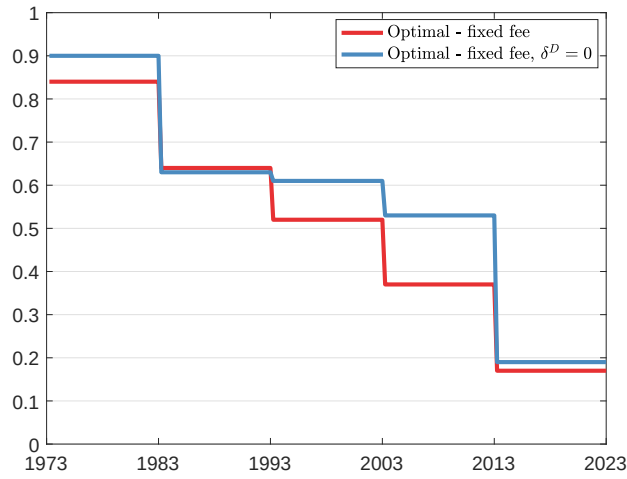
Notes. This figure plots time-series patterns of two untargeted moments. Panels A and B plot firm concentration growth and export intensity over time, respectively. In Panel A, to compare firm concentration growth between the model and the data, in the model we compute leaders' sales as a share of total gross output; in the data, we compute sales shares of firms that have ever engaged in adoption or innovation and compare the two shares.

Figure E.5: Adoption Expenditure Share in the Model and the Data



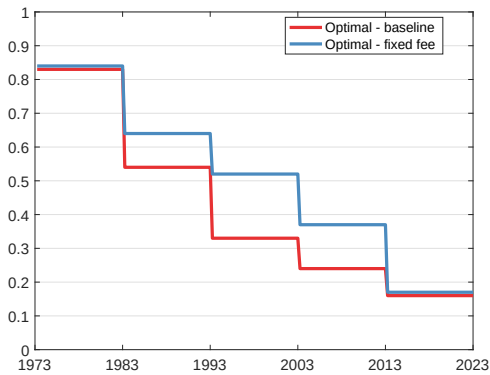
Notes. This figure plots the adoption fee expenditure / (adoption fee + innovation cost) in the model and the data.

Figure E.6: Role of Domestic Diffusion in the Optimal Subsidy Path



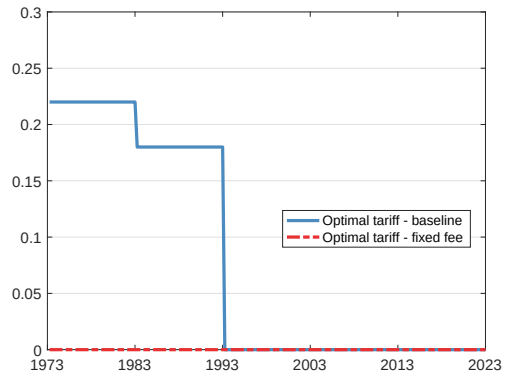
Notes. The figure plots optimal adoption subsidy share paths under fixed adoption fees, which are calibrated to match the average adoption rate under $\delta^D > 0$ and $\delta^D = 0$, respectively. The subsidy-to-GDP ratio is held fixed as is the actual policy.

Figure E.7: Optimal Policy under Endogenous vs. Fixed Fees



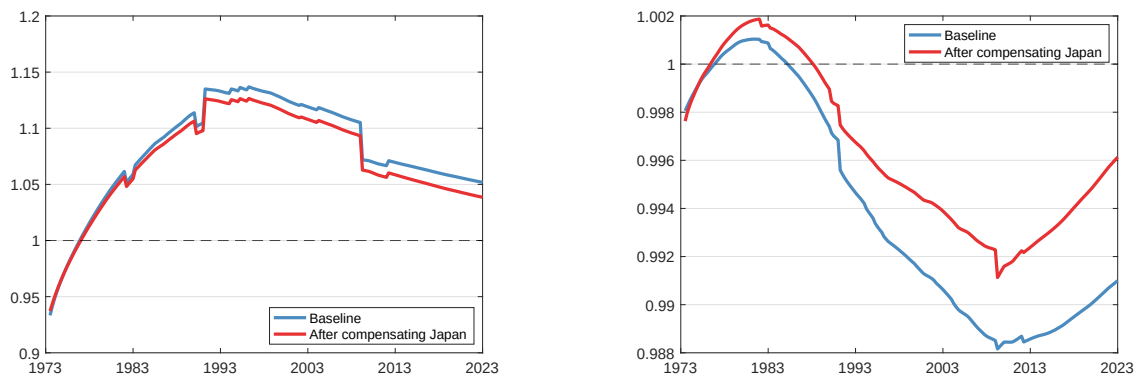
A. Optimal Subsidy Share

Notes. Panel A plots the optimal subsidy share path under endogenous and fixed fees, holding tariffs at their actual values. Panel B plots the optimal tariff path, holding subsidy rates at their actual values.



B. Optimal Tariff

Figure E.8: Consumption Paths under the Baseline Actual Policy vs. the Scenario in which Japan's Profit Loss Is Compensated by Korea through a Lump-Sum Transfer

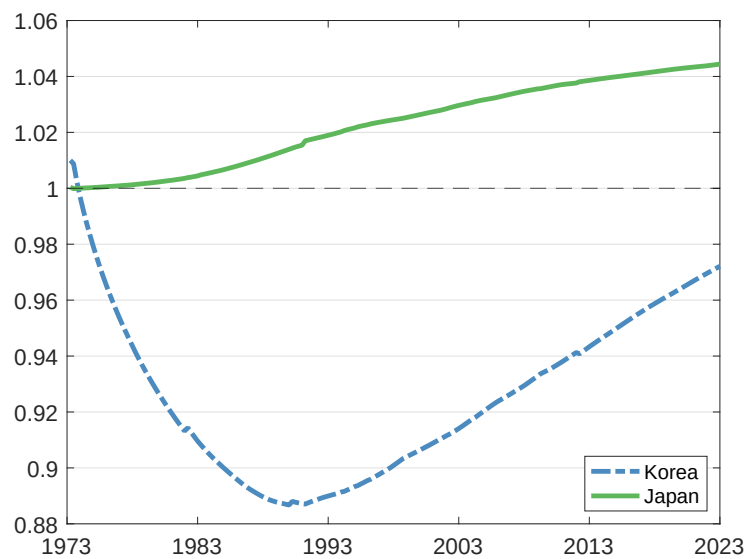


A. Korea

B. Japan

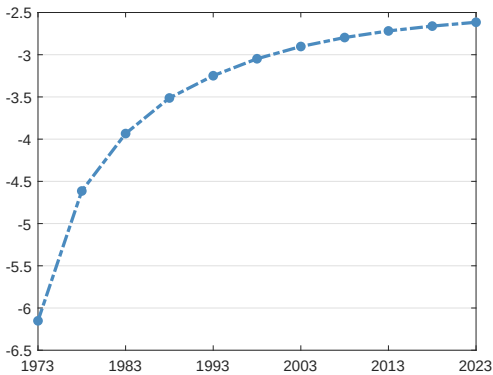
Notes. The figure plots consumption paths for Korea and Japan under the actual policy baseline and the scenario in which Japan's profit shares remain constant through lump-sum transfers from Korea to Japan.

Figure E.9: Consumption Paths under a Foreign Ban on Technology Transfers

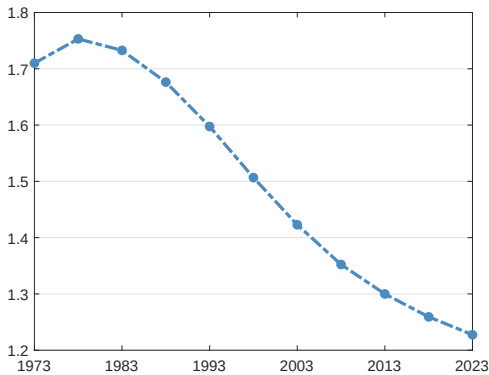


Notes. The figure plots consumption paths for Korea and Japan relative to the actual policy baseline. In this counterfactual, all technology transfers from Japan to Korea are banned, and Korea reallocates the same subsidy-to-GDP ratio to innovation.

Figure E.10: Welfare Effects of a Foreign Ban on Technology Transfers by Timing



A. Δ Welfare of Korea (%)



B. Δ Welfare of Japan (%)

Notes. This figure plots the welfare effects of a foreign ban on technology transfers relative to the actual policy baseline. Transfers are permanently banned from the listed year onward, and Korea reallocates the same subsidy-to-GDP ratio to innovation.

Table E.1: Sensitivity Checks

	Δ Welfare (%)		
	Innovation only	Adoption only	Actual policy
Baseline	3.63	4.62	6.22
<i>Panel A. Discount factor (Baseline $\rho = 0.03$)</i>			
$\rho = 0.02$	3.63	2.63	5.07
$\rho = 0.04$	2.49	4.61	5.52
<i>Panel B. Deadweight cost (Baseline $\theta = 1$)</i>			
$\theta = 0$	5.33	6.37	7.99
$\theta = 2$	1.92	2.88	4.46
<i>Panel C. Elasticity of substitution (Baseline $\sigma = 8$)</i>			
$\sigma = 6$	3.41	4.62	5.93
$\sigma = 10$	3.33	4.36	5.91
<i>Panel D. Convexity of R&D cost (Baseline $\gamma = 1.5$)</i>			
$\gamma = 1.45$	2.92	3.55	5.17
$\gamma = 1.55$	3.79	4.72	6.34
<i>Panel E. Bargaining power of adopter (Baseline $\xi = 0.85$)</i>			
$\xi = 0.7$	2.68	2.88	4.55
$\xi = 1$	4.51	6.98	8.20
<i>Panel F. Iceberg trade cost (Baseline $\tau = 1.50$)</i>			
$\tau = 1.3$	3.58	3.58	5.54
$\tau = 1.7$	3.37	5.39	6.65
<i>Panel G. Exogenous international diffusion (Baseline $\delta^F = 0.029$)</i>			
$\delta^F = 0.025$	4.51	6.39	7.91
$\delta^F = 0.033$	3.02	3.33	4.97
<i>Panel H. Exogenous domestic diffusion (Baseline $\delta^D = 0.044$)</i>			
$\delta^D = 0.034$	3.50	4.79	6.19
$\delta^D = 0.054$	3.77	4.47	6.28
<i>Panel I. Maximum technology gap (Baseline $\bar{m} = 30, \tilde{m} = 3$)</i>			
$\bar{m} = 35$	3.60	4.66	6.24
$\tilde{m} = 5$	3.58	3.37	5.39
<i>Panel J. Endogenous labor supply</i>			
	4.76	5.67	7.30

Notes. Panels A–I report sensitivity checks for alternative parameter values. Panel J reports results with endogenous labor supply (see Section D.4 for details).

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